

# Likelihood

## 4.1 Likelihood

### 4.1.1 Definition and examples

Suppose we have observed the value  $y$  of a random variable  $Y$ , whose probability density function is supposed known up to the value of a parameter  $\theta$ . We write  $f(y; \theta)$  to emphasize that the density is a function of both data and a parameter. In general both  $y$  and  $\theta$  will be vectors whose respective elements we denote by  $y_j$  and  $\theta_r$ . The parameter takes values in a *parameter space*  $\Theta$ , and the data  $Y$  take values in a *sample space*  $\mathcal{Y}$ . Our goal is to make statements about the distribution of  $Y$ , based on the observed data  $y$ . The assumption that  $f$  is known apart from uncertainty about  $\theta$  reduces the problem to making statements about what range of values of  $\theta$  within  $\Theta$  is plausible, given that  $y$  has been observed.

A fundamental tool is the *likelihood* for  $\theta$  based on  $y$ , which is defined to be

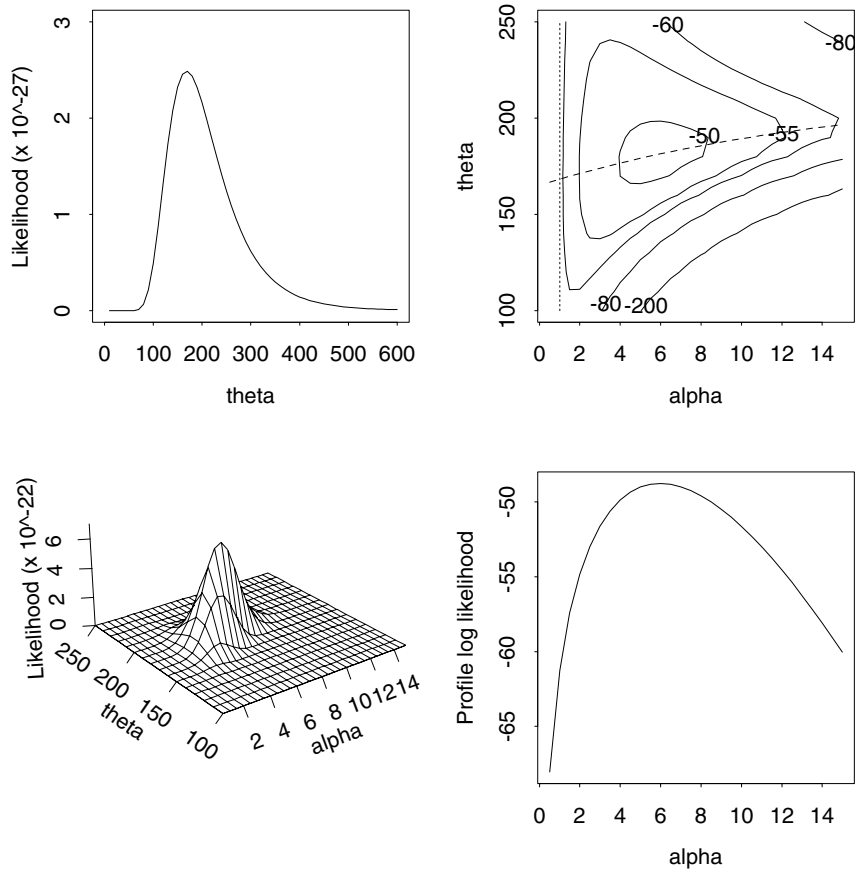
$$L(\theta) = f(y; \theta), \quad \theta \in \Theta, \quad (4.1)$$

regarded as a function of  $\theta$  for fixed  $y$ . Our interest in this is motivated by the idea that it will be relatively larger for values of  $\theta$  near that which generated the data. When  $Y$  is discrete we use  $f(y; \theta) = \Pr(Y = y; \theta)$ , while if  $Y$  is continuous, we take  $f(y; \theta)$  to be its probability density function. Owing to rounding, the recorded  $y$  is always discrete in practice, and occasional minor difficulties can be avoided by taking this into account, as we shall see in Example 4.42. However in constructing (4.1) for continuous  $Y$  we almost always use its density function. When  $y = (y_1, \dots, y_n)$  is a collection of independent observations the likelihood is

$$L(\theta) = f(y; \theta) = \prod_{j=1}^n f(y_j; \theta). \quad (4.2)$$

**Example 4.1 (Poisson distribution)** Suppose that  $y$  consists of a single observation from the Poisson density (2.6). Here the data and the parameter are both scalars, and  $L(\theta) = \theta^y e^{-\theta} / y!$ . The parameter space is  $\{\theta : \theta > 0\}$  and the sample space is

**Figure 4.1** Likelihoods for the spring failure data at stress 950 N/mm<sup>2</sup>. The upper left panel is the likelihood for the exponential model, and below it is a perspective plot of the likelihood for the Weibull model. The upper right panel shows contours of the log likelihood for the Weibull model; the exponential likelihood is obtained by setting  $\alpha = 1$ , that is, slicing  $L$  along the vertical dotted line. The lower right panel shows the profile log likelihood for  $\alpha$ , which corresponds to the log likelihood values along the dashed line in the panel above, plotted against  $\alpha$ .



$\{0, 1, 2, \dots\}$ . If  $y = 0$ ,  $L(\theta)$  is a monotonic decreasing function of  $\theta$ , whereas if  $y > 0$  it has a maximum at  $\theta = y$ , and limit zero as  $\theta$  approaches zero or infinity. ■

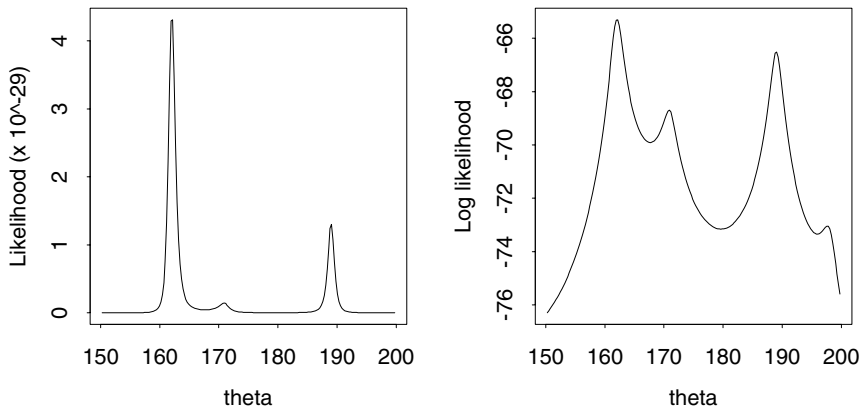
**Example 4.2 (Exponential distribution)** Let  $y$  be a random sample  $y_1, \dots, y_n$  from the exponential density  $f(y; \theta) = \theta^{-1}e^{-y/\theta}$ ,  $y > 0, \theta > 0$ . The parameter space is  $\Theta = \mathbb{R}_+$  and the sample space the Cartesian product  $\mathbb{R}_+^n$ . Here (4.2) gives

$$L(\theta) = \prod_{j=1}^n \theta^{-1} e^{-y_j/\theta} = \theta^{-n} \exp\left(-\frac{1}{\theta} \sum_{j=1}^n y_j\right), \quad \theta > 0. \quad (4.3)$$

The spring failure times at stress 950 N/mm<sup>2</sup> in Example 1.2 are

225, 171, 198, 189, 189, 135, 162, 135, 117, 162,

and the top left panel of Figure 4.1 shows the likelihood (4.3). The function is unimodal and is maximized at  $\theta \doteq 168$ ;  $L(168) \doteq 2.49 \times 10^{-27}$ . At  $\theta = 150$ ,  $L(\theta)$  equals  $2.32 \times 10^{-27}$ , so that 150 is  $2.32/2.49 = 0.93$  times less likely than  $\theta = 168$  as an explanation for the data. If we were to declare that any value of  $\theta$  for which



**Figure 4.2** Cauchy likelihood and log likelihood for the spring failure data at stress 950N/mm<sup>2</sup>.

$L(\theta) > cL(168)$  was “plausible” based on these data, values of  $\theta$  in the range  $(120, 260)$  or so would be plausible when  $c = \frac{1}{2}$ . ■

**Example 4.3 (Cauchy distribution)** The Cauchy density centered at  $\theta$  is  $f(y; \theta) = [\pi\{1 + (y - \theta)^2\}]^{-1}$ , where  $y \in \mathbb{R}$  and  $\theta \in \mathbb{R}$ . Hence the likelihood for a random sample  $y_1, \dots, y_n$  is

$$L(\theta) = \prod_{j=1}^n \frac{1}{\pi\{1 + (y_j - \theta)^2\}}, \quad -\infty < \theta < \infty.$$

The sample space is  $\mathbb{R}^n$  and the parameter space is  $\mathbb{R}$ .

The left panel of Figure 4.2 shows  $L(\theta)$  for the spring data in Example 4.2. There seem to be three local maxima in the range for which  $L(\theta)$  is plotted, with a global maximum at  $\theta \doteq 162$ . We can see more detail in the log likelihood  $\log L(\theta)$  shown in the right panel of the figure. There are at least four local maxima — apparently one at each observation, with a more prominent one when observations are duplicated. By contrast with the previous example, for some values of  $c$  a “plausible” set for  $\theta$  here consists of disjoint intervals. ■

**Example 4.4 (Weibull distribution)** The Weibull density is

$$f(y; \theta, \alpha) = \frac{\alpha}{\theta} \left(\frac{y}{\theta}\right)^{\alpha-1} \exp\left\{-\left(\frac{y}{\theta}\right)^\alpha\right\}, \quad y > 0, \quad \theta, \alpha > 0. \quad (4.4)$$

When  $\alpha = 1$  this is the exponential density of Example 4.2; the exponential model is nested within the Weibull model, the parameter space for which is  $\mathbb{R}_+^2$ , and the sample space for which is  $\mathbb{R}_+^n$ .

A random sample  $y = (y_1, \dots, y_n)$  from (4.4) has joint density

$$f(y; \theta, \alpha) = \prod_{j=1}^n f(y_j; \theta, \alpha) = \prod_{j=1}^n \left[ \frac{\alpha}{\theta} \left(\frac{y_j}{\theta}\right)^{\alpha-1} \exp\left\{-\left(\frac{y_j}{\theta}\right)^\alpha\right\} \right]$$

and hence the likelihood is

$$L(\theta, \alpha) = \frac{\alpha^n}{\theta^{n\alpha}} \left( \prod_{j=1}^n y_j \right)^{\alpha-1} \exp \left\{ - \sum_{j=1}^n \left( \frac{y_j}{\theta} \right)^\alpha \right\}, \quad \theta, \alpha > 0. \quad (4.5)$$

The lower left panel of Figure 4.1 shows  $L(\theta, \alpha)$  for the data of Example 4.2. The likelihood is maximized at  $\theta \doteq 181$  and  $\alpha \doteq 6$ , and  $L(181, 6)$  equals  $6.7 \times 10^{-22}$ . This is  $2.7 \times 10^5$  times greater than the largest value for the exponential model. The top right panel shows contours of the log likelihood,  $\log L(\theta, \alpha)$ . The dotted line indicates the slice corresponding to the exponential density obtained when  $\alpha = 1$ . The factor  $2.5 \times 10^5$  gives a difference of  $\log(2.7 \times 10^5) = 12.5$  between the maximum log likelihoods. This big improvement suggests that the Weibull model fits the data better. However, if we judge model fit by the maximum likelihood value, the Weibull model is bound to fit at least as well as the exponential, because  $\max_{\theta, \alpha} L(\theta, \alpha) \geq \max_{\theta} L(\theta, 1)$ , with equality only if the maximum occurs on the line  $\alpha = 1$ . ■

The examples above involve random samples, but (4.1) and (4.2) apply also to more complex situations.

**Example 4.5 (Challenger data)** Consider the data in Table 1.3 on O-ring thermal distress. For now we ignore the effect of pressure, and treat the temperature  $x_1$  at launch as fixed and the number of O-rings with thermal distress as binomial variables with denominator  $m$  and probability  $\pi$ , giving

$$\Pr(R = r) = \frac{m!}{r!(m-r)!} \pi^r (1-\pi)^{m-r}, \quad r = 0, 1, \dots, m.$$

If  $\pi$  depends on temperature through the relation

$$\pi = \frac{\exp(\beta_0 + \beta_1 x_1)}{1 + \exp(\beta_0 + \beta_1 x_1)},$$

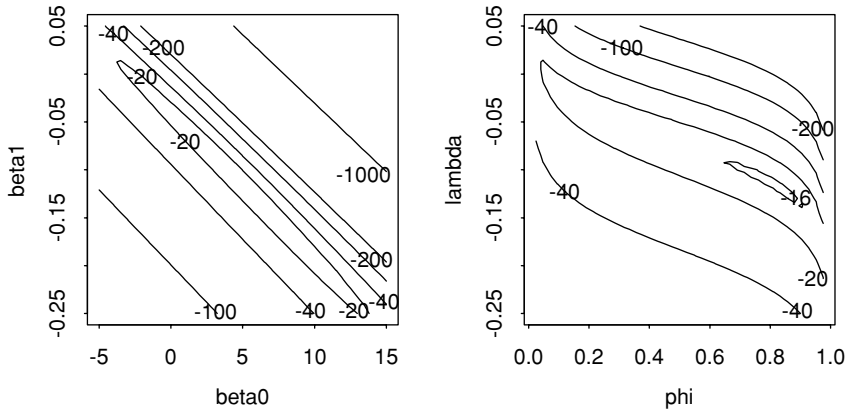
then the parameter  $\beta_0$  determines the probability of thermal distress when  $x_1 = 0^\circ\text{F}$ , which is  $e^{\beta_0}/(1 + e^{\beta_0})$ . The parameter  $\beta_1$  determines how  $\pi$  depends on temperature; we expect that  $\beta_1 < 0$ , since  $\pi$  decreases with increasing  $x_1$ .

If the data for the  $j$ th flight consist of  $r_j$  O-rings with thermal distress at launch temperature  $x_{1j}$ ,  $j = 1, \dots, n$ , and  $n = 23$  and  $m = 6$ , we have

$$\begin{aligned} \Pr(R_j = r_j; \beta_0, \beta_1) &= \frac{m!}{r_j!(m-r_j)!} \left\{ \frac{e^{\beta_0 + \beta_1 x_{1j}}}{1 + e^{\beta_0 + \beta_1 x_{1j}}} \right\}^{r_j} \left\{ \frac{1}{1 + e^{\beta_0 + \beta_1 x_{1j}}} \right\}^{m-r_j} \\ &= \frac{m!}{r_j!(m-r_j)!} \frac{\exp\{r_j(\beta_0 + \beta_1 x_{1j})\}}{\{1 + \exp(\beta_0 + \beta_1 x_{1j})\}^m}. \end{aligned}$$

If the  $R_j$  are independent, the likelihood for the entire set of data is

$$\begin{aligned} L(\beta_0, \beta_1) &= \prod_{j=1}^n \Pr(R_j = r_j; \beta_0, \beta_1) \\ &= \prod_{j=1}^n \frac{m!}{r_j!(m-r_j)!} \times \frac{\exp(\beta_0 \sum_{j=1}^n r_j + \beta_1 \sum_{j=1}^n r_j x_{1j})}{\prod_{j=1}^n \{1 + \exp(\beta_0 + \beta_1 x_{1j})\}^m}. \quad (4.6) \end{aligned}$$



**Figure 4.3** Log likelihoods for a binomial model for the O-ring thermal distress data. The probability of thermal distress is taken to be  $\psi = \exp(\beta_0 + \beta_1 x_1) / \{1 + \exp(\beta_0 + \beta_1 x_1)\}$ . The left panel gives the log likelihood for parameters  $\beta_0$  and  $\beta_1$ , and the right panel the log likelihood for the probability of thermal distress at 31°F,  $\psi = \exp(\beta_0 + 31\beta_1) / \{1 + \exp(\beta_0 + 31\beta_1)\}$  and  $\lambda = \beta_1$ .

The left panel of Figure 4.3 shows contours of this function, which is largest at  $\beta_0 \doteq 5$  and  $\beta_1 \doteq -0.1$ . However it is difficult to interpret because of the strong negative association between  $\beta_0$  and  $\beta_1$ : the values of  $\beta_1$  most plausible for  $\beta_0 \doteq 0$  are different from those most plausible when  $\beta_0 \doteq 10$ . ■

### Dependent data

In the examples above the data are assumed independent, though not necessarily identically distributed. In more complicated problems the dependence structure of the data may be very complex, making it hard to write down  $f(y; \theta)$  explicitly. Matters simplify when the data are recorded in time order, so that  $y_1$  precedes  $y_2$  precedes  $y_3, \dots$ . Then it can help to write

$$f(y; \theta) = f(y_1, \dots, y_n; \theta) = f(y_1; \theta) \prod_{j=2}^n f(y_j | y_1, \dots, y_{j-1}; \theta). \quad (4.7)$$

For example, if the data arise from a Markov process, (4.7) becomes

$$f(y; \theta) = f(y_1; \theta) \prod_{j=2}^n f(y_j | y_{j-1}; \theta), \quad (4.8)$$

where we have used the Markov property, that given the “present”  $Y_{j-1}$ , the ‘future’,  $Y_j, Y_{j+1}, \dots$ , is independent of the ‘past’,  $\dots, Y_{j-3}, Y_{j-2}$ .

**Example 4.6 (Poisson birth process)** Suppose that  $Y_0, \dots, Y_n$  are such that, given that  $Y_j = y_j$ , the conditional density of  $Y_{j+1}$  is Poisson with mean  $\theta y_j$ . That is,

$$f(y_{j+1} | y_j; \theta) = \frac{(\theta y_j)^{y_{j+1}}}{y_{j+1}!} \exp(-\theta y_j), \quad y_{j+1} = 0, 1, \dots, \quad \theta > 0.$$

If  $Y_0$  is Poisson with mean  $\theta$ , the joint density of data  $y_0, \dots, y_n$  is

$$f(y_0; \theta) \prod_{j=1}^n f(y_j | y_{j-1}; \theta) = \frac{\theta^{y_0}}{y_0!} \exp(-\theta) \prod_{j=0}^{n-1} \frac{(\theta y_j)^{y_{j+1}}}{y_{j+1}!} \exp(-\theta y_j),$$

This is sometimes called the prediction decomposition.

so the likelihood (4.8) equals

$$L(\theta) = \left( \prod_{j=0}^n y_j! \right)^{-1} \exp(s_0 \log \theta - s_1 \theta), \quad \theta > 0,$$

where  $s_0 = \sum_{j=0}^n y_j$  and  $s_1 = 1 + \sum_{j=0}^{n-1} y_j$ . ■

### 4.1.2 Basic properties

It can be convenient to plot the likelihood on a logarithmic scale. This scale is also mathematically convenient, and we define the *log likelihood* to be

$$\ell(\theta) = \log L(\theta).$$

Statements about relative likelihoods become statements about differences between log likelihoods. When  $y$  has independent components,  $y_1, \dots, y_n$ , we can write

$$\ell(\theta) = \sum_{j=1}^n \log f(y_j; \theta) = \sum_{j=1}^n \ell_j(\theta), \quad (4.9)$$

where  $\ell_j(\theta) \equiv \ell(\theta; y_j) = \log f(y_j; \theta)$  is the contribution to the log likelihood from the  $j$ th observation. The arguments of  $f$  and  $\ell$  are reversed to stress that we are primarily interested in  $f$  as a function of  $y$ , and in  $\ell$  as a function of  $\theta$ .

To combine the likelihoods for two independent sets of data  $y$  and  $z$  that both carry information about  $\theta$ , note that their joint probability density is just the product of their individual densities, and therefore the likelihood based on  $y$  and  $z$  is the product of the individual likelihoods:

$$L(\theta; y, z) = f(y; \theta)f(z; \theta) = L(\theta; y)L(\theta; z),$$

say, where for clarity the data are an additional argument in the likelihoods.

An important property of likelihood is its invariance to known transformations of the data. Suppose that there are two observers of the same experiment, and that one records the value  $y$  of a continuous random variable,  $Y$ , while the other records the value  $z$  of  $Z$ , where  $Z$  is a known 1–1 transformation of  $Y$ . Then the probability density function of  $Z$  is

$$f_Z(z; \theta) = f_Y(y; \theta) \left| \frac{dy}{dz} \right|, \quad (4.10)$$

where  $y$  is regarded as a function of  $z$ , and  $|dy/dz|$  is the Jacobian of the transformation from  $Y$  to  $Z$ . As (4.10) differs from (4.1) only by a constant that does not depend on the parameter, the log likelihood based on  $z$  equals that based on  $y$  plus a constant: the relative likelihoods of different values of  $\theta$  are the same. This implies that within a particular model  $f$  the absolute value of the likelihood is irrelevant to inference about  $\theta$ . When the maximum value of the likelihood is finite we define the *relative likelihood* of  $\theta$  to be

$$RL(\theta) = \frac{L(\theta)}{\max_{\theta'} L(\theta')}.$$

This takes values between one and zero, and its logarithm takes values between zero and minus infinity. As the absolute value of  $L(\theta)$ , or equivalently  $\ell(\theta)$ , is irrelevant to inference about  $\theta$ , we can neglect constants and use whatever version of  $L$  we wish. Henceforth we use the notation  $\equiv$  to indicate that constants have been ignored in defining a log likelihood. However we may not neglect constants if our goal is to compare models from different families of distributions.

**Example 4.7 (Spring failure data)** We can compare the Cauchy and Weibull models for the data in Examples 4.2–4.4 in terms of the maximum likelihood value achieved. Under this criterion, the Weibull model, for which the largest log likelihood is about  $-48$ , is a much better model than is the Cauchy, for which the maximum log likelihood is about  $-66$ . Evidently it makes no sense to add a constant to one of these and not to the other. ■

Suppose that the distribution of  $Y$  is determined by  $\psi$ , which is a 1–1 transformation of  $\theta$ , so that  $\theta = \theta(\psi)$ . Then the likelihood for  $\psi$ ,  $L^*(\psi)$ , and the likelihood for  $\theta$ ,  $L(\theta)$ , are related by the expression  $L^*(\psi) = L\{\theta(\psi)\}$ . The value of  $L$  is not changed by this transformation, so the likelihood is invariant to 1–1 reparametrization. We can use a parametrization that has a direct interpretation in terms of our particular problem.

**Example 4.8 (Challenger data)** We focus on the probability of thermal distress at  $31^\circ\text{F}$ , expressed in terms of the original parameters as

$$\psi = \frac{\exp(\beta_0 + 31\beta_1)}{1 + \exp(\beta_0 + 31\beta_1)}.$$

If we reparametrize  $L$  in terms of  $\psi$  and  $\lambda = \beta_1$ , we have  $\beta_0(\psi, \lambda) = \log\{\psi/(1 - \psi)\} - 31\lambda$ , and  $L^*(\psi, \lambda) = L\{\beta_0(\psi, \lambda), \lambda\}$ . The plot of the log likelihood  $\ell^*(\psi, \lambda) = \log L^*(\psi, \lambda)$  in the right panel of Figure 4.3 is easier to interpret than the plot of  $\ell(\beta_0, \beta_1)$  in the left panel, because the plausible range of values for  $\psi$  changes more slowly with  $\lambda$ . The contours in the left panel seem roughly elliptical, but those in the right are not. The most plausible range of values for  $\psi$  is  $(0.7, 0.9)$ , throughout which the value of  $\lambda$  is roughly  $-0.1$ . ■

### Interpretation

When there is a particular parametric model for a set of data, likelihood provides a natural basis for assessing the plausibility of different parameter values, but how should it be interpreted? One viewpoint is that values of  $\theta$  can be compared using a scale such as

$$\begin{aligned} 1 &\geq RL(\theta) > \frac{1}{3}, & \theta \text{ strongly supported,} \\ \frac{1}{3} &\geq RL(\theta) > \frac{1}{10}, & \theta \text{ supported,} \\ \frac{1}{10} &\geq RL(\theta) > \frac{1}{100}, & \theta \text{ weakly supported,} \\ \frac{1}{100} &\geq RL(\theta) > \frac{1}{1000}, & \theta \text{ poorly supported,} \\ \frac{1}{1000} &\geq RL(\theta) > 0, & \theta \text{ very poorly supported.} \end{aligned} \quad (4.11)$$

Under this *pure likelihood* approach, values of  $\theta$  are compared solely in terms of relative likelihoods. A scale such as (4.11) is simple and directly interpretable, but as it has the disadvantages that the numbers  $\frac{1}{3}$ ,  $\frac{1}{10}$  and so forth are arbitrary and take no account of the dimension of  $\theta$ , this interpretation is not the most common one in practice. We discuss repeated sampling calibration of likelihood values in Section 4.5.

## Exercises 4.1

- 1 Sketch the Cauchy likelihood for the observations 1.1, 2.3, 1.5, 1.4. Show that the distribution function of the *two-parameter Cauchy density*,

$$f(u; \theta, \sigma) = \frac{\sigma}{\pi \{\sigma^2 + (u - \theta)^2\}}, \quad -\infty < u < \infty, \sigma > 0, -\infty < \theta < \infty,$$

is  $F(u) = \frac{1}{2} + \pi^{-1} \tan^{-1}\{(u - \theta)/\sigma\}$ . Hence find  $\Pr(|Y - \theta| < 20)$  when  $\sigma = 1$ , and with hindsight explain why the model in Example 4.3 fits poorly.

- 2 Find the likelihood for a random sample  $y_1, \dots, y_n$  from the geometric density  $\Pr(Y = y) = \pi(1 - \pi)^y$ ,  $y = 0, 1, \dots$ , where  $0 < \pi < 1$ .
- 3 Verify that the likelihood for  $f(y; \lambda) = \lambda \exp(-\lambda y)$ ,  $y, \lambda > 0$ , is invariant to the reparametrization  $\psi = 1/\lambda$ .
- 4 Show that the log likelihood for two independent sets of data is the sum of their log likelihoods.
- 5 Let  $A_n \subset A_{n-1} \subset \dots \subset A_1$  be events on the same probability space. Show that

$$\Pr(A_n) = \Pr(A_n | A_{n-1})\Pr(A_{n-1}) = \Pr(A_n | A_{n-1}) \cdots \Pr(A_2 | A_1)\Pr(A_1)$$

and hence establish (4.7).

## 4.2 Summaries

### 4.2.1 Quadratic approximation

In a problem with one or two parameters, the likelihood can be visualized. However models with a few dozen parameters are commonplace, and sometimes there are many more, so we often need to summarize the likelihood.

A key idea is that in many cases the log likelihood is approximately quadratic as a function of the parameter. To illustrate this, the left panel of Figure 4.4 shows log likelihoods for random samples of size  $n = 5, 10, 20, 40$  and  $80$  from an exponential density,  $\theta^{-1} \exp(-u/\theta)$ ,  $\theta > 0, u > 0$ . In each case the sample has average  $\bar{y} = e^{-1}$ . The panel has two general features. First, the maximum of each log likelihood is at  $\theta = e^{-1}$ . To see why, note that (4.3) implies that

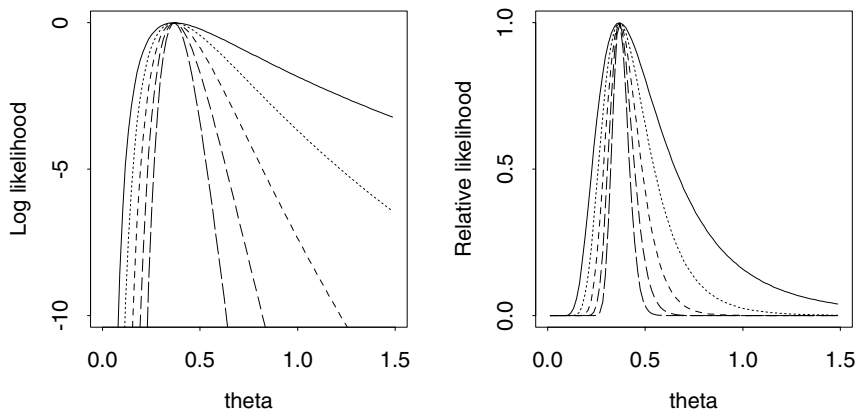
$$\ell(\theta) = -n \log \theta - \theta^{-1} \sum_{j=1}^n y_j = -n (\log \theta + \bar{y}/\theta),$$

which is maximized when  $d\ell(\theta)/d\theta = 0$ , that is, when  $\theta = \bar{y}$ . Now

$$\frac{d^2\ell(\theta)}{d\theta^2} = -n \left( -\frac{1}{\theta^2} + \frac{2\bar{y}}{\theta^3} \right)$$

For example, an image of  $512 \times 512$  pixels may have a parameter for each pixel.

As usual,  $\bar{y} = n^{-1} \sum y_j$ .



**Figure 4.4** Log likelihoods and relative likelihoods for exponential samples with sample sizes  $n = 5, 10, 20, 40, 80$ . The curvature of the functions increases with  $n$ , so the highest curve in each panel is for  $n = 5$ , and the lowest is for  $n = 80$ .

takes the value  $-n/\bar{y}^2$  at  $\theta = \bar{y}$ , so  $\bar{y}$  gives the unique maximum of  $\ell$ . The value of  $\theta$  for which  $L$ , or equivalently  $\ell$ , is greatest is called the *maximum likelihood estimate*,  $\hat{\theta}$ . In this case  $\hat{\theta} = \bar{y}$ . For future reference, note that the values of  $-n^{-1}d^2\ell(\theta)/d\theta^2$  and its derivative  $-n^{-1}d^3\ell(\theta)/d\theta^3$  are bounded in a neighbourhood  $\mathcal{N} = \{\theta : |\theta - \hat{\theta}| < \delta\}$  of  $\hat{\theta}$ , provided  $\mathcal{N}$  excludes  $\theta = 0$ .

Second, the curvature of the log likelihood at the maximum increases with  $n$ , because the second derivative of  $\ell$ , which measures its curvature as a function of  $\theta$ , is a linear function of  $n$ . The function  $-d^2\ell(\theta)/d\theta^2$  is called the *observed information*. In this case its value at  $\hat{\theta}$  is  $n/\bar{y}^2 = n/\hat{\theta}^2$ .

The right panel of Figure 4.4 shows the relative likelihoods corresponding to the left panel. The effect of increasing  $n$  is that the likelihood becomes more concentrated about the maximum, and so it becomes relatively less and less plausible that each value of  $\theta$  a fixed distance from  $\hat{\theta}$  generated the data. To express this algebraically, we write the log relative likelihood,  $\log RL(\theta)$ , as  $\ell(\theta) - \ell(\hat{\theta})$  and expand  $\ell(\theta)$  in a Taylor series about  $\hat{\theta}$  to obtain

$$\log RL(\theta) = \ell(\hat{\theta}) + (\theta - \hat{\theta})\ell'(\hat{\theta}) + \frac{1}{2}(\theta - \hat{\theta})^2\ell''(\theta_1) - \ell(\hat{\theta}) = \frac{1}{2}(\theta - \hat{\theta})^2\ell''(\theta_1); \quad (4.12)$$

$\theta_1$  lies between  $\theta$  and  $\hat{\theta}$ . We denote differentiation with respect to  $\theta$  by a prime, thus  $\ell'(\theta) = d\ell(\theta)/d\theta$ , and so forth; note that  $\ell'(\hat{\theta}) = 0$ . Each derivative of  $\ell$  is a sum of  $n$  terms. As  $n$  increases, we see that the bound on  $-n^{-1}\ell''(\theta_1)$  implies that (4.12) will become increasingly negative except at  $\theta = \hat{\theta}$ . Hence  $RL(\theta)$  tends to zero unless  $\theta = \hat{\theta}$ , while  $RL(\hat{\theta}) = 1$  for all  $n$ .

To examine the behaviour of the log likelihood more closely, we take another term in the Taylor expansion leading to (4.12), to find that

$$\log RL(\theta) = \frac{1}{2}(\theta - \hat{\theta})^2\ell''(\hat{\theta}) + \frac{1}{6}(\theta - \hat{\theta})^3\ell'''(\theta_2),$$

where  $\theta_2$  lies between  $\theta$  and  $\hat{\theta}$ . Now consider what happens, not at a fixed distance from  $\hat{\theta}$ , but at  $\theta = \hat{\theta} + n^{-1/2}\delta$ . As  $n$  increases this corresponds to “zooming in” and

examining the region around  $\hat{\theta}$  ever more closely. Now

$$\log RL(\hat{\theta} + n^{-1/2}\delta) = \frac{1}{2}\delta^2 n^{-1} \ell''(\hat{\theta}) + \frac{1}{6}\delta^3 n^{-3/2} \ell'''(\theta_2), \quad (4.13)$$

and crucially, both  $\ell''(\theta)$  and  $\ell'''(\theta)$  are linear functions of  $n$ . The bound on  $-n^{-1}\ell'''(\theta)$  implies that the last term on the right of (4.13) disappears as  $n \rightarrow \infty$ , but the quadratic term becomes  $-\frac{1}{2}\delta^2\{-n^{-1}\ell''(\hat{\theta})\}$ , which in this case is  $-\frac{1}{2}\delta^2/\bar{y}^2$ . Thus in large samples the likelihood close to the maximum is a quadratic function and can be summarized in terms of the maximum likelihood estimate  $\hat{\theta}$  and the observed information  $-\ell''(\hat{\theta})$ . One implication of this is that if we restrict ourselves to parameter values that are plausible relative to the maximum likelihood estimate, say those values of  $\theta$  such that  $RL(\theta) > c$ , we find  $\log RL(\theta) > \log c$ . Comparison with (4.13) shows that our range of ‘plausible’  $\theta$  is decreasing with  $n$  and has length roughly proportional to  $n^{-1/2}$ .

This discussion concerns a scalar parameter, but extends to higher dimensions, where  $d^2\ell/d\theta^2$  is replaced by the matrix of second derivatives of  $\ell$ .

Whether a quadratic approximation to  $\ell$  is useful depends on the problem. To summarize the log likelihood in Figure 4.2 in such terms would be very misleading, unless a summary was required only very close to the maximum. If feasible, it is sensible to plot the likelihood.

**Example 4.9 (Uniform distribution)** Suppose we are presented with a random sample  $y_1, \dots, y_n$  from the uniform density on  $(0, \theta)$ :

$$f(u; \theta) = \begin{cases} \theta^{-1}, & 0 < u < \theta, \\ 0, & \text{otherwise.} \end{cases}$$

The likelihood is

$$L(\theta) = \prod_j f(y_j; \theta) = \begin{cases} \theta^{-n}, & 0 < y_1, \dots, y_n < \theta, \\ 0, & \text{otherwise.} \end{cases}$$

It is maximized at  $\hat{\theta} = \max(y_j)$ , but  $d\ell(\hat{\theta})/d\theta \neq 0$  and  $-d^2\ell(\hat{\theta})/d\theta^2 = -n/\hat{\theta}^2 < 0$ , and  $\ell(\theta)$  becomes increasingly spikey as  $n \rightarrow \infty$  and is not approximately quadratic near  $\hat{\theta}$  for any  $n$ . ■

### 4.2.2 Sufficient statistics

In well-behaved problems and with large samples the likelihood may be summarized in terms of the maximum likelihood estimate and observed information, though Examples 4.3 and 4.9 show that this can fail. A better approach rests on the fact that the likelihood often depends on the data only through some low-dimensional function  $s(y)$  of the  $y_j$ , and then a suitable summary can be given in terms of this. Thus in Examples 4.2 and 4.9 the likelihoods depend on the data through  $(n, \sum y_j)$  and  $(n, \max y_j)$  respectively. If we believe that our model is correct, we need only these functions to calculate the likelihoods for any value of  $\theta$ . These functions are examples of *sufficient statistics*.

Suppose that we have observed data,  $y$ , generated by a distribution whose density is  $f(y; \theta)$ , and that the statistic  $s(y)$  is a function of  $y$  such that the conditional density of the corresponding random variable  $Y$ , given that  $S = s(Y)$ , is independent of  $\theta$ . That is,

$$f_{Y|S}(y | s; \theta) \quad (4.14)$$

does not depend on  $\theta$ . Then  $S$  is said to be a *sufficient statistic for  $\theta$  based on  $Y$* , or just a *sufficient statistic for  $\theta$* . The idea is that any extra information in  $Y$  but not in  $S$  is given by the conditional density (4.14), and if this conditional density is free of  $\theta$ ,  $Y$  contains no more information about  $\theta$  than does  $S$ . We shall see later that  $S$  is not unique.

Definition (4.14) is hard to use, because we must guess that a given statistic  $S$  is sufficient before we can calculate the conditional density. An equivalent and more useful definition is via the *factorization criterion*. This states that a necessary and sufficient condition for a statistic  $S$  to be a sufficient statistic for a parameter  $\theta$  in a family of probability density functions  $f(y; \theta)$  is that the density of  $Y$  can be expressed as

$$f(y; \theta) = g\{s(y); \theta\}h(y). \quad (4.15)$$

Thus the density of  $Y$  factorizes into a function  $g$  of  $s(y)$  and  $\theta$ , and a function of  $y$ ,  $h$ , that does not depend on  $\theta$ .

The equivalence of these two definitions is almost self-evident. First note that if  $S$  is a sufficient statistic, the conditional distribution of  $Y$  given  $S$  is independent of  $\theta$ , that is,

$$f_{Y|S}(y | s) = \frac{f_{Y,S}(y, s; \theta)}{f_S(s; \theta)} \quad (4.16)$$

is free of  $\theta$ . But as  $S$  is a function  $s(Y)$  of  $Y$ , the joint density of  $S$  and  $Y$  is zero except where  $S = s(Y)$ , and so the numerator of the right-hand side of (4.16) is just  $f_Y(y; \theta)$ . Rearrangement of (4.16) implies that if  $S$  is sufficient, (4.15) holds with  $g(\cdot) = f_S(\cdot)$  and  $h(\cdot) = f_{Y|S}(\cdot)$ .

Conversely, if (4.15) holds, we find the density of  $S$  at  $s$  by summing or integrating (4.15) over the range of  $y$  for which  $s(y) = s$ . In the discrete case

$$f_S(s; \theta) = \sum g\{s(y); \theta\}h(y) = g\{s; \theta\} \sum h(y),$$

because the sum is over those  $y$  for which  $s(y)$  equals  $s$ . Therefore the conditional density of  $Y$  given  $S$  is

$$\frac{f_Y(y; \theta)}{f_S(s; \theta)} = \frac{g\{s(y); \theta\}h(y)}{g\{s; \theta\} \sum h(y)} = \frac{h(y)}{\sum h(y)},$$

which shows that  $S$  is sufficient.

**Example 4.10 (Bernoulli distribution)** A Bernoulli random variable  $Y$  records the ‘success’ or ‘failure’ of a binary trial. Thus

$$\Pr(Y = 1) = 1 - \Pr(Y = 0) = \pi, \quad 0 \leq \pi \leq 1,$$

Proof in the continuous case would replace the sum here by an integral, but a detailed proof is not simple because all elements of the parametric model must be dominated by a single measure. See for example Theorem 2.21 of Schervish (1995).

with  $Y = 1$  representing success and  $Y = 0$  failure. The likelihood contribution from a single trial with outcome  $Y = y$  may be written  $\pi^y(1 - \pi)^{1-y}$ , and hence the likelihood for  $\pi$  based on the outcomes of  $n$  independent trials is

$$L(\pi) = \prod_{j=1}^n \pi^{y_j} (1 - \pi)^{1-y_j} = \pi^r (1 - \pi)^{n-r},$$

say, where  $r = \sum y_j$  is the number of successes in the  $n$  trials. The distribution of the corresponding random variable,  $R = \sum Y_j$ , is binomial with probability  $\pi$  and denominator  $n$ , that is,

$$\Pr(R = r) = \binom{n}{r} \pi^r (1 - \pi)^{n-r}, \quad r = 0, \dots, n.$$

Hence the distribution of  $Y_1, \dots, Y_n$  conditional on  $R$  is

$$\Pr(Y_1 = y_1, \dots, Y_n = y_n \mid \sum Y_j = r) = \frac{1}{\binom{n}{r}},$$

which puts equal probability on each of the  $\binom{n}{r}$  permutations of  $r$  1's and  $n - r$  0's. This conditional distribution does not depend on  $\pi$ , so  $R$  is sufficient for  $\pi$ , as is intuitively clear.

Although there is no loss of information about  $\pi$  when  $Y_1, \dots, Y_n$  is reduced to  $R$ , the original data are more useful for some purposes. For example, if  $y_1, \dots, y_n$  consisted of a sequence of zeros followed by a sequence of ones, we might want to revise our belief that the trials were independent, but we could not know this if only  $\sum y_j$  had been reported. ■

**Example 4.11 (Exponential distribution)** Suppose that  $Y_1$  and  $Y_2$  are independently exponentially distributed. Then their joint density is

$$\begin{aligned} f(y; \lambda) &= \lambda e^{-\lambda y_1} \cdot \lambda e^{-\lambda y_2}, \quad y_1, y_2 > 0, \\ &= \lambda^2 \exp\{-\lambda(y_1 + y_2)\} \\ &= \lambda^2 \exp(-\lambda s) \cdot 1, \end{aligned}$$

which factorizes into a function of  $s = y_1 + y_2$  and the constant 1. Therefore  $S = Y_1 + Y_2$  is sufficient, using the factorization criterion (4.15).

To verify this using the original definition (4.14), note that  $S$  is a sum of two independent exponential random variables, and so has the gamma distribution with density

$$f(s; \lambda) = \lambda^2 s \exp(-\lambda s), \quad s > 0.$$

Thus the conditional density of  $Y_1$  and  $Y_2$  given that  $S = Y_1 + Y_2 = s$  is

$$\frac{f(y_1, y_2; \lambda)}{f(s; \lambda)} = \frac{\lambda^2 \exp\{-\lambda(y_1 + y_2)\}}{\lambda^2 s \exp(-\lambda s)} = \frac{1}{s}, \quad y_1 + y_2 = s > 0.$$

This, the uniform density on  $(0, s)$ , is free of  $\lambda$ . Thus given the particular value  $s$  for the line  $Y_1 + Y_2 = s$  on which the point  $(Y_1, Y_2)$  lies, the position of  $(Y_1, Y_2)$  on the line conveys no extra information about  $\lambda$ . ■

**Example 4.12 (Random sample)** Let  $Y_1, \dots, Y_n$  be a random sample of scalar observations from a density  $f(y; \theta)$ . Now as all the observations are on an equal footing, their order is irrelevant. It follows that the order statistics  $Y_{(1)}, \dots, Y_{(n)}$  are sufficient for  $\theta$ . To see this, note that we saw at (2.25) that the joint density of the order statistics is

$$n! f(y_{(1)}; \theta) \times \cdots \times f(y_{(n)}; \theta), \quad y_{(1)} \leq \cdots \leq y_{(n)}.$$

Hence the conditional density of  $Y_1, \dots, Y_n$  given  $Y_{(1)}, \dots, Y_{(n)}$  is  $1/n!$ , provided that  $Y_{(1)}, \dots, Y_{(n)}$  is a permutation of  $Y_1, \dots, Y_n$ , and is zero otherwise. Evidently this conditional density is free of  $\theta$ , and hence the order statistics are a sufficient statistic of dimension  $n$  for  $\theta$ .

If we are willing to make more specific assumptions about  $f(y; \theta)$ , we can reduce the data further. For the exponential density, for example, the likelihood is  $\theta^{-n} \exp(-\theta^{-1} \sum y_j)$ , so it follows that  $(N, \sum Y_j)$  is also sufficient for  $\theta$ . Thus there can be different sufficient statistics for a single model. ■

**Example 4.13 (Capture-recapture model)** Capture-recapture models are widely used to estimate the sizes of animal populations and survival rates from one year to the next. The idea is to capture animals on a number of separate occasions, to mark them, and to return them to the wild after each occasion. The proportion of marked animals seen on the second and subsequent occasions gives an idea of the quantities of interest. For example, if the population is large and only a small proportion of it is seen on the first occasion, then few of the animals captured next time will already be marked.

Suppose there are three capture occasions (years) labelled 0, 1, and 2, that the probability of survival from one occasion to the next is  $\psi$ , and that, for an animal alive in year  $s$ , the probability of recapture is  $\lambda_s$ . Then the possible capture histories and their probabilities are

|                                                                        |                                        |
|------------------------------------------------------------------------|----------------------------------------|
| 111 $\psi \lambda_1 \times \psi \lambda_2$ ,                           | 011 $\psi \lambda_2$ ,                 |
| 110 $\psi \lambda_1 \times \{1 - \psi + \psi(1 - \lambda_2)\}$ ,       | 010 $1 - \psi + \psi(1 - \lambda_2)$ , |
| 101 $\psi(1 - \lambda_1) \times \psi \lambda_2$ ,                      | 001 1,                                 |
| 100 $1 - \psi + \psi(1 - \lambda_1)\{1 - \psi + \psi(1 - \lambda_2)\}$ |                                        |

where, for example, 110 represents an animal seen in years 0 and 1, but not 2. The probability of being alive and seen in year 1 is  $\psi \lambda_1$ , and conditional on being alive in year 1, the animal may be dead in year 2, with probability  $1 - \psi$ , or alive but not seen, with probability  $\psi(1 - \lambda_2)$ . Without further assumptions we can say nothing about animals with history 000, which we never see.

If animals are assumed independent, the likelihood is a product of such terms, and we notice that, for example, there is a contribution  $\psi \lambda_1$  from animals with history 111 or 110, a contribution  $\psi \lambda_2$  from animals with history 111 or 011, and so on. Thus the likelihood may be written as

$$(\psi \lambda_1)^{r_{01}} \{\psi(1 - \lambda_1) \psi \lambda_2\}^{r_{02}} \{1 - \psi \lambda_1 - \psi(1 - \lambda_1) \psi \lambda_2\}^{m_0 - r_{01} - r_{02}} \\ \times (\psi \lambda_2)^{r_{11}} (1 - \psi \lambda_2)^{m_1 - r_{11}},$$

**Table 4.1** Sufficient statistics and probabilities for capture-recapture model.

| Year | Number captured | Number first recaptured in year |          | Number never recaptured |
|------|-----------------|---------------------------------|----------|-------------------------|
|      |                 | 1                               | 2        |                         |
| 0    | $m_0$           | $r_{01}$                        | $r_{02}$ | $m_0 - r_{01} - r_{02}$ |
| 1    | $m_1$           |                                 | $r_{11}$ | $m_1 - r_{11}$          |

| Year | Probability first recaptured in year |                                    | Probability never recaptured                           |
|------|--------------------------------------|------------------------------------|--------------------------------------------------------|
|      | 1                                    | 2                                  |                                                        |
| 0    | $\psi\lambda_1$                      | $\psi(1 - \lambda_1)\psi\lambda_2$ | $1 - \psi\lambda_1 - \psi(1 - \lambda_1)\psi\lambda_2$ |
| 1    |                                      | $\psi\lambda_2$                    | $1 - \psi\lambda_2$                                    |

where  $m_s$  is the number of animals seen in year  $s$ , of whom  $r_{st}$  are first seen again in year  $t$ . Evidently the quantities  $m_s$  and  $r_{st}$  are sufficient statistics. We lay out these and the corresponding probabilities in Table 4.1, which is a standard representation for such data. With  $k$  occasions the number of individual histories is  $2^k - 1$  but the table contains just  $\frac{1}{2}(k + 2)(k - 1)$  elements, so the reduction can be considerable, but more importantly the data structure is clearer in terms of the sufficient statistics. ■

### Minimal sufficiency

Even for a single model, sufficient statistics are not unique. Apart from the possibility that different functions  $s(Y)$  might satisfy the factorization criterion, the data themselves form a sufficient statistic. Moreover it is easy to see from (4.15) that any known 1–1 function of a sufficient statistic is itself sufficient. What is unique to each sufficient statistic is the partition that it induces on the sample space.

To see this, we say that two samples  $Y_1$  and  $Y_2$  with corresponding sufficient statistics  $S_1 = s(Y_1)$  and  $S_2 = s(Y_2)$  are equivalent if  $S_1 = S_2$ . This evidently satisfies the three properties of an equivalence relation:

- reflexivity,  $Y$  is equivalent to itself;
- symmetry,  $Y_1$  is equivalent to  $Y_2$  if  $Y_2$  is equivalent to  $Y_1$ ; and
- transitivity,  $Y_1$  is equivalent to  $Y_3$  whenever  $Y_1$  is equivalent to  $Y_2$  and  $Y_2$  is equivalent to  $Y_3$ .

Therefore the sample space is partitioned by the relation into equivalence classes, corresponding to each of the distinct values that  $S$  can take. Unlike the sufficient statistic itself, this partitioning is invariant under 1–1 transformation of  $S$ . By the factorization criterion it has the property that the conditional density of the data  $Y$  given that  $Y$  falls into a particular equivalence class is independent of the parameter, and hence is called a *sufficient partition*. Such a partition has the property that if we are told into which of its equivalence classes the data fall, we can reconstruct the log likelihood up to additive constants. A mathematical discussion of sufficiency would

be in terms of sufficient partitions rather than sufficient statistics. However it is more natural to think in terms of sufficient statistics, and we mostly do so.

As sufficient statistics are not unique, we can choose which to use. The biggest reduction of the data is obtained by taking a sufficient statistic whose dimension is as small as possible, that is, a *minimal sufficient statistic*. A sufficient statistic is said to be minimal if it is a function of any other sufficient statistic. This corresponds to the coarsest sufficient partition of the sample space, while the data generate the finest sufficient partition. To find a minimal sufficient statistic, we return to the likelihood. Suppose that the likelihoods of two sets of data,  $y$  and  $z$ , are the same up to a constant. Then  $L(\theta; y)/L(\theta; z)$  does not depend on  $\theta$ , and the partition that this equivalence relation generates is minimally sufficient. Thus a minimal sufficient statistic is obtained by examining the likelihood to see on what functions of the data it depends.

**Example 4.14 (Exponential distribution)** In Example 4.11 the sample space into which  $(Y_1, Y_2)$  falls is  $\mathbb{R}_+^2$ , and this is partitioned by the lines  $y_1 + y_2 = s$ ,  $s > 0$ , each of which corresponds to an equivalence class.

In order to find a minimal sufficient statistic, note that the likelihood based on data  $y_1, y_2$  is  $\lambda^2 \exp\{-\lambda(y_1 + y_2)\}$ , whereas the likelihood based on  $x_1, \dots, x_m$  would be  $\lambda^m \exp\{-\lambda(x_1 + \dots + x_m)\}$ . The ratio of these would be independent of  $\lambda$  only if  $m = 2$  and  $x_1 + x_2 = y_1 + y_2$ . Hence a minimal sufficient statistic is  $(N, S)$ , the number of observations in the sample, and their sum. Usually  $N$  is chosen without regard to  $\lambda$ , and  $S$  alone is regarded as minimal sufficient. ■

**Example 4.15 (Poisson birth process)** We saw in Example 4.6 that the likelihood based on data  $y_0, \dots, y_n$  from such a process is

$$L(\theta) = \left( \prod_{j=0}^n y_j! \right)^{-1} \exp(s_0 \log \theta - s_1 \theta), \quad \theta > 0,$$

where  $s_0 = \sum_{j=0}^n y_j$  and  $s_1 = 1 + \sum_{j=0}^{n-1} y_j$ . The factorization criterion shows that a sufficient statistic is  $(S_0, S_1)$ , but equally so is  $(S_0, Y_n)$ , since  $S_1 = S_0 + 1 - Y_n$ . Evidently either of these is also minimal sufficient. ■

**Example 4.16 (Logistic regression)** Suppose that independent binomial random variables  $R_j$  have denominators  $m_j$  and probabilities  $\pi_j$ , where

$$\pi_j = \frac{\exp(\beta_0 + \beta_1 x_{1j})}{1 + \exp(\beta_0 + \beta_1 x_{1j})}, \quad j = 1, \dots, n,$$

and the  $x_{1j}$  are known constants. The likelihood is (4.6), and on applying the factorization criterion we see that a minimal sufficient statistic for  $(\beta_0, \beta_1)$  is  $S = (\sum R_j, \sum R_j x_{1j})$ . Although the  $m_j, x_{1j}$ , and  $n$  are needed to calculate the likelihood, they are non-random and not included in  $S$ . ■

## Exercises 4.2

- 1 Find the maximum likelihood estimate and observed information in Example 4.1. Find also the maximum likelihood estimate of  $\Pr(Y = 0)$ .

- 2 Find maximum likelihood estimates for  $\theta$  based on a random sample of size  $n$  from the densities (i)  $\theta y^{\theta-1}$ ,  $0 < y < 1$ ,  $\theta > 0$ ; (ii)  $\theta^2 y e^{-\theta y}$ ,  $y > 0$ ,  $\theta > 0$ ; and (iii)  $(\theta + 1)y^{-\theta-2}$ ,  $y > 1$ ,  $\theta > 0$ ;
- 3 Plot the likelihood for  $\theta$  based on a random sample  $y_1, \dots, y_n$  from the density

$$f(x; \theta) = \begin{cases} 1/(2c), & \theta - c < x < \theta + c, \\ 0, & \text{otherwise,} \end{cases}$$

where  $c$  is a known constant. Find a maximum likelihood estimate, and show that it is not unique.

- 4 In the discussion following (4.13), show that if the log likelihood was exactly quadratic and we agreed that values of  $\theta$  such that  $RL(\theta) > c$  were 'plausible', the range of plausible  $\theta$  would be  $\hat{\theta} \pm \{2 \log c / \ell''(\hat{\theta})\}^{1/2}$ .
- 5 Data are available from  $n$  independent experiments concerning a scalar parameter  $\theta$ . The log likelihood for the  $j$ th experiment may be summarized as a quadratic function,  $\ell_j(\theta) \doteq \hat{\ell}_j - \frac{1}{2} J_j(\hat{\theta}_j)(\theta - \hat{\theta}_j)^2$ , where  $\hat{\theta}_j$  is the maximum likelihood estimate and  $J_j(\hat{\theta}_j)$  is the observed information. Show that the overall log likelihood may be summarized as a quadratic function of  $\theta$ , and find the overall maximum likelihood estimate and observed information.
- 6 In a *first-order autoregressive process*,  $Y_0, \dots, Y_n$ , the conditional distribution of  $Y_j$  given the previous observations,  $Y_1, \dots, Y_{j-1}$ , is normal with mean  $\alpha y_{j-1}$  and variance one. The initial observation  $Y_0$  has the normal distribution with mean zero and variance one. Show that the log likelihood is proportional to  $y_0^2 + \sum_{j=1}^n (y_j - \alpha y_{j-1})^2$ , and hence find the maximum likelihood estimate of  $\alpha$  and the observed information.
- 7 Find a minimal sufficient statistic for  $\theta$  based on a random sample  $Y_1, \dots, Y_n$  from the Poisson density (2.6).
- 8 Let  $Y_1, \dots, Y_n$  be a random sample from the  $N(\mu, \sigma^2)$  distribution.
  - (a) Use the factorization criterion to show that  $(\sum Y_j, \sum Y_j^2)$  is sufficient for  $(\mu, \sigma^2)$ . Say, giving your reasons, which of the following are also sufficient: (i)  $(\bar{Y}, S^2)$ ; (ii)  $(\bar{Y}^2, S)$ ; (iii) the order statistics  $Y_{(1)} < \dots < Y_{(n)}$ .
  - (b) If  $\sigma^2 = 1$ , show that the sample average is minimal sufficient for  $\mu$ .
  - (c) Suppose that  $\mu$  equals the known value  $\mu_0$ . Show that  $S = \sum (Y_j - \mu_0)^2$  is a minimal sufficient statistic for  $\sigma^2$ , and give its distribution. Show that  $S$  is a function of the minimal sufficient statistic when both parameters are unknown.
- 9 Find the minimal sufficient statistic based on a random sample  $Y_1, \dots, Y_n$  from the gamma density (2.7).
- 10 Use the factorization criterion to show that the maximum likelihood estimate and observed information based on  $f(y; \theta)$  are functions of data  $y$  only through a sufficient statistic  $s(y)$ .
- 11 Verify that the relation ' $y_1$  is equivalent to  $y_2$ ' if  $L(\theta; y_1)/L(\theta; y_2)$  is independent of  $\theta$  is an equivalence relation and that the corresponding partition is sufficient. Deduce that the likelihood itself is minimal sufficient.

## 4.3 Information

### 4.3.1 Expected and observed information

In a model with log likelihood  $\ell(\theta)$ , the *observed information* is defined to be

$$J(\theta) = -\frac{d^2 \ell(\theta)}{d\theta^2}.$$

When  $\ell(\theta)$  is a sum of  $n$  components, so too is  $J(\theta)$ , because (4.9) implies that

$$J(\theta) = -\frac{d^2 \ell(\theta)}{d\theta^2} = -\frac{d^2}{d\theta^2} \sum_{j=1}^n \ell_j(\theta) = \sum_{j=1}^n -\frac{d^2 \log f(y_j; \theta)}{d\theta^2}. \quad (4.17)$$

We saw in Section 4.2.1 that when the log likelihood is roughly quadratic, the relative plausibility of parameter values near the maximum likelihood estimate is determined by the observed information. High information, or equivalently high curvature, will pin down  $\theta$  more tightly than if the observed information is low. The amount of information is typically related to the size of the dataset, a fact useful in planning experiments. Before we conduct an experiment it is valuable to assess what information there will be in the data, to see if the proposed sample is large enough. Otherwise we may need more data or a more informative experiment. Before the experiment is performed we have no data, so we cannot obtain the observed information. However we can calculate the *expected* or *Fisher information*,

$$I(\theta) = E \left\{ -\frac{d^2 \ell(\theta)}{d\theta^2} \right\},$$

which is the mean information the data will contain when collected, if the model is correct and the true parameter value is  $\theta$ .

If the data are a random sample, (4.17) implies that  $I(\theta) = ni(\theta)$ , where  $i(\theta)$  is the information from a single observation,

$$i(\theta) = E \left\{ -\frac{d^2 \log f(Y_j; \theta)}{d\theta^2} \right\}.$$

When  $\theta$  is a  $p \times 1$  vector, the information matrices are

$$J(\theta) = -\frac{\partial^2 \ell(\theta)}{\partial \theta \partial \theta^T}, \quad I(\theta) = -E \left\{ \frac{\partial^2 \ell(\theta)}{\partial \theta \partial \theta^T} \right\};$$

these are symmetric  $p \times p$  matrices whose  $(r, s)$  elements are respectively

$$-\frac{\partial^2 \ell(\theta)}{\partial \theta_r \partial \theta_s}, \quad E \left\{ -\frac{\partial^2 \ell(\theta)}{\partial \theta_r \partial \theta_s} \right\}.$$

**Example 4.17 (Binomial distribution)** The likelihood for a binomial variable  $R$  with denominator  $m$  and probability of success  $0 < \pi < 1$  is  $L(\pi) = \binom{m}{r} \pi^r (1 - \pi)^{m-r}$ , so  $\ell(\pi) \equiv r \log \pi + (m - r) \log(1 - \pi)$  and

$$J(\pi) = -\frac{d^2 \ell(\pi)}{d\pi^2} = \frac{r}{\pi^2} + \frac{m - r}{(1 - \pi)^2},$$

given an observed value  $r$  of  $R$ . Before the experiment has been performed the value of  $r$  is unknown, and we replace it by the corresponding random variable  $R$ . In this

For a  $p \times 1$  vector  $\theta$  we use  $\partial \ell / \partial \theta$  to denote the  $p \times 1$  vector whose  $r$ th element is  $\partial \ell / \partial \theta_r$ , and  $\partial^2 \ell / \partial \theta \partial \theta^T$  to denote the  $p \times p$  matrix whose  $(r, s)$  element is  $\partial^2 \ell / \partial \theta_r \partial \theta_s$ .

case  $J(\pi)$  too is random, and

$$\begin{aligned} I(\pi) &= E\{J(\pi)\} \\ &= E\left\{\frac{R}{\pi^2} + \frac{m-R}{(1-\pi)^2}\right\} \\ &= \frac{m\pi}{\pi^2} + \frac{m(1-\pi)}{(1-\pi)^2} = \frac{m}{\pi(1-\pi)}, \end{aligned}$$

since  $E(R) = m\pi$ . The expected information  $I(\pi)$  increases linearly with  $m$  and is symmetric in  $\pi$ , for  $0 < \pi < 1$ . ■

**Example 4.18 (Normal distribution)** The density function of a normal random variable with mean  $\mu$  and variance  $\sigma^2$  is (3.5), so the log likelihood for a random sample  $y_1, \dots, y_n$  is

$$\ell(\mu, \sigma) \equiv -\frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{j=1}^n (y_j - \mu)^2.$$

Its first derivatives are

$$\frac{\partial \ell}{\partial \mu} = \sigma^{-2} \sum (y_j - \mu), \quad \frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum (y_j - \mu)^2,$$

and the elements of the observed information matrix  $J(\mu, \sigma^2)$  are given by

$$\frac{\partial^2 \ell}{\partial \mu^2} = -\frac{n}{\sigma^2}, \quad \frac{\partial^2 \ell}{\partial \mu \partial \sigma^2} = -\frac{n}{\sigma^4} (\bar{y} - \mu), \quad \frac{\partial^2 \ell}{\partial (\sigma^2)^2} = -\frac{n}{2\sigma^4} + \frac{1}{\sigma^6} \sum (y_j - \mu)^2.$$

On replacing  $y_j$  with  $Y_j$  and taking expectations, we get

$$I(\mu, \sigma^2) = \begin{pmatrix} n/\sigma^2 & 0 \\ 0 & n/(2\sigma^4) \end{pmatrix}, \quad (4.18)$$

because  $E(Y_j) = \mu$  and  $E\{(Y_j - \mu)^2\} = \sigma^2$ . ■

### 4.3.2 Efficiency

Suppose that we might adopt one of two sampling schemes, and we wish to see which is most efficient in the sense of needing least data to pin down the parameter to a given range. One way to do this is to compare the information in each likelihood. If  $\theta$  is scalar, the *asymptotic efficiency* of sampling scheme A relative to sampling scheme B is

$$\frac{I_A(\theta)}{I_B(\theta)}, \quad (4.19)$$

where  $I_A(\theta)$  and  $I_B(\theta)$  are the expected information quantities for schemes A and B. In simple random samples (4.19) equals  $n_A i_A(\theta) / \{n_B i_B(\theta)\}$ , where  $n_A$  and  $n_B$  observations are used by the sampling schemes. The information from both schemes is equal if

$$\frac{n_B}{n_A} = \frac{i_A(\theta)}{i_B(\theta)} \quad (4.20)$$

and we see that  $i_A(\theta)/i_B(\theta)$  can be interpreted as the number of observations an observer using scheme B would need in order to get the information in a single observation sampled under scheme A, when the parameter value is  $\theta$ . Expression (4.19) is called the asymptotic efficiency because this use of the information rests on the quadratic likelihoods usually entailed by large samples.

**Example 4.19 (Poisson process)** Over short periods the times at which vehicles pass an observer on a country road might be modelled as a Poisson process of rate  $\lambda$  vehicles/hour. Observer A decides to estimate  $\lambda$  by counting how many cars pass in a period of  $t_0$  minutes. Observer B, who is more diligent, records the times at which they pass.

The total number of events,  $N$ , when a Poisson process of rate  $\lambda$  is observed for a period of length  $t_0$  has the Poisson distribution with mean  $\lambda t_0$ . Hence A bases her inference on the likelihood

$$L_A(\lambda) = \frac{(\lambda t_0)^N}{N!} e^{-\lambda t_0}, \quad \lambda > 0,$$

for which the observed and expected information quantities are

$$J_A(\lambda) = N/\lambda^2, \quad I_A(\lambda) = t_0/\lambda,$$

since  $E(N) = \lambda t_0$ .

The times between events in a Poisson process of rate  $\lambda$  have independent exponential distributions with density  $\lambda e^{-\lambda u}$ ,  $u > 0$ . Therefore if observer B records cars passing at times  $0 < t_1 < \dots < t_N < t_0$ , his likelihood is

$$\lambda e^{-\lambda t_1} \times \lambda e^{-\lambda(t_2-t_1)} \times \dots \times \lambda e^{-\lambda(t_N-t_{N-1})} \times e^{-\lambda(t_0-t_N)},$$

where the final term corresponds to observing no cars in the interval  $(t_N, t_0)$ . Thus B bases his inference on

$$L_B(\lambda) = \lambda^N e^{-\lambda t_0},$$

for which the observed and expected information quantities are the same as those for A. Thus the efficiency of A relative to B is  $I_A(\lambda)/I_B(\lambda) = 1$ : no information is lost by recording only the number of cars. This is because  $L_A(\lambda) \propto L_B(\lambda)$ ; under either sampling scheme, the statistic  $N$  is sufficient for  $\lambda$ .

Inference for Poisson processes is discussed in Section 6.5.1. ■

**Example 4.20 (Censoring)** A widget has lifetime  $T$ , but trials to estimate widget lifetimes finish after a known time  $c$  when the vice president for widget testing has a tea break. The available data are the observed lifetime  $Y = \min(T, c)$ , and  $D = I(T \leq c)$ , where  $D$  indicates whether  $T$  has been observed. If  $T > c$  then  $T$  is said to be *right-censored*: we know only that its value exceeds  $c$ .

If  $T$  has density and distribution functions  $f(t; \theta)$  and  $F(t; \theta)$ , the likelihood contribution from  $(Y, D)$  is

$$f(Y; \theta)^D \{1 - F(c; \theta)\}^{1-D},$$

$I(\cdot)$  is the indicator function of the event ' $\cdot$ '.

so the likelihood for a random sample of data  $(y_1, d_1), \dots, (y_n, d_n)$  is

$$\prod_{j=1}^n [f(y_j; \theta)^{d_j} \{1 - F(y_j; \theta)\}^{1-d_j}] = \prod_{\text{uncens}} f(y_j; \theta) \times \prod_{\text{cens}} \{1 - F(c; \theta)\},$$

where the first product is over uncensored data, and the second is over censored data.

The likelihood for a random sample with exponential density  $f(u; \lambda) = \lambda e^{-\lambda u}$ ,  $u > 0$ ,  $\lambda > 0$ , and distribution  $F(u; \lambda) = 1 - e^{-\lambda u}$ ,  $u > 0$ , is

$$\prod_{\text{uncens}} \lambda e^{-\lambda y_j} \times \prod_{\text{cens}} e^{-\lambda c} = \exp \left( \sum_{j=1}^n d_j \log \lambda - \lambda \sum_{j=1}^n y_j \right).$$

The observed information is  $J(\lambda) = \sum d_j / \lambda^2$ , which decreases as  $\sum d_j$  decreases: if  $n$  is known, there is information only in observations that were seen to fail. To find the expected information  $I_c(\lambda)$  when there is censoring at  $c$ , note that

$$E \left( \sum_{j=1}^n D_j \right) = n \Pr(Y \leq c) = n(1 - e^{-\lambda c}),$$

so that  $I_c(\lambda) = n(1 - e^{-\lambda c})/\lambda^2$ . By letting  $c \rightarrow \infty$  we can obtain the expected information when there is no censoring,  $I_\infty(\lambda) = n/\lambda^2$ . Therefore the relative efficiency when there is censoring at  $c$  is

$$\frac{I_c(\lambda)}{I_\infty(\lambda)} = \frac{n(1 - e^{-\lambda c})/\lambda^2}{n/\lambda^2} = 1 - e^{-\lambda c}.$$

This equals the proportion of uncensored data, which is unsurprising, as we saw above that censored observations do not contribute to  $J(\lambda)$ . As one would anticipate, the loss of information becomes more severe as  $c$  decreases.

Inference for censored data is discussed in Sections 5.4 and 10.8. ■

When  $\theta$  is a  $p \times 1$  vector, we replace (4.19) by the ratio

$$\left\{ \frac{|I_A(\theta)|}{|I_B(\theta)|} \right\}^{1/p},$$

which preserves the interpretation of efficiency given at (4.20) in terms of numbers of observations. This is an overall measure of the efficiency of the schemes, but often in practice one may want to compare the efficiency of estimation for a single component of  $\theta$ , say  $\theta_r$ . For reasons to be given in Section 4.4.2, the appropriate measure is then  $I_B^{rr}(\theta)/I_A^{rr}(\theta)$ , where  $I_A^{rr}(\theta)$  is the  $(r, r)$ th element of the inverse matrix  $I_A(\theta)^{-1}$ .

**Example 4.21 (Rounding)** What information is lost when the sample 2.71828, 3.14159, ... is rounded to 2.7, 3.1, ...? Let  $Y$  denote a real-valued continuous random variable with distribution function  $F(y; \theta)$ . In recording the data,  $Y$  is rounded to  $X$ , the nearest multiple of  $\delta$ . Thus  $X = k\delta$  if  $(k - \frac{1}{2})\delta \leq Y < (k + \frac{1}{2})\delta$ , an event with probability

$$\pi_k(\theta) = F \left\{ \left( k + \frac{1}{2} \right) \delta; \theta \right\} - F \left\{ \left( k - \frac{1}{2} \right) \delta; \theta \right\}.$$

$|C|$  is the determinant of the  $p \times p$  matrix  $C$ .

| $\delta/\sigma$           | 0.001 | 0.01 | 0.1  | 0.2  | 0.5  | 1    | 1.5  | 2    | 3    |
|---------------------------|-------|------|------|------|------|------|------|------|------|
| Overall efficiency        | 100   | 100  | 99.9 | 99.5 | 97.0 | 88.9 | 77.9 | 64.0 | 37.5 |
| Efficiency for $\mu$      | 100   | 100  | 99.9 | 99.7 | 98.0 | 92.3 | 84.2 | 75.5 | 54.2 |
| Efficiency for $\sigma^2$ | 100   | 100  | 99.8 | 99.3 | 96.0 | 85.5 | 72.0 | 54.2 | 25.9 |

**Table 4.2** Efficiency (%) of likelihood inference when  $N(0, \sigma^2)$  data are rounded to the nearest  $\delta$ .

The density of a single rounded observation may be written  $\prod_k \pi_k(\theta)^{I(X=k\delta)}$ , so the log likelihood for  $\theta$  based on  $X$  is

$$\ell(\theta) = \sum_{k=-\infty}^{\infty} I(X = k\delta) \log \pi_k(\theta).$$

On differentiation we find that

$$\frac{\partial^2 \ell(\theta)}{\partial \theta_r \partial \theta_s} = \sum_{k=-\infty}^{\infty} I(X = k\delta) \left\{ \frac{1}{\pi_k} \frac{\partial^2 \pi_k}{\partial \theta_r \partial \theta_s} - \left( \frac{1}{\pi_k} \frac{\partial \pi_k}{\partial \theta_r} \right) \left( \frac{1}{\pi_k} \frac{\partial \pi_k}{\partial \theta_s} \right) \right\},$$

and as  $\sum_k \pi_k(\theta) = 1$  for all  $\theta$  and  $E\{I(X = k\delta)\} = \pi_k(\theta)$ , the  $(r, s)$  element of the expected information matrix for a random sample  $X_1, \dots, X_n$  is

$$n \sum_{k=-\infty}^{\infty} \frac{1}{\pi_k(\theta)} \frac{\partial \pi_k(\theta)}{\partial \theta_r} \frac{\partial \pi_k(\theta)}{\partial \theta_s}. \quad (4.21)$$

For concreteness, suppose that  $Y$  is normally distributed with mean  $\mu$  and variance  $\sigma^2$ , in which case  $\pi_k(\mu, \sigma^2) = \Phi(z_{k+1}) - \Phi(z_k)$  and

$$\frac{\partial \pi_k}{\partial \mu} = -\frac{1}{\sigma} \{\phi(z_{k+1}) - \phi(z_k)\}, \quad \frac{\partial \pi_k}{\partial \sigma^2} = -\frac{1}{2\sigma^2} \{z_{k+1}\phi(z_{k+1}) - z_k\phi(z_k)\}, \quad (4.22)$$

where  $z_k = \sigma^{-1}\{(k - \frac{1}{2})\delta - \mu\}$ . With  $\mu = 0$  it turns out that the expected information may be written as

$$n \begin{pmatrix} \sigma^{-2} I_{\mu\mu}(\delta/\sigma) & 0 \\ 0 & (4\sigma^4)^{-1} I_{\sigma\sigma}(\delta/\sigma) \end{pmatrix},$$

where the elements are given by substituting (4.22) into (4.21). On comparing this with (4.18), we see that the overall efficiency for the two parameters is  $\{I_{\mu\mu}(\delta/\sigma)I_{\sigma\sigma}(\delta/\sigma)/2\}^{1/2}$ , while the efficiencies for  $\mu$  and  $\sigma^2$  separately are  $I_{\mu\mu}(\delta/\sigma)$  and  $\frac{1}{2}I_{\sigma\sigma}(\delta/\sigma)$ . Table 4.2 shows that these are remarkably high even with quite heavy rounding. When  $\delta = \sigma = 1$ , rounding  $Y$  to  $X$  gives a discrete distribution with almost all its probability on the seven values  $-3, -2, \dots, 3$ , but a sample  $x_1, \dots, x_{100}$  of such values gives almost the same efficiency as 89 of the corresponding  $y$ s: the overall loss of efficiency is only 11%. If the data are rounded to the equivalent of one decimal place,  $\delta = 0.1\sigma$ , there is effectively no information lost. with  $\delta = 1.5\sigma$  or more the loss is more dramatic, particularly for estimation of  $\sigma$ , and with  $\delta = 3\sigma$  the data are almost binary.

Although suggestive, these results should be regarded with caution for two reasons. First, they apply to large samples, and the efficiency loss might be different in small

samples. Second, they rest on the assumption that the multinomial likelihood based on the  $x_j$  is used, but in practice the rounded data would usually be treated as continuous and inference based on the (incorrect) log likelihood  $\sum_j \log f(x_j; \theta)$ . Practical 4.1 considers the effect of this. ■

### Exercises 4.3

- 1 (a) Show that the log likelihood for a random sample from density (2.7) is

$$\ell(\lambda, \kappa) = n\kappa \log \lambda + (\kappa - 1) \sum \log y_j - \lambda \sum y_j - n \log \Gamma(\kappa),$$

deduce that the observed information is

$$J(\lambda, \kappa) = n \begin{pmatrix} \kappa/\lambda^2 & -1/\lambda \\ -1/\lambda & d^2 \log \Gamma(\kappa)/d\kappa^2 \end{pmatrix},$$

and find the expected information  $I(\lambda, \kappa)$ .

(b) Suppose that we write  $\lambda = \kappa/\mu$ , where  $\mu$  is the distribution mean. Find the log likelihood in terms of  $\mu$  and  $\kappa$ , and show that  $J(\mu, \kappa)$  is random and  $I(\mu, \kappa) = n \text{diag}\{2\kappa/\mu^2, d^2 \log \Gamma(\kappa)/d\kappa^2 - 1/\kappa\}$ .

- 2 Check the details of Example 4.19.

- 3  $Y_1, \dots, Y_n$  are independent normal random variables with unit variances and means  $E(Y_j) = \beta x_j$ , where the  $x_j$  are known quantities in  $(0, 1]$  and  $\beta$  is an unknown parameter. Show that  $\ell(\beta) \equiv -\frac{1}{2} \sum (y_j - x_j \beta)^2$  and find the expected information  $I(\beta)$  for  $\beta$ .

Suppose that  $n = 10$  and that an experiment to estimate  $\beta$  is to be designed by choosing the  $x_j$  appropriately. Show that  $I(\beta)$  is maximized when all the  $x_j$  equal 1. Is this design sensible if there is any possibility that  $E(Y_j) = \alpha + \beta x_j$ , with  $\alpha$  unknown?

- 4 Use (4.21) and (4.22) to give expressions for the quantities  $I_{\mu\mu}(\delta/\sigma)$  and  $I_{\sigma\sigma}(\delta/\sigma)$  in Example 4.21. Show that  $I_{\mu\sigma}(\delta/\sigma) = 0$  when  $\mu = 0$ .

- 5 Find the expected information for  $\theta$  based on a random sample  $Y_1, \dots, Y_n$  from the geometric density

$$f(y; \theta) = \theta(1 - \theta)^{y-1}, \quad y = 1, 2, 3, \dots, \quad 0 < \theta < 1.$$

A statistician has a choice between observing random samples from the Bernoulli or geometric densities with the same  $\theta$ . Which will give the more precise inference on  $\theta$ ?

- 6 Suppose a random sample  $Y_1, \dots, Y_n$  from the exponential density is rounded down to the nearest  $\delta$ , giving  $\delta Z_j$ , where  $Z_j = \lfloor Y_j/\delta \rfloor$ . Show that the likelihood contribution from a rounded observation can be written  $(1 - e^{-\lambda\delta})e^{-Z_j\lambda\delta}$ , and deduce that the expected information for  $\lambda$  based on the entire sample is  $n\delta^2 \exp(-\lambda\delta)\{1 - \exp(-\lambda\delta)\}^{-2}$ . Show that this has limit  $n/\lambda^2$  as  $\delta \rightarrow 0$ , and that if  $\lambda = 1$ , the loss of information when data are rounded down to the nearest integer rather than recorded exactly, is less than 10%. Find the loss of information when  $\delta = 0.1$ , and comment briefly.

## 4.4 Maximum Likelihood Estimator

### 4.4.1 Computation

The maximum likelihood estimate of  $\theta$ ,  $\hat{\theta}$ , is a value of  $\theta$  that maximizes the likelihood, or equivalently the log likelihood. Suppose  $\psi = \psi(\theta)$  is a 1–1 function of  $\theta$ . Then in terms of  $\psi$  the likelihood is

$$L^*(\psi) = L^*\{\psi(\theta)\} = L(\theta),$$

A sketch may help.

so the largest values of  $L^*$  and  $L$  coincide, and the maximum likelihood estimate of  $\psi$  is  $\hat{\psi} = \psi(\hat{\theta})$ . This simplifies calculation of maximum likelihood estimates, as we can compute them in the most convenient parametrization, and then transform them to the scale of interest.

Often, though not invariably,  $\hat{\theta}$  satisfies the *likelihood equation*

$$\frac{\partial \ell(\hat{\theta})}{\partial \theta} = 0. \quad (4.23)$$

If  $\theta$  is a  $p \times 1$  vector, (4.23) is a  $p \times 1$  system of equations that must be solved simultaneously for the components of  $\hat{\theta}$ . We check that  $\hat{\theta}$  gives a local maximum by verifying that  $-d^2 \ell(\hat{\theta})/d\theta^2 > 0$ , or in the vector case that the observed information matrix  $J(\theta) = -d^2 \ell(\theta)/d\theta d\theta^T$  is positive definite at  $\hat{\theta}$ . If there are several solutions to (4.23), in principle we find them all, check which are maxima, and then evaluate  $\ell(\theta)$  at each local maximum, thereby obtaining the global maximum. If there are numerous local maxima, as in Figure 4.2, doubt is cast on the usefulness of summarizing  $\ell(\theta)$  in terms of  $\hat{\theta}$  and  $J(\hat{\theta})$ , but many log likelihoods can be shown to be strictly concave. Then a local maximum is also the global maximum, so there is a unique maximum; moreover if there is a solution to (4.23), it is unique and gives the maximum.

**Example 4.22 (Normal distribution)** The likelihood equation for a random sample  $y_1, \dots, y_n$  from the normal distribution with mean  $\mu$  and variance  $\sigma^2$  is (Example 4.18)

$$\begin{pmatrix} \frac{\partial \ell(\mu, \sigma^2)}{\partial \mu} \\ \frac{\partial \ell(\mu, \sigma^2)}{\partial \sigma^2} \end{pmatrix} = \begin{pmatrix} \sigma^{-2} \sum (y_j - \mu) \\ -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum (y_j - \mu)^2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

The first of these has the sole solution  $\hat{\mu} = \bar{y}$  for all values of  $\sigma^2$ , and  $\ell(\hat{\mu}, \sigma^2)$  is unimodal with maximum at  $\hat{\sigma}^2 = n^{-1} \sum (y_j - \bar{y})^2$ . At the point  $(\hat{\mu}, \hat{\sigma}^2)$ , the observed information matrix  $J(\mu, \sigma^2)$  is diagonal with elements  $\text{diag}\{n/\hat{\sigma}^2, n/(2\hat{\sigma}^4)\}$ , and so is positive definite. Hence  $\bar{y}$  and  $n^{-1} \sum (y_j - \bar{y})^2$  are the sole solutions to the likelihood equation, and therefore are the maximum likelihood estimates.

If we wish to estimate the mean of  $\exp(Y)$ , which is  $\psi = \exp(\mu + \sigma^2/2)$ , then rather than reparametrize in terms of  $\psi$  and  $\mu$ , say, and maximizing directly, we use the earlier results on transformations to see that the maximum likelihood estimate of  $\psi$  is  $\hat{\psi} = \exp(\hat{\mu} + \hat{\sigma}^2/2)$ . ■

In most realistic cases (4.23) must be solved iteratively, and often variants of the Newton–Raphson algorithm can be used. Given a starting-value  $\theta^\dagger$ , we expand (4.23) by Taylor series about  $\theta^\dagger$  to obtain

$$0 = \frac{\partial \ell(\hat{\theta})}{\partial \theta} \doteq \frac{\partial \ell(\theta^\dagger)}{\partial \theta} + \frac{\partial^2 \ell(\theta^\dagger)}{\partial \theta \partial \theta^T} (\hat{\theta} - \theta^\dagger). \quad (4.24)$$

On rearranging (4.24) we obtain

$$\hat{\theta} \doteq \theta^\dagger + J(\theta^\dagger)^{-1} U(\theta^\dagger), \quad (4.25)$$

where  $U(\theta) = \partial \ell(\theta)/\partial \theta$  is called the *score statistic* or *score vector*, and  $J(\theta)$  is the observed information (4.17). In the vector case  $\hat{\theta}$ ,  $\theta^\dagger$  and  $U(\theta^\dagger)$  are  $p \times 1$  vectors and

$J(\theta^\dagger)$  is a  $p \times p$  matrix. The log likelihood is usually maximized in a few iterations of (4.25), using  $\hat{\theta}$  from one iteration as  $\theta^\dagger$  for the next. In doubtful cases it is wise to try several initial values of  $\theta^\dagger$ .

The iteration (4.25) gives  $\hat{\theta}$  in one step if  $\ell(\theta)$  is actually quadratic, so convergence is accelerated by choosing a parametrization in which  $\ell(\theta)$  is as close to quadratic as possible. Often it helps to transform components of  $\theta$  to take values in the real line, for example removing the restrictions  $\lambda > 0$  and  $0 < \pi < 1$  by maximizing in terms of  $\log \lambda$  and  $\log\{\pi/(1 - \pi)\}$ . This also avoids steps that take  $\hat{\theta}$  outside the parameter space. Another simple trick is to use a variable step-length in (4.25). We replace  $J(\theta^\dagger)^{-1}U(\theta^\dagger)$  by  $cJ(\theta^\dagger)^{-1}U(\theta^\dagger)$ , choose  $c$  to maximize  $\ell$  along this line, then recalculate  $U$  and  $J$ , and try again. Many standard models are readily fitted with a few lines of code in statistical packages, but fitting more adventurous models may involve writing special programs.

**Example 4.23 (Weibull distribution)** The log likelihood for a random sample from the Weibull density (4.4) is

$$\ell(\theta, \alpha) = n \log \alpha - n \log \theta + (\alpha - 1) \sum_{j=1}^n \log \left( \frac{y_j}{\theta} \right) - \sum_{j=1}^n \left( \frac{y_j}{\theta} \right)^\alpha,$$

the score function is

$$U(\theta, \alpha) = \begin{pmatrix} \partial \ell / \partial \theta \\ \partial \ell / \partial \alpha \end{pmatrix} = \begin{pmatrix} -n\alpha/\theta + \alpha\theta^{-1} \sum (y_j/\theta)^\alpha \\ n/\alpha + \sum \log(y_j/\theta) - \sum (y_j/\theta)^\alpha \log(y_j/\theta) \end{pmatrix},$$

and the likelihood equation (4.23) cannot be solved analytically. The observed information matrix  $J(\theta, \alpha)$  is

$$\begin{pmatrix} \alpha(\alpha + 1)/\theta^2 \sum (y_j/\theta)^\alpha - n\alpha\theta^{-2} & \theta^{-1} \sum [1 - (y_j/\theta)^\alpha \{1 + \alpha \log(y_j/\theta)\}] \\ \theta^{-1} \sum [1 - (y_j/\theta)^\alpha \{1 + \alpha \log(y_j/\theta)\}] & n/\alpha^2 + \sum (y_j/\theta)^\alpha \{\log(y_j/\theta)\}^2 \end{pmatrix},$$

and to obtain maximum likelihood estimates we would iterate (4.25) until it converged. Suitable starting-values could be obtained by setting  $\alpha^\dagger = 1$ , in which case  $\theta^\dagger = \bar{y}$ . If trouble arose in using (4.25), it would be sensible to write the problem in terms of  $\psi = (\log \theta, \log \alpha)^\top$ , and iterate based on  $\hat{\psi} = \psi^\dagger + J(\psi^\dagger)^{-1}U(\psi^\dagger)$ .

In this case a two-dimensional maximization can be avoided by noticing that for fixed  $\alpha$  the unique maximum likelihood estimate of  $\theta$  is

$$\hat{\theta}_\alpha = \left( n^{-1} \sum_{j=1}^n y_j^\alpha \right)^{1/\alpha}.$$

The dashed line in the upper right panel of Figure 4.1 shows the curve traced out by  $\hat{\theta}_\alpha$  as a function of  $\alpha$ . The value of  $\ell$  along this curve, the *profile log likelihood* for  $\alpha$ ,

$$\ell_p(\alpha) = \max_{\theta} \ell(\theta, \alpha) = \ell(\hat{\theta}_\alpha, \alpha),$$

is shown in the lower right panel of the figure. This function is unimodal, and from it we see that  $\hat{\alpha} \doteq 6$ . More precise estimates are obtained maximizing  $\ell_p(\alpha)$  numerically over  $\alpha$ , to obtain  $\hat{\alpha}$  and hence  $\hat{\theta} = \hat{\theta}_{\hat{\alpha}}$ . ■

A variant of the Newton–Raphson method, *Fisher scoring*, replaces  $J(\theta^\dagger)$  in (4.25) with the expected information  $I(\theta^\dagger)$ . This is useful when  $J(\theta^\dagger)$  is badly behaved — for example, not positive definite — but typically (4.25) works well. It has the advantage that it can be implemented in an automatic way using numerical first and second derivatives of  $\ell(\theta)$ . In simple problems where minimizing programming time is more important than saving computing time it is generally simplest to maximize the log likelihood directly using a packaged routine.

#### 4.4.2 Large-sample distribution

Thus far we have treated the maximum likelihood estimate as a summary of a likelihood based on a given sample  $y_1, \dots, y_n$ , rather than as a random variable. Evidently, however, we may consider its properties when samples are repeatedly taken from the model. Suppose we have a random sample  $Y_1, \dots, Y_n$  from a density  $f(y; \theta)$  that satisfies the regularity conditions:

- the true value  $\theta^0$  of  $\theta$  is interior to the parameter space  $\Theta$ , which has finite dimension and is compact;
- the densities defined by any two different values of  $\theta$  are distinct;
- there is a neighbourhood  $\mathcal{N}$  of  $\theta^0$  within which the first three derivatives of the log likelihood with respect to  $\theta$  exist almost surely, and for  $r, s, t = 1, \dots, p$ ,  $n^{-1}E\{|\partial^3 \ell(\theta)/\partial \theta_r \partial \theta_s \partial \theta_t|\}$  is uniformly bounded for  $\theta \in \mathcal{N}$ ; and
- within  $\mathcal{N}$ , the Fisher information matrix  $I(\theta)$  is finite and positive definite, and its elements satisfy

$$I(\theta)_{rs} = E \left\{ \frac{\partial \ell(\theta)}{\partial \theta_r} \frac{\partial \ell(\theta)}{\partial \theta_s} \right\} = E \left\{ - \frac{\partial^2 \ell(\theta)}{\partial \theta_r \partial \theta_s} \right\}, \quad r, s = 1, \dots, p.$$

We shall see below that this implies that  $I(\theta)$  is the variance matrix of the score vector.

Some cases where these conditions fail are described in Section 4.6. If they do hold, the main results below also apply to many situations where the data are neither independent nor identically distributed.

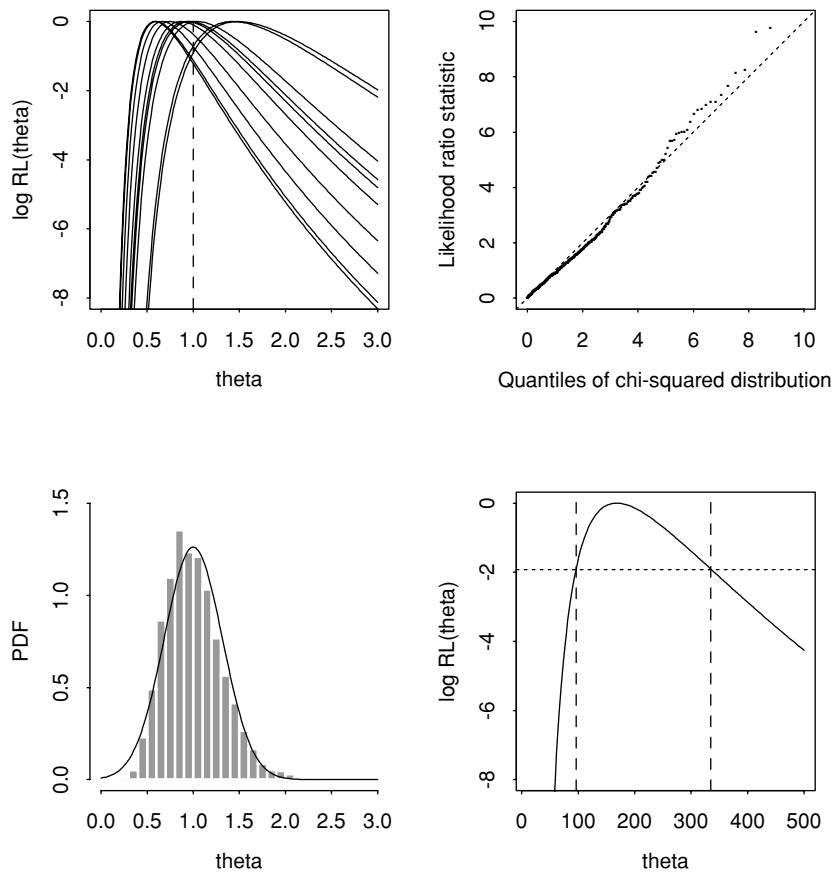
At the end of this section we establish two key results. First, as  $n \rightarrow \infty$  there is a value  $\hat{\theta}$  of  $\theta$  such that  $\ell(\hat{\theta})$  is a local maximum of  $\ell(\theta)$  and  $\Pr(\hat{\theta} \rightarrow \theta^0) = 1$ ; this is a strongly consistent estimator of  $\theta$ . Second,

$$I(\theta^0)^{1/2}(\hat{\theta} - \theta^0) \xrightarrow{D} Z \quad \text{as } n \rightarrow \infty, \quad (4.26)$$

where  $Z$  has the  $N_p(0, I_p)$  distribution. The first holds very generally, but the second requires smoothness of certain log likelihood derivatives. The condition  $n \rightarrow \infty$  can often be replaced by  $I(\theta^0) \rightarrow \infty$ .

Another way to express (4.26) is to say that for large  $n$ ,  $\hat{\theta} \sim N(\theta^0, I(\theta^0)^{-1})$ , and this explains our definition of asymptotic relative efficiency for components of vector parameters, on page 113: we compare asymptotic variances of two different estimators of  $\theta^0$ .

**Figure 4.5** Repeated sampling likelihood inference for the exponential mean. The upper left panel shows the functions  $\log RL(\theta)$  for ten random samples of size  $n = 10$  from the exponential distribution with mean  $\theta^0 = 1$ ; the dashed line shows  $\theta^0$ . The lower left panel shows a histogram of 5000 maximum likelihood estimates  $\hat{\theta}$ , together with their approximate normal density. The upper right panel shows a probability plot of 5000 replicates of  $W(\theta^0) = -2 \log RL(\theta^0)$  against quantiles of the  $\chi^2_1$  distribution. The lower right panel shows the construction of a 95% confidence region for the value of  $\theta$  using ten observations from the spring failure data. The region is the set of all  $\theta$  such that  $\log RL(\theta) \geq -\frac{1}{2}c_1(0.95)$ , where  $c_1(0.95)$  is the 0.95 quantile of the  $\chi^2_1$  distribution; the dotted horizontal line shows  $\frac{1}{2}c_1(0.95)$  and the limits of the region are the dashed vertical lines.



We illustrate (4.26) with random samples of size  $n = 10$  from the exponential distribution with true mean  $\theta^0 = 1$ . As we saw in Section 4.2.1, the log likelihood for a random sample  $y_1, \dots, y_n$  is  $\ell(\theta) \equiv -n(\log \theta + \bar{y}/\theta)$ , and the maximum likelihood estimate is  $\hat{\theta} = \bar{y}$ . The observed information and expected information are  $J(\theta) = n(2\bar{y}/\theta^3 - 1/\theta^2)$  and  $I(\theta) = n/\theta^2$ . The upper left panel of Figure 4.5 shows the log relative likelihoods for ten such samples. Each curve is asymmetric about its maximum, so the distribution of  $\hat{\theta}$  is skewed; see the lower left panel. The density of  $\hat{\theta}$  is roughly normal with mean  $\theta^0 = 1$  and variance  $I(\theta^0)^{-1} = 1/10$ , but this is a poor approximation. In fact  $\bar{Y}$  has an exact gamma density with shape parameter 10 and unit mean.

On replacing  $I(\theta^0)$  in (4.26) by  $I(\hat{\theta})$ , we obtain the approximation

$$\hat{\theta} \sim N_p(\theta^0, V), \quad (4.27)$$

where  $V = I(\hat{\theta})^{-1}$  is the inverse expected information. Provided (4.26) is true, replacement of  $I(\theta^0)$  by  $I(\hat{\theta})$  or  $J(\hat{\theta})$  is justified by the fact that both converge in probability to  $I(\theta^0)$ , so we can apply Slutsky's lemma (2.15). The main use of (4.27) is to construct confidence regions for components of  $\theta^0$ .

### Scalar parameter

If  $\theta$  is scalar, (4.27) boils down to

$$I(\hat{\theta})^{1/2}(\hat{\theta} - \theta^0) \sim N(0, 1).$$

Thus  $I(\hat{\theta})^{1/2}(\hat{\theta} - \theta^0)$  is an approximate pivot from which to find confidence intervals for  $\theta^0$ . That is,

$$\begin{aligned} 1 - 2\alpha &= \Pr\{z_\alpha \leq I(\hat{\theta})^{1/2}(\hat{\theta} - \theta^0) \leq z_{1-\alpha}\} \\ &= \Pr\{\hat{\theta} - z_{1-\alpha}I(\hat{\theta})^{-1/2} \leq \theta^0 \leq \hat{\theta} - z_\alpha I(\hat{\theta})^{-1/2}\}, \end{aligned}$$

$z_\alpha$  is the  $\alpha$  quantile of the standard normal distribution.

giving the  $(1 - 2\alpha)$  confidence interval for  $\theta^0$ ,

$$(\hat{\theta} - z_{1-\alpha}I(\hat{\theta})^{-1/2}, \hat{\theta} - z_\alpha I(\hat{\theta})^{-1/2}). \quad (4.28)$$

The corresponding interval using the observed information  $J(\hat{\theta})$ ,

$$(\hat{\theta} - z_{1-\alpha}J(\hat{\theta})^{-1/2}, \hat{\theta} - z_\alpha J(\hat{\theta})^{-1/2}), \quad (4.29)$$

is easier to calculate than (4.28) because it requires no expectations, and moreover its coverage probability is often closer to the nominal level. Both intervals are symmetric about  $\hat{\theta}$ .

**Example 4.24 (Spring failure data)** We reconsider the exponential model fitted to the data of Example 4.2, for which  $n = 10$  and  $\hat{\theta} = \bar{y} = 168.3$ . For this model  $I(\hat{\theta}) = J(\hat{\theta}) = n/\bar{y}^2$ , so the 95% confidence intervals (4.28) and (4.29) for the true mean both equal  $\bar{y} \pm z_{0.025}n^{-1/2}\bar{y}$ , that is, (64.0, 272.6). ■

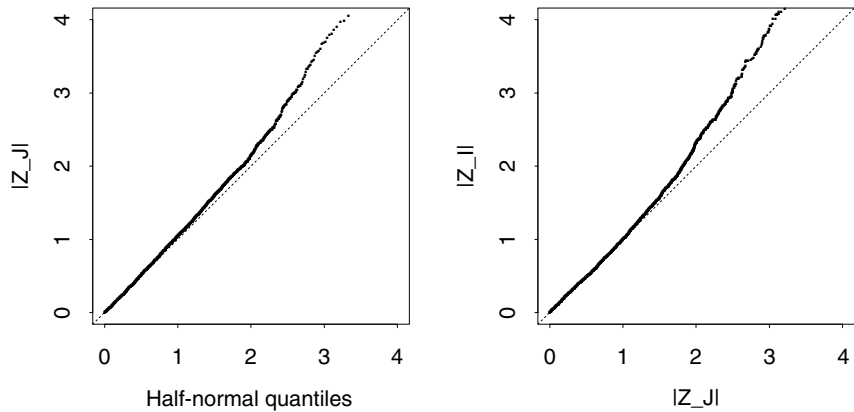
**Example 4.25 (Cauchy data)** To see the quality of these confidence intervals, we take samples of size  $n$  from the Cauchy density (2.16), for which

$$\ell(\theta) \equiv -\sum_{j=1}^n \log\{1 + (y_j - \theta)^2\}, \quad J(\theta) = 2 \sum_{j=1}^n \frac{1 - (y_j - \theta)^2}{\{1 + (y_j - \theta)^2\}^2}, \quad I(\theta) = \frac{1}{2}n;$$

we take  $\theta^0 = 0$ . The basis of (4.28) and (4.29) is large-sample normality of  $Z_I = I(\hat{\theta})^{1/2}(\hat{\theta} - \theta^0)$  and  $Z_J = J(\hat{\theta})^{1/2}(\hat{\theta} - \theta^0)$ , and to assess this we compare  $Z_I$  and  $Z_J$  with a standard normal variable  $Z$ . Symmetry of the Cauchy density about  $\theta^0$  implies that  $Z_I$  and  $Z_J$  are distributed symmetrically about the origin, so the left panel of Figure 4.6 compares quantiles of  $|Z_J|$  with those of  $|Z|$  in a half-normal plot (Practical 3.1), for 5000 simulated Cauchy samples of size  $n = 15$ . Evidently the distribution of  $Z_J$  is close to normal; its empirical 0.9, 0.95, 0.975 and 0.99 quantiles are 1.34, 1.76, 2.08 and 2.55, compared with 1.28, 1.65, 1.96 and 2.33 for  $Z$ . With  $\alpha = 0.025$ , (4.29) has estimated coverage probability 0.93, close to the nominal 0.95. The right panel shows that  $Z_I$  has heavier tails than  $Z_J$ ; the coverage probability for (4.28) with  $\alpha = 0.025$  is 0.91. Use of observed information is preferable, but the large-sample approximations seem accurate enough for practical use even with  $n = 15$ .

Just one of the 5000 log likelihoods had two local maxima, compared to 36 for 5000 samples with  $n = 10$ ; the rest appeared unimodal. Thus  $\hat{\theta}$  was almost invariably the sole solution to the likelihood equation. ■

**Figure 4.6** Inference based on observed and expected information in samples of  $n = 15$  Cauchy observations. Left: half-normal plot of  $|Z_J| = J(\hat{\theta})^{1/2}|\hat{\theta} - \theta^0|$ ; the dotted line shows the ideal, so  $Z_J$  is slightly heavier-tailed than normal. Right: comparison of  $|Z_I| = I(\hat{\theta})^{1/2}|\hat{\theta} - \theta^0|$  with  $|Z_J|$ .  $|Z_I|$  has heavier tails.



### Vector parameter

When  $\theta$  is a vector, confidence sets for the  $r$ th element of  $\theta^0$ ,  $\theta_r^0$ , may be based on the fact that the corresponding maximum likelihood estimator,  $\hat{\theta}_r$ , has approximately the  $N(\theta_r^0, v_{rr})$  distribution, where  $v_{rr}$  is the  $(r, r)$  element of  $V = I(\hat{\theta})^{-1}$  or  $J(\hat{\theta})^{-1}$ . This gives intervals (4.28) and (4.29), but with  $\hat{\theta}$ ,  $I(\hat{\theta})^{-1}$ , and  $J(\hat{\theta})^{-1}$  replaced by  $\hat{\theta}_r$ ,  $v_{rr}$ , and the  $(r, r)$  element of  $J(\hat{\theta})^{-1}$ .

**Example 4.26 (Normal distribution)** In Examples 4.18 and 4.22 we saw that the maximum likelihood estimates of the mean and variance of the normal distribution based on a random sample  $y_1, \dots, y_n$  are  $\hat{\mu} = \bar{y}$  and  $\hat{\sigma}^2 = n^{-1} \sum (y_j - \bar{y})^2$ , and that the expected information matrix is  $\text{diag}\{n/\sigma^2, n/(2\sigma^4)\}$ . Hence  $V = \text{diag}\{n^{-1}\hat{\sigma}^2, n^{-1}2\hat{\sigma}^4\}$ , and the  $(1 - 2\alpha)$  confidence intervals for  $\mu$  and  $\sigma^2$  based on the large-sample results above are

$$\bar{y} \pm n^{-1/2}\hat{\sigma}z_{\alpha}, \quad \hat{\sigma}^2 \pm (2/n)^{1/2}\hat{\sigma}^2z_{\alpha}.$$

The asymptotic approximation gives an interval for  $\mu$  with the same form as the exact interval,  $\bar{y} \pm n^{-1/2}st_{n-1}(\alpha)$ , but with  $s$  replaced by  $\hat{\sigma}$  and the  $t$  quantile replaced by the corresponding normal quantile. Provided that  $n > 20$  or so, these alterations will typically have little effect on the interval. Larger samples are needed for the interval for  $\sigma^2$  to be good, because normal approximation to the distribution of  $\hat{\sigma}^2$  is poorer than to the distribution of  $\hat{\mu}$ . ■

The use of (4.27) to give confidence regions for the whole of  $\theta$  rests on the fact that (4.27) entails  $(\hat{\theta} - \theta^0)^T V^{-1}(\hat{\theta} - \theta^0) \sim \chi_p^2$ . Hence an approximate  $(1 - 2\alpha)$  confidence region is

$$\{\theta : (\hat{\theta} - \theta)^T V^{-1}(\hat{\theta} - \theta) \leq c_p(1 - 2\alpha)\};$$

an ellipsoid centred at  $\hat{\theta}$ , with shape determined by the elements of  $V$  and volume determined by  $c_p(1 - 2\alpha)$ . Another version replaces  $I(\hat{\theta})^{-1}$  with  $J(\hat{\theta})^{-1}$ .

$s^2 = n\hat{\sigma}^2/(n - 1)$  is the unbiased estimate of  $\sigma^2$ .

**Example 4.27 (Challenger data)** Examples 4.5 and 4.8 discuss a model for the Challenger data, where the probability of O-ring thermal distress depends on the launch temperature. The maximum likelihood estimates for this model are  $\hat{\beta}_0 = 5.084$  and  $\hat{\beta}_1 = -0.116$ , and the inverse observed information is

$$J(\hat{\beta}_0, \hat{\beta}_1)^{-1} = \begin{pmatrix} 9.289 & -0.142 \\ -0.142 & 0.00220 \end{pmatrix},$$

yielding standard errors  $9.289^{1/2} = 3.048$  and  $0.00220^{1/2} = 0.0469$ . The estimated correlation of  $\hat{\beta}_0$  and  $\hat{\beta}_1$ ,  $-0.142/(9.289 \times 0.00220)^{1/2}$ , equals  $-0.993$ , and we see that the matrix  $J(\hat{\beta}_0, \hat{\beta}_1)$  is close to singular. In view of the left panel of Figure 4.3 this is not surprising.

A joint 95% confidence region for  $(\beta_0, \beta_1)$  is the ellipsoid given by

$$(\beta_0 - 5.084, \beta_1 + 0.116)J(\hat{\beta}_0, \hat{\beta}_1) \begin{pmatrix} \beta_0 - 5.084 \\ \beta_1 + 0.116 \end{pmatrix} \leq c_2(0.95) = 5.99.$$

■

Often we focus on a scalar parameter  $\psi = \psi(\theta)$ , estimated by  $\hat{\psi} = \psi(\hat{\theta})$ . To approximate the variance of  $\hat{\psi}$  we apply the delta method (2.19), giving

$$\psi(\hat{\theta}) \doteq \psi(\theta^0) + \frac{\partial \psi(\theta^0)}{\partial \theta^\top} (\hat{\theta} - \theta^0).$$

Consequently

$$\text{var}\{\psi(\hat{\theta})\} \doteq \frac{\partial \psi(\theta^0)}{\partial \theta^\top} \text{var}(\hat{\theta}) \frac{\partial \psi(\theta^0)}{\partial \theta} \doteq \frac{\partial \psi(\hat{\theta})}{\partial \theta^\top} J(\hat{\theta})^{-1} \frac{\partial \psi(\hat{\theta})}{\partial \theta},$$

where  $\partial \psi(\hat{\theta})/\partial \theta$  is the  $p \times 1$  vector of derivatives of  $\psi$  evaluated at  $\hat{\theta}$ . Thus an approximate  $(1 - 2\alpha)$  confidence interval for  $\psi$  is

$$\psi(\hat{\theta}) \pm z_\alpha \{ \partial \psi(\hat{\theta})/\partial \theta^\top J(\hat{\theta})^{-1} \partial \psi(\hat{\theta})/\partial \theta \}^{1/2}. \quad (4.30)$$

**Example 4.28 (Challenger data)** One quantity of particular interest is the probability of failure at 31°F,  $\psi = e^{\beta_0 + 31\beta_1}/(1 + e^{\beta_0 + 31\beta_1})$ . Its maximum likelihood estimate and derivatives are

$$\hat{\psi} = \frac{e^{\hat{\beta}_0 + 31\hat{\beta}_1}}{1 + e^{\hat{\beta}_0 + 31\hat{\beta}_1}} = 0.816, \quad \frac{\partial \psi}{\partial \beta_0} = \psi(1 - \psi), \quad \frac{\partial \psi}{\partial \beta_1} = 31\psi(1 - \psi).$$

The 95% confidence interval (4.30) for  $\psi$  is  $0.816 \pm 1.96 \times 0.242 = (0.34, 1.29)$ . As this contains values greater than one it is less than satisfactory, so we need a better approach, such as the one described in Section 4.5.2. ■

#### Consistency of $\hat{\theta}$

We now obtain the key convergence results for maximum likelihood estimation of a scalar, subject to the regularity conditions on page 118.

Let  $h : \mathbb{R} \rightarrow \mathbb{R}$  be convex. Then for any real  $x_1, x_2$ ,

$$h\{\pi x_1 + (1 - \pi)x_2\} \leq \pi h(x_1) + (1 - \pi)h(x_2), \quad 0 \leq \pi \leq 1.$$

If  $X$  is a real-valued random variable, then *Jensen's inequality* says that  $E\{h(X)\} \geq h\{E(X)\}$ , with equality if and only if  $X$  is degenerate.

Let  $Y_1, \dots, Y_n$  be a random sample from a density  $f(y; \theta)$ , where  $\theta$  is scalar with true value  $\theta^0$ , and let  $\bar{\ell}(\theta) = n^{-1} \sum \log f(Y_j; \theta)$ . Now

$$\begin{aligned} E\{\bar{\ell}(\theta) - \bar{\ell}(\theta^0)\} &= E \left[ \log \left\{ \frac{f(Y_1; \theta)}{f(Y_1; \theta^0)} \right\} \right] \\ &\leq \log E \left\{ \frac{f(Y_1; \theta)}{f(Y_1; \theta^0)} \right\} \\ &= \log \int \frac{f(y; \theta)}{f(y; \theta^0)} f(y; \theta^0) dy = 0, \end{aligned} \quad (4.31)$$

where we have applied Jensen's inequality to the convex function  $-\log x$ . The inequality is strict unless the density ratio is constant, so that the densities are the same, and according to our regularity conditions this may occur only if  $\theta = \theta^0$ . As  $n \rightarrow \infty$ , the weak law of large numbers applies to the average  $\bar{\ell}(\theta) - \bar{\ell}(\theta^0)$ , which converges in probability to

$$\int \log \left\{ \frac{f(y; \theta)}{f(y; \theta^0)} \right\} f(y; \theta^0) dy = -D(f_\theta, f_{\theta^0}),$$

say. This is negative unless  $\theta = \theta^0$ . The quantity  $D(f, g) \geq 0$  is known as the *Kullback–Leibler discrepancy* between  $f$  and  $g$ ; it is minimized when  $f = g$ . In fact this convergence is almost sure, that is,  $\bar{\ell}(\theta) - \bar{\ell}(\theta^0)$  converges to  $-D(f_\theta, f_{\theta^0})$  with probability one. This shores up our earlier informal discussion of Figure 4.4, for we see that if  $\theta \neq \theta^0$ , then

$$\bar{\ell}(\theta) - \bar{\ell}(\theta^0) \sim nD(f_\theta, f_{\theta^0}) \rightarrow -\infty$$

with probability one as  $n \rightarrow \infty$ .

Now for any  $\delta > 0$ ,  $\bar{\ell}(\theta^0 - \delta) - \bar{\ell}(\theta^0)$  and  $\bar{\ell}(\theta^0 + \delta) - \bar{\ell}(\theta^0)$  converge with probability one to the negative quantities  $-D(f_{\theta^0-\delta}, f_{\theta^0})$  and  $-D(f_{\theta^0+\delta}, f_{\theta^0})$ . Hence for any sequence of random variables  $Y_1, \dots, Y_n$  there is an  $n'$  such that for  $n > n'$ ,  $\bar{\ell}(\theta)$  has a local maximum in the interval  $(\theta^0 - \delta, \theta^0 + \delta)$ . If we let  $\hat{\theta}$  denote the value at which this local maximum occurs, then  $\Pr(\hat{\theta} \rightarrow \theta^0) = 1$  and  $\hat{\theta}$  is said to be a *strongly consistent* estimate of  $\theta^0$ . This implies  $\hat{\theta} \xrightarrow{P} \theta^0$ , so  $\hat{\theta}$  is consistent in our usual, weaker, sense.

As this proof does not require  $f(y; \theta)$  to be smooth it is very general. It says nothing about uniqueness of  $\hat{\theta}$ , merely that a strongly consistent local maximum exists, but if  $\bar{\ell}(\theta)$  has just one maximum, then  $\hat{\theta}$  must also be the global maximum. A more delicate argument is needed when  $\theta$  is vector, because it is then not enough to consider only the two values  $\theta^0 \pm \delta$ .

Solomon Kullback (1907–1994) was born and educated in New York. He had careers in the US Defense Department and then at George Washington University. His main scientific contribution is to information theory. Richard Arthur Leibler (1914–) has spent much of his life working in the US defense community. Their definition of information was published in 1951.

### Asymptotic normality of $\hat{\theta}$

To prove asymptotic normality of  $\hat{\theta}$ , we assume that  $\hat{\theta}$  satisfies the likelihood equation and consider the score statistic,  $U(\theta) = d\ell(\theta)/d\theta$ . Its mean and variance are

$$\begin{aligned} E\{U(\theta)\} &= \sum_{j=1}^n E\left\{\frac{d \log f(Y_j; \theta)}{d\theta}\right\} = n E\{u(\theta)\}, \\ \text{var}\{U(\theta)\} &= \sum_{j=1}^n \text{var}\left\{\frac{d \log f(Y_j; \theta)}{d\theta}\right\} = n \text{var}\{u(\theta)\}, \end{aligned}$$

where  $u(\theta) = d \log f(Y_j; \theta)/d\theta$  is the score function for a single random variable. Provided the order of differentiation and integration may be interchanged, the mean of  $u(\theta)$  is

$$E\{u(\theta)\} = \int \frac{d \log f(y; \theta)}{d\theta} f(y; \theta) dy = \int \frac{df(y; \theta)}{d\theta} dy = \frac{d}{d\theta} \int f(y; \theta) dy = 0, \quad (4.32)$$

because  $f(y; \theta)$  has integral one for each value of  $\theta$ . Furthermore

$$\begin{aligned} 0 &= \frac{d}{d\theta} \int \frac{d \log f(y; \theta)}{d\theta} f(y; \theta) dy \\ &= \int \frac{d^2 \log f(y; \theta)}{d\theta^2} f(y; \theta) dy + \int \left\{ \frac{d \log f(y; \theta)}{d\theta} \right\}^2 f(y; \theta) dy, \end{aligned}$$

and so

$$\text{var}\{u(\theta)\} = E\{u(\theta)^2\} = - \int \frac{d^2 \log f(y; \theta)}{d\theta^2} f(y; \theta) dy = i(\theta), \quad (4.33)$$

the expected information from a single observation.

Now both  $U(\theta^0)$  and  $J(\theta^0) = - \sum d^2 \ell(\theta^0)/d\theta^2$  are sums of  $n$  independent random variables, and  $E\{U(\theta^0)\} = 0$ ,  $\text{var}\{U(\theta^0)\} = I(\theta^0) = ni(\theta^0)$ , while  $E\{J(\theta^0)\} = I(\theta^0) = ni(\theta^0)$ . Hence the central limit theorem (2.11) and the weak law of large numbers imply that

$$I(\theta^0)^{-1/2} U(\theta^0) \xrightarrow{D} Z, \quad I(\theta^0)^{-1} J(\theta^0) \xrightarrow{P} 1, \quad (4.34)$$

where  $Z$  has the standard normal distribution.

If the log likelihood is sufficiently smooth to allow Taylor series expansion, then  $\hat{\theta}$  satisfies the likelihood equation

$$0 = U(\hat{\theta}) \doteq U(\theta^0) + \frac{d^2 \ell(\theta^0)}{d\theta^2} (\hat{\theta} - \theta^0),$$

rearrangement of which gives

$$\hat{\theta} - \theta^0 \doteq J(\theta^0)^{-1} U(\theta^0),$$

where  $J(\theta^0)$  is the observed information and we require that the missing terms of the Taylor series are asymptotically small enough to be ignored. If so,

$$\begin{aligned} I(\theta^0)^{1/2}(\hat{\theta} - \theta^0) &\doteq I(\theta^0)^{1/2}J(\theta^0)^{-1}U(\theta^0) \\ &= I(\theta^0)^{1/2}J(\theta^0)^{-1}I(\theta^0)^{1/2} \times I(\theta^0)^{-1/2}U(\theta^0) \\ &\xrightarrow{D} Z, \end{aligned}$$

by (4.34) and Slutsky's lemma (2.15). Replacement of  $I(\theta^0)$  by  $I(\hat{\theta})$  or  $J(\hat{\theta})$  is justified by the fact that both converge in probability to  $I(\theta^0)$  as  $n \rightarrow \infty$ .

This argument is generalized to vector  $\theta$  by interpreting the score as a  $p \times 1$  vector and the information quantities as  $p \times p$  matrices, with  $Z$  having a  $N_p(0, I_p)$  distribution.

## Exercises 4.4

- 1 In Example 4.23, show that  $\hat{\alpha}$  is the solution of the equation

$$\hat{\alpha} = \left\{ \frac{\sum_j y_j^{\hat{\alpha}} \log y_j}{\sum_j y_j^{\hat{\alpha}}} - n^{-1} \sum_j \log y_j \right\}^{-1}.$$

- 2 If the log likelihood for a  $p \times 1$  vector of parameters is  $\ell(\theta) = a + b^T \theta - \frac{1}{2} \theta^T C \theta$ , where the constants  $a$ ,  $b$  and  $C$  are respectively scalar, a  $p \times 1$  vector, and a  $p \times p$  symmetric positive definite matrix, show that the score statistic can be written  $b - C\theta$ . Find the observed information  $J(\theta)$ , and show that  $\hat{\theta}$  is attained in one step of (4.25) from any initial value of  $\theta$ .
- 3 The *Laplace* or *double exponential* distribution has density

$$f(y; \mu, \sigma) = \frac{1}{2\sigma} \exp(-|y - \mu|/\sigma), \quad -\infty < y < \infty, \quad -\infty < \mu < \infty, \sigma > 0.$$

Sketch the log likelihood for a typical sample, and explain why the maximum likelihood estimate is only unique when the sample size is odd. Derive the score statistic and observed information. Is maximum likelihood estimation regular for this distribution?

- 4 Eggs are thought to be infected with a bacterium *salmonella enteriditis* so that the number of organisms,  $Y$ , in each has a Poisson distribution with mean  $\mu$ . The value of  $Y$  cannot be observed directly, but after a period it becomes certain whether the egg is infected ( $Y > 0$ ) or not ( $Y = 0$ ). Out of  $m$  such eggs,  $r$  are found to be infected. Find the maximum likelihood estimator  $\hat{\mu}$  of  $\mu$  and its asymptotic variance. Is the exact variance of  $\hat{\mu}$  defined?
- 5 If  $Y_1, \dots, Y_n$  is a random sample from density  $\theta^{-1}e^{-x/\theta}$ , show that the maximum likelihood estimator  $\hat{\theta}$  has an asymptotic normal distribution with mean  $\theta$  and variance  $\theta^2/n$ . Deduce that an approximate  $(1 - 2\alpha)$  confidence interval for  $\theta$  is

$$\frac{\hat{\theta}}{1 + z_{\alpha} n^{-1/2}} \geq \theta \geq \frac{\hat{\theta}}{1 + z_{1-\alpha} n^{-1/2}}.$$

Show that  $\hat{\theta}/\theta$  is an exact pivot, having the gamma distribution with unit mean and shape parameter  $\kappa = n$ . Hence find an exact confidence interval for  $\theta$ , and compare it with the approximate one when  $n = 10$  and  $\hat{\theta} = 100$ .

- 6 If  $Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} N(\mu, c\mu^2)$ , where  $c$  is a known constant, show that the minimal sufficient statistic for  $\mu$  is the same as for the  $N(\mu, \sigma^2)$  distribution. Find the maximum likelihood estimate of  $\mu$  and give its large-sample standard error. Show that the distribution of  $\bar{Y}^2/S^2$  does not depend on  $\mu$ .

$z_{\alpha}$  is the  $\alpha$  quantile of the standard normal distribution.

## 4.5 Likelihood Ratio Statistic

### 4.5.1 Basic ideas

Suppose that our model is determined by a parameter  $\theta$  of dimension  $p$ , whose true but unknown value is  $\theta^0$ , and for which the maximum likelihood estimate is  $\hat{\theta}$ . Then provided the model satisfies the conditions for asymptotic normality of the maximum likelihood estimator given in the previous section, in large samples the *likelihood ratio statistic*

$$W(\theta^0) = -2 \log RL(\theta^0) = 2\{\ell(\hat{\theta}) - \ell(\theta^0)\} \quad (4.35)$$

has an approximate chi-squared distribution on  $p$  degrees of freedom under repeated sampling of data from the model. That is, as  $I(\theta^0) \rightarrow \infty$ ,

$$W(\theta^0) \xrightarrow{D} \chi_p^2, \quad (4.36)$$

so  $W(\theta^0) \sim \chi_1^2$  when  $\theta$  is scalar. In practice this result is used to generate approximations for finite samples. It is illustrated in the upper right panel of Figure 4.5, which compares 5000 simulated values of  $W(\theta^0)$ , based on exponential samples of size  $n = 10$ , with quantiles of the  $\chi_1^2$  distribution. Here  $p = 1$ ,  $W(\theta^0) = 2n\{\bar{Y}/\theta^0 - 1 - \log(\bar{Y}/\theta^0)\}$ , and  $\theta^0 = 1$ . This approximation seems better than that for  $\hat{\theta}$ .

To establish (4.36), we note that  $d\ell(\hat{\theta})/d\theta = 0$  and make a Taylor series expansion of  $W(\theta^0)$ , giving

$$\begin{aligned} W(\theta^0) &= 2\{\ell(\hat{\theta}) - \ell(\theta^0)\} \\ &\doteq 2 \left\{ \ell(\hat{\theta}) - \ell(\hat{\theta}) - (\theta^0 - \hat{\theta})^T \frac{\partial \ell(\hat{\theta})}{\partial \theta} - \frac{1}{2} (\theta^0 - \hat{\theta})^T \frac{\partial^2 \ell(\hat{\theta})}{\partial \theta \partial \theta^T} (\theta^0 - \hat{\theta}) \right\} \\ &= (\hat{\theta} - \theta^0)^T J(\hat{\theta})(\hat{\theta} - \theta^0) \\ &\doteq (\hat{\theta} - \theta^0)^T I(\theta^0)(\hat{\theta} - \theta^0), \end{aligned}$$

and the limiting normal distribution for  $\hat{\theta}$  at (4.26) and the relation (3.23) linking this to the chi-squared distribution yield (4.36).

Expression (4.36) shows that  $W(\theta^0)$  is an approximate pivot which may be used to provide confidence regions for  $\theta^0$ . For if  $W(\theta^0) \sim \chi_p^2$ , then

$$\Pr\{W(\theta^0) \leq c_p(1 - 2\alpha)\} \doteq 1 - 2\alpha,$$

$c_p(\alpha)$  denotes the  $\alpha$  quantile of the  $\chi_p^2$  distribution.

and hence values of  $\theta$  for which  $W(\theta) \leq c_p(1 - 2\alpha)$  may be regarded as ‘plausible’ at the  $(1 - 2\alpha)$  level. Equivalently, the set

$$\left\{ \theta : \ell(\theta) \geq \ell(\hat{\theta}) - \frac{1}{2} c_p(1 - 2\alpha) \right\} \quad (4.37)$$

is a  $(1 - 2\alpha)$  confidence region for the unknown  $\theta^0$ . We use  $(1 - 2\alpha)$  here for consistency with our earlier discussion of confidence intervals.

These ‘plausible’ sets of  $\theta$  based on  $W(\theta^0)$  under repeated sampling have the same form as those for the pure likelihood approach described at the end of Section 4.1.2,

since the condition  $RL(\theta) \geq c$  is equivalent to  $W(\theta) \leq -2 \log c$ . Here however the constant  $-2 \log c$  is replaced by  $c_p(1 - 2\alpha)$ , chosen with respect to the approximate distribution of  $W(\theta^0)$  under repeated sampling. Often  $\alpha$  is taken to be 0.05, 0.025 or 0.005, values that correspond to regions containing  $\theta^0$  with approximate probabilities 0.9, 0.95 and 0.99.

**Example 4.29 (Spring failure data)** The likelihood ratio statistic for the exponential model in Example 4.2 is  $W(\theta) = 2n\{\bar{y}/\theta - 1 - \log(\bar{y}/\theta)\}$ . As  $c_1(0.95) = 3.84$ , a 95% confidence region for  $\theta$  based on  $W(\theta)$  is the set

$$\{\theta : 2n\{\bar{y}/\theta - 1 - \log(\bar{y}/\theta)\} \leq 3.84\}.$$

This set is found by plotting the log likelihood and reading off the values of  $\theta$  for which  $\ell(\theta) \geq \ell(\hat{\theta}) - \frac{1}{2} \times 3.84$ . The lower right panel of Figure 4.5 shows this region, (96, 335), which is not symmetric about the maximum likelihood estimate  $\bar{y} = 168.3$ .

We saw in Example 4.24 that the 95% confidence interval for  $\theta$  based on the asymptotic normal distribution of  $\hat{\theta}$ , (64, 273), is symmetric about  $\hat{\theta}$ . The difference between intervals based on  $W(\theta)$  and  $\hat{\theta}$  would vanish in sufficiently large samples, but it can be important to capture the asymmetry of  $\ell(\theta)$  when  $n$  is small or moderate. Regions defined by (4.37) need not be connected, unlike those based on normal approximation to the distribution of  $\hat{\theta}$ , which may be problematic when  $\ell(\theta)$  is multimodal. ■

When  $\theta$  is vector, confidence regions for  $\theta^0$  can in principle be obtained from (4.37) through contour plots of  $\ell$ . This seems infeasible when  $p$  exceeds three. We discuss one resolution of this in the next section.

### 4.5.2 Profile log likelihood

In the previous section we treated all elements of  $\theta$  equally, but in practice some are more important than others. We write  $\theta^T = (\psi^T, \lambda^T)$ , where  $\psi$  is a  $p \times 1$  vector of *parameters of interest*, and  $\lambda$  is a  $q \times 1$  vector of *nuisance parameters*. Our enquiry centres on  $\psi$ , but we cannot avoid including  $\lambda$  in the model. We may wish to check whether a particular value  $\psi^0$  of  $\psi$  is consistent with the data, or to find a plausible range of values for  $\psi$ , but in either case the value of  $\lambda$  is irrelevant or of at most secondary interest. The division into  $\psi$  and  $\lambda$  may change in the course of an investigation.

Two models are said to be *nested* if one reduces to the other when certain parameters are fixed. Thus a model with parameters  $(\psi^0, \lambda)$  is nested within the more general model with parameters  $(\psi, \lambda)$ ; the corresponding parameter spaces are  $\{\psi^0\} \times \Lambda$  and  $\Psi \times \Lambda$ , where  $\psi^0 \in \Psi$ . Under the more restrictive model the value of  $\lambda$  that maximizes the log likelihood  $\ell(\psi^0, \lambda)$  is  $\hat{\lambda}_{\psi^0}$ , whereas the overall maximum likelihood estimate,  $(\hat{\psi}, \hat{\lambda})$ , maximizes  $\ell$  over both parameters. Of course,  $\ell(\hat{\psi}, \hat{\lambda}) \geq \ell(\psi^0, \hat{\lambda}_{\psi^0})$ .

**Example 4.30 (Weibull distribution)** The Weibull density (4.4) has two parameters  $\alpha$  and  $\theta$ , and reduces to the exponential density when  $\alpha = 1$ . In terms of our general discussion we set  $\alpha = \psi$  and  $\lambda = \theta$ , with  $\psi^0 = 1$ ,  $\Lambda = \mathbb{R}_+$ , and  $\Psi = \mathbb{R}_+$ . Then the

vertical dotted line in the upper right panel of Figure 4.1 corresponds to  $\{\psi^0\} \times \Lambda$ , while the entire upper right quadrant of the plane is  $\Psi \times \Lambda$ . Evidently the likelihood reaches its maximum away from the exponential submodel. The maximum likelihood estimates under the submodel are (1, 168), while overall they are roughly (6, 181); the difference of log likelihoods is 12.5. ■

A natural statistic with which to compare two nested models is the log ratio of maximized likelihoods,

$$W_p(\psi^0) = 2\{\ell(\widehat{\psi}, \widehat{\lambda}) - \ell(\psi^0, \widehat{\lambda}_{\psi^0})\}. \quad (4.38)$$

This is sometimes called the *generalized likelihood ratio statistic* because it generalizes (4.35), but as (4.38) is the version almost invariably used in practice we shall refer to both simply as likelihood ratio statistics. At the end of this section we show that for regular models (4.36) generalizes to

$$W_p(\psi^0) \xrightarrow{D} \chi_p^2. \quad (4.39)$$

That is, even though nuisance parameters are estimated, the likelihood ratio statistic has an approximate chi-squared distribution in large samples.

Often the parameter of interest,  $\psi$ , is scalar or has much smaller dimension than the nuisance parameter,  $\lambda$ , and we wish to form a confidence region for its true value  $\psi^0$  regardless of  $\lambda$ . To do so we use the *profile log likelihood*,

$$\ell_p(\psi) = \max_{\lambda} \ell(\psi, \lambda) = \ell(\psi, \widehat{\lambda}_{\psi}),$$

where  $\widehat{\lambda}_{\psi}$  is the maximum likelihood estimate of  $\lambda$  for fixed  $\psi$ . The above result for  $W_p(\psi^0)$  implies that confidence regions for  $\psi^0$  can be based on  $\ell_p$  for regular models. A  $(1 - 2\alpha)$  confidence region for  $\psi^0$  is the set

$$\left\{ \psi : \ell_p(\psi) \geq \ell_p(\widehat{\psi}) - \frac{1}{2}c_p(1 - 2\alpha) \right\}. \quad (4.40)$$

This is our primary approach to finding confidence regions from likelihoods. It often yields good approximations to standard intervals.

When  $\psi$  is scalar we define the *signed likelihood ratio statistic*

$$Z(\psi^0) = \text{sign}(\widehat{\psi} - \psi^0)[2\{\ell(\widehat{\psi}, \widehat{\lambda}) - \ell(\psi^0, \widehat{\lambda}_{\psi^0})\}]^{1/2}.$$

This is sometimes called the *directed deviance statistic*.

The relation between the normal and chi-squared distributions implies that  $c_1(1 - 2\alpha) = z_{\alpha}^2 = z_{1-\alpha}^2$ , so

$$\begin{aligned} 1 - 2\alpha &\doteq \Pr\{W_p(\psi^0) \leq c_1(1 - 2\alpha)\} \\ &= \Pr\{Z(\psi^0) \leq z_{1-\alpha}\} - \Pr\{Z(\psi^0) \leq z_{\alpha}\}, \end{aligned}$$

and  $Z(\psi^0)$  may be regarded as having an approximate standard normal distribution and is an approximate pivot on which inference for  $\psi^0$  may be based; when  $p = 1$ , a different way of writing (4.40) is

$$\{\psi : z_{\alpha} \leq Z(\psi) \leq z_{1-\alpha}\}. \quad (4.41)$$

We have briefly mentioned the effect of reparametrization on likelihood. If  $\psi$  is of central interest, inference should be invariant to *interest-preserving transformations*, under which  $\psi, \lambda \mapsto \eta(\psi), \zeta(\psi, \lambda)$ , where the map  $\psi \rightarrow \eta$  is one-one for each value of  $\psi$ , and so too is the map  $\lambda \mapsto \zeta$ . For such a reparametrization,  $\ell_p(\eta) = \ell_p(\psi)$ , so  $W_p(\psi)$  is invariant; so too is  $Z(\psi)$  apart from a possible change in sign.

**Example 4.31 (Normal distribution)** The log likelihood for a normal sample  $y_1, \dots, y_n$  is

$$\ell(\mu, \sigma^2) \equiv -\frac{1}{2} \left\{ n \log \sigma^2 + \frac{1}{\sigma^2} \sum_{j=1}^n (y_j - \mu)^2 \right\}.$$

To use the profile log likelihood to find a confidence region for  $\mu$ , we set  $\psi = \mu$ ,  $\lambda = \sigma^2$ , and note that for fixed  $\mu$ , the maximum likelihood estimate of  $\sigma^2$  is

$$\begin{aligned} \hat{\sigma}_\mu^2 &= n^{-1} \sum (y_j - \mu)^2 \\ &= n^{-1} \left\{ \sum (y_j - \bar{y})^2 + n(\bar{y} - \mu)^2 \right\} \\ &= \frac{n-1}{n} s^2 \left\{ 1 + \frac{t(\mu)^2}{n-1} \right\}, \end{aligned}$$

where  $t(\mu) = (\bar{y} - \mu)/(s^2/n)^{1/2}$  is the observed value of the  $t$  statistic (3.16) and  $s^2 = (n-1)^{-1} \sum (y_j - \bar{y})^2$ . Thus the profile log likelihood for  $\mu$  is

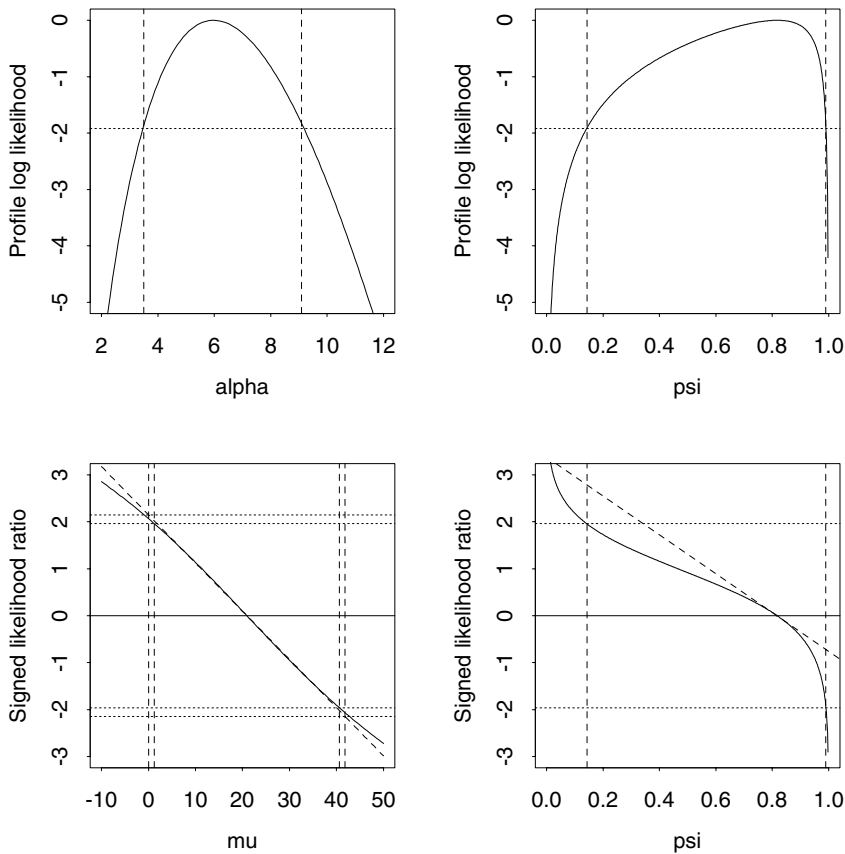
$$\ell_p(\mu) = \ell(\mu, \hat{\sigma}_\mu^2) \equiv -\frac{n}{2} \log[s^2 \{1 + t(\mu)^2/(n-1)\}],$$

and as the overall maximum likelihood estimate of  $\mu$  is  $\hat{\mu} = \bar{y}$ ,  $t(\hat{\mu}) = 0$  and

$$W_p(\mu) = n \log \left\{ 1 + \frac{T(\mu)^2}{n-1} \right\}, \quad Z(\mu) = \text{sign}(\bar{Y} - \mu) \left[ n \log \left\{ 1 + \frac{T(\mu)^2}{n-1} \right\} \right]^{1/2},$$

whose values are large when  $T(\mu) = (\bar{Y} - \mu)/(S^2/n)^{1/2}$  is large, that is, when  $\mu$  differs from  $\bar{Y}$  in either direction. Evidently the confidence interval (4.40) has the form  $T(\mu)^2 \leq c$  and may be written  $\bar{Y} \pm n^{-1/2} S c^{1/2}$ . The usual  $(1 - 2\alpha)$  confidence interval, based on the exact distribution of  $T(\mu)$ , sets  $c^{1/2}$  to be a quantile of the Student  $t$  distribution,  $t_{n-1}(1 - \alpha)$ . For  $n = 15$  and  $\alpha = 0.025$ ,  $t_{n-1}(1 - \alpha) = 2.14$ , while the value of  $c^{1/2}$  from (4.40) is 2.05. This close agreement is not surprising, as Taylor series expansion shows that  $W_p(\mu) \doteq nT(\mu)^2/(n-1)$ ,  $T(\mu)^2$  has the  $F_{1, n-1}$  distribution, and the  $F_{1, v_2}$  distribution approaches the  $\chi_1^2$  distribution when  $v_2 \rightarrow \infty$ .

The lower left panel of Figure 4.7 shows  $z(\mu) = \text{sign}(\bar{y} - \mu)w_p(\mu)^{1/2}$  for the differences between cross- and self-fertilized plant heights in Table 1.1, for which  $n = 15$ ,  $\bar{y} = 20.93$ , and  $s^2 = 1424.6$ . The function  $z(\mu)$  differs only slightly from the straight line  $t(\mu) = (\bar{y} - \mu)/(s^2/n)^{1/2}$ . The inner dotted lines at  $z_\alpha, z_{1-\alpha} = \pm 1.96$  lead to the confidence set (4.41), here (1.23, 40.63), shown by the inner vertical dashed lines. This is only slightly narrower than the exact interval (0.03, 41.84) obtained by solving  $t(\mu) = \pm t_{14}(0.025)$ ; this interval is shown by the outer dotted and dashed lines.



**Figure 4.7** Inference from likelihood ratio statistics. Top left and right: profile log likelihoods for the shape parameter of the Weibull model for the springs failure data, and for the probability,  $\psi$ , of O-ring thermal distress at 31°F for the Challenger data. The dashed vertical lines show 95% confidence intervals based on the approximate distribution of the likelihood ratio statistic, that is, the set of  $\psi$  such that  $\ell_p(\psi) \geq \ell_p(\hat{\psi}) - \frac{1}{2}c_1(0.95)$ , with the horizontal dotted line at  $-\frac{1}{2}c_1(0.95)$ . Bottom left and right: signed likelihood ratio statistics for the maize data and the Challenger data probability  $\psi$ . The solid curves are  $Z(\mu)$  and  $Z(\psi)$ , and the dotted horizontal lines are at  $z_\alpha, z_{1-\alpha} = \pm 1.96$ ; the dashed vertical lines show 95% confidence intervals. The dashed diagonal line in the right panel shows  $(0.816 - \psi)/0.242$  and corresponds to using approximate normality of  $\hat{\psi}$  to set a confidence interval. The dashed diagonal line in the left panel shows the Student  $t$  quantity  $t(\mu)$ , with the outer dotted lines showing  $\pm t_{14}(0.025)$ , from which the  $t$  confidence interval shown by the outer dashed lines is read off.

In practice the exact interval would be used, but such results build confidence in use of (4.40) and (4.41) when there is no exact interval. ■

**Example 4.32 (Weibull distribution)** For the data in Example 4.4, we saw that the difference of maximized likelihoods for the Weibull and exponential models is roughly 12.5, and so  $W_p(\alpha^0) = 2\{\ell(\hat{\theta}, \hat{\alpha}) - \ell(\hat{\theta}_{\alpha^0}, \alpha^0)\} \doteq 25$ . If  $\alpha^0 = 1$  was the true value for  $\alpha$ , (4.39) implies that the distribution of  $W_p(\alpha^0)$  would be approximately  $\chi_1^2$ . However the 0.95 and 0.99 quantiles of this distribution are respectively  $c_1(0.95) = 3.84$  and  $c_1(0.99) = 6.635$ , and a value as large as 25 is very unlikely to arise by chance. Thus the Weibull model fits the data appreciably better than the exponential one.

A 95% confidence region for the true value of  $\alpha$  based on the profile log likelihood is the set of  $\alpha$  such that  $\ell_p(\alpha) \geq \ell_p(\hat{\alpha}) - \frac{1}{2} \times 3.84$ ; we read this off from the top left panel of Figure 4.7 and obtain (3.5, 9.2). As we would expect, this interval does not contain  $\alpha = 1$ . ■

**Example 4.33 (Challenger data)** Examples 4.5, 4.8, and 4.27 concern likelihood analysis of a binomial model for the data in Table 1.3. Our model is that at temperature  $x_1$  and pressure  $x_2$ , the number of O-rings suffering thermal distress is binomial with

denominator  $m = 6$  and probability

$$\pi(\beta_0, \beta_1, \beta_2) = \frac{\exp(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}{1 + \exp(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}.$$

Apart from a constant, the corresponding log likelihood is

$$\beta_0 \sum_{j=1}^n r_j + \beta_1 \sum_{j=1}^n r_j x_{1j} + \beta_2 \sum_{j=1}^n r_j x_{2j} - m \sum_{j=1}^n \log\{1 + \exp(\beta_0 + \beta_1 x_{1j} + \beta_2 x_{2j})\}.$$

We maximize this first as it is, then with  $\beta_2$  held equal to zero, and then with both  $\beta_1$  and  $\beta_2$  held equal to zero, and obtain  $-15.05$ ,  $-15.82$  and  $-18.90$ . To check whether there is a pressure effect when temperature is included, we calculate the corresponding likelihood ratio statistic,  $2 \times \{-15.05 - (-15.82)\} = 1.54$ . This is smaller than the 0.95 quantile of the  $\chi_1^2$  distribution,  $c_1(0.95) = 3.84$ , so any pressure effect is slight. Assuming no pressure effect, the likelihood ratio statistic for no temperature effect is  $2 \times \{-15.82 - (-18.90)\} = 6.16$ , which we again compare to the  $\chi_1^2$  distribution. But  $\Pr(\chi_1^2 \geq 6.16) = 0.013$ , so 6.16 is unlikely to occur by chance if the true value of  $\beta_1$  is zero: there seems to be a temperature effect.

The focus in this problem is the probability of thermal distress at temperature  $x_1 = 31^\circ\text{F}$ , and if there is an effect of temperature but not of pressure this probability is  $\psi = \pi(\beta_0, \beta_1, 0)$ , for which we would like confidence intervals. In Example 4.28 we saw how to apply the delta method to  $\hat{\psi}$ , but it gave the unsatisfactory 95% confidence interval  $(0.34, 1.29)$ .

The upper right panel of Figure 4.7 shows the profile log likelihood  $\ell_p(\psi)$ . A 95% confidence interval based on this is  $(0.14, 0.99)$ ; unlike intervals based on normal approximation to  $\hat{\psi}$ , this is guaranteed to be a subset of  $(0, 1)$ . The panel below shows the signed likelihood ratio statistic, which is far from a straight line because the profile log likelihood is far from quadratic in  $\psi$ . The dashed diagonal line shows how the interval based on the normal distribution of  $\hat{\psi}$  contains values outside the interval  $[0, 1]$ ; an interval symmetric about  $\hat{\psi}$  is wholly inappropriate. ■

In both the preceding examples the profile log likelihood is asymmetric. Particularly in the second example, the profile log likelihood or equivalently  $W_p(\psi)$  or  $Z(\psi)$ , provide better confidence intervals than normal approximation to the distribution of the maximum likelihood estimate.

### 4.5.3 Model fit

So far we have supposed that the model is known apart from parameter values, but this is rarely the case in practice and it is essential to check model fit. Graphs play an important role in this, with variants of probability plots (Section 2.1.4) particularly useful. A more formal approach is to nest the model in a larger one, and then to assess whether the expanded model fits the data appreciably better. If its log likelihood is  $\ell(\psi, \lambda)$  and the original model restricts  $\psi$  to  $\psi_0$ , the two may be compared using a likelihood ratio statistic. The usefulness of this approach depends on the expanded

model: if it is uninteresting, so too will be the comparison. We have already seen an application of this in Example 4.33.

**Example 4.34 (Generalized gamma distribution)** A random variable  $Y$  with the generalized gamma distribution has density function

$$f(y; \lambda, \alpha, \kappa) = \frac{\alpha \lambda^\kappa y^{\alpha\kappa-1}}{\Gamma(\kappa)} \exp(-\lambda y^\alpha), \quad y > 0, \quad \lambda, \alpha, \kappa > 0. \quad (4.42)$$

This arises on supposing that for some  $\alpha$ ,  $Y^\alpha$  has a gamma distribution, and reduces to the gamma density (2.7) when  $\alpha = 1$ , to the Weibull density (4.4) with  $\theta = \lambda^{-1/\alpha}$  when  $\kappa = 1$ , and to the exponential density when  $\alpha = \kappa = 1$ ; it is a flexible generalization of these models. In terms of our general discussion  $\psi = \alpha$ , with  $\psi^0 = 1$ , and  $\lambda = (\kappa, \lambda)^\top$ .

When applied to the data in Table 2.1, the maximized log likelihoods are  $-250.65$  for the generalized gamma model,  $-251.12$  for the gamma model, and  $-251.17$  for the Weibull model. The likelihood ratio statistic for comparison of the gamma and generalized gamma densities is  $2 \times \{-250.65 - (-251.12)\} = 0.94$ , to be treated as  $\chi_1^2$ . There is no evidence that (4.42) fits better than the gamma density, which fits about equally as well as the Weibull density. ■

One useful approach in this context is a *score test*. Suppose that  $\psi$  and  $\lambda$  have dimensions  $p \times 1$  and  $q \times 1$ , and let  $I_{\lambda\psi} = E(-\partial^2 \ell / \partial \lambda \partial \psi^\top)$ , and so forth. The idea is that if the restricted model is adequate, then the maximized log likelihood  $\ell(\psi^0, \hat{\lambda}_{\psi^0})$  will not increase sharply in the  $\psi$ -direction, so its gradient  $\partial \ell(\psi, \lambda) / \partial \psi$  evaluated at  $(\psi^0, \hat{\lambda}_{\psi^0})$  should be modest. We show at the end of this section that

$$\frac{\partial \ell(\psi^0, \hat{\lambda}_{\psi^0})}{\partial \psi} \sim N_p(0, I_{\psi\psi} - I_{\psi\lambda} I_{\lambda\lambda}^{-1} I_{\lambda\psi}),$$

implying that if the simpler model is adequate, then

$$S = \frac{\partial \ell(\psi^0, \hat{\lambda}_{\psi^0})}{\partial \psi^\top} (I_{\psi\psi} - I_{\psi\lambda} I_{\lambda\lambda}^{-1} I_{\lambda\psi})^{-1} \frac{\partial \ell(\psi^0, \hat{\lambda}_{\psi^0})}{\partial \psi} \sim \chi_p^2, \quad (4.43)$$

where  $S$  is evaluated at  $(\psi^0, \hat{\lambda}_{\psi^0})$ . When  $p = 1$  the signed square root of  $S$  should have an approximate standard normal distribution. The statistic  $S$  is asymptotically equivalent to the likelihood ratio statistic  $W_p(\psi^0)$ , but is more convenient because it involves maximization only under the simpler model. Expected information quantities may be replaced by observed information quantities.

**Example 4.35 (Spring failure data)** We illustrate the score test by checking whether  $\alpha = 1$  for the spring failure data. In terms of our general discussion,  $\psi = \alpha$ , with  $\psi^0 = 1$ , and  $\lambda = \theta$ . The score and observed information are given in Example 4.23. When  $\alpha = 1$ ,  $\hat{\theta} = \bar{y} = 168.3$ . At  $(\hat{\theta}, 1)$ , we have  $\partial \ell(\theta, \alpha) / \partial \alpha = 9.64$  and  $(J_{\alpha\alpha} - J_{\alpha\theta} J_{\theta\theta}^{-1} J_{\theta\alpha})^{-1} = 0.097$ , so  $S$  takes value 8.99. Compared to the  $\chi_1^2$  distribution this gives strong evidence that  $\alpha \neq 1$ . ■

*Chi-squared statistics*

Sometimes it is useful to assess fit without a specific alternative in mind. One approach is to group the data and to use a *chi-squared statistic*.

Suppose we have  $n$  independent observations that fall into categories  $1, \dots, k$ , with  $Y_i$  denoting the number of observations in category  $i$ . The probability that a single observation falls into this category is  $\pi_i$ , where  $0 < \pi_i < 1$  and  $\sum_{i=1}^k \pi_i = 1$ , but as  $\pi_k = 1 - \pi_1 - \dots - \pi_{k-1}$ , the parameter space is the interior of a simplex in  $k$  dimensions, that is, the set

$$\left\{ (\pi_1, \dots, \pi_k) : \sum_{i=1}^k \pi_i = 1, 0 < \pi_1, \dots, \pi_k < 1 \right\} \quad (4.44)$$

of dimension  $k - 1$ . The model whose fit we wish to assess is that category  $i$  has probability  $\pi_i(\lambda)$ , where  $\sum_i \pi_i(\lambda) = 1$  for each  $\lambda$  and the parameter  $\lambda$  has dimension  $p$ . This is multinomial with probabilities  $\pi_1, \dots, \pi_k$  and denominator  $n$ ; see Example 2.36. We suppose that there is a 1–1 mapping between  $\pi = (\pi_1, \dots, \pi_{k-1})^T$  and  $(\psi, \lambda)$ , and that setting  $\psi = \psi_0$  corresponds to the restricted model  $\pi(\lambda) = (\pi_1(\lambda), \dots, \pi_{k-1}(\lambda))^T$ . Thus our model of interest restricts  $\pi$  to a  $p$ -dimensional subset of (4.44), where  $p < k - 1$ , and is nested within the full multinomial model with  $k - 1$  parameters.

Given data  $y_1, \dots, y_k$ , the likelihood under the general model is

$$L(\pi) = \frac{n!}{y_1! \cdots y_k!} \pi_1^{y_1} \times \cdots \times \pi_k^{y_k}, \quad \sum_{i=1}^k \pi_i = 1, 0 < \pi_1, \dots, \pi_k < 1,$$

where  $\sum_i y_i = n$ , so the log likelihood is

$$\ell(\pi) \equiv \sum_{i=1}^{k-1} y_i \log \pi_i + y_k \log(1 - \pi_1 - \cdots - \pi_{k-1}), \quad (4.45)$$

resulting in score vector and observed information matrix with components

$$\begin{aligned} \frac{\partial \ell(\pi)}{\partial \pi_i} &= \frac{y_i}{\pi_i} - \frac{y_k}{1 - \pi_1 - \cdots - \pi_{k-1}}, \\ -\frac{\partial^2 \ell(\pi)}{\partial \pi_i \partial \pi_j} &= \begin{cases} \frac{y_i}{\pi_i^2} + \frac{y_k}{(1 - \pi_1 - \cdots - \pi_{k-1})^2}, & i = j, \\ \frac{y_k}{(1 - \pi_1 - \cdots - \pi_{k-1})^2}, & i \neq j, \end{cases} \end{aligned} \quad (4.46)$$

where  $i$  and  $j$  run over  $1, \dots, k - 1$ . Manipulation of the likelihood equations shows that the maximum likelihood estimators are  $\hat{\pi}_i = Y_i/n$  (Exercise 4.5.4). The expected information matrix involves  $E(Y_i)$ , which may be calculated by noting that if we regard an observation in category  $i$  as a ‘success’,  $Y_i$  is the number of successes out of  $n$  independent trials, so its marginal distribution is binomial with denominator  $n$  and probability  $\pi_i$  and mean  $n\pi_i$ ; see Example 2.36. The expected information is the

$(k-1) \times (k-1)$  matrix

$$I(\pi) = n \begin{pmatrix} 1/\pi_1 + 1/\pi_k & 1/\pi_k & \cdots & 1/\pi_k \\ 1/\pi_k & 1/\pi_2 + 1/\pi_k & \cdots & 1/\pi_k \\ \vdots & \vdots & \ddots & \vdots \\ 1/\pi_k & 1/\pi_k & \cdots & 1/\pi_{k-1} + 1/\pi_k \end{pmatrix}, \quad (4.47)$$

and it is straightforward to verify that its inverse is

$$I(\pi)^{-1} = n^{-1} \begin{pmatrix} \pi_1(1-\pi_1) & -\pi_1\pi_2 & \cdots & -\pi_1\pi_{k-1} \\ -\pi_2\pi_1 & \pi_2(1-\pi_2) & \cdots & -\pi_2\pi_{k-1} \\ \vdots & \vdots & \ddots & \vdots \\ -\pi_{k-1}\pi_1 & -\pi_{k-1}\pi_2 & \cdots & \pi_{k-1}(1-\pi_{k-1}) \end{pmatrix};$$

this is unsurprising, because  $\hat{\pi}_i = Y_i/n$ . Provided none of the  $\pi_i$  equals zero or one, the usual large-sample properties of maximum likelihood estimates are satisfied as  $n \rightarrow \infty$ , and in particular  $\hat{\pi}$  has a limiting normal distribution.

We now return to the restricted model, whose log likelihood is

$$\ell(\lambda) = \ell\{\pi(\lambda)\} \equiv \sum_{i=1}^{k-1} y_i \log \pi_i(\lambda) + y_k \log \{1 - \pi_1(\lambda) - \cdots - \pi_{k-1}(\lambda)\},$$

maximization of which gives the maximum likelihood estimator  $\hat{\lambda}$ . The first and second derivatives of  $\ell(\lambda)$  are

$$\begin{aligned} \frac{\partial \ell(\lambda)}{\partial \lambda_r} &= \sum_{i=1}^{k-1} \frac{\partial \pi_i}{\partial \lambda_r} \frac{\partial \ell(\pi)}{\partial \pi_i}, \\ \frac{\partial^2 \ell(\lambda)}{\partial \lambda_r \partial \lambda_s} &= \sum_{i=1}^{k-1} \frac{\partial^2 \pi_i}{\partial \lambda_r \partial \lambda_s} \frac{\partial \ell(\pi)}{\partial \pi_i} + \sum_{i=1}^{k-1} \sum_{j=1}^{k-1} \frac{\partial \pi_i}{\partial \lambda_r} \frac{\partial \pi_j}{\partial \lambda_s} \frac{\partial^2 \ell(\pi)}{\partial \pi_i \partial \pi_j}, \end{aligned}$$

and as  $E\{\partial \ell(\pi)/\partial \pi_i\} = 0$ , the expected information for  $\lambda$  is the  $p \times p$  matrix

$$I(\lambda) = \frac{\partial \pi^\top}{\partial \lambda} E \left\{ -\frac{\partial^2 \ell(\pi)}{\partial \pi \partial \pi^\top} \right\} \frac{\partial \pi}{\partial \lambda^\top} = \frac{\partial \pi^\top}{\partial \lambda} I(\pi) \frac{\partial \pi}{\partial \lambda^\top},$$

where  $\partial \pi^\top / \partial \lambda$  is the  $p \times (k-1)$  matrix of partial derivatives of the  $\pi_i$  with respect to the  $\lambda_r$ , and  $I(\pi)$  is given by (4.47); see Problem 4.2. Thus provided  $\partial \pi^\top / \partial \lambda \neq 0$ , the parameter  $\lambda$  has a large-sample normal distribution under the restricted model, and the general results in Section 4.5.2 imply that the likelihood ratio statistic used to compare the two models satisfies

$$W = 2 \sum_{i=1}^k y_i \log \left\{ \frac{\hat{\pi}_i}{\pi_i(\hat{\lambda})} \right\} = 2 \sum_{i=1}^k y_i \log \left\{ \frac{y_i}{n\pi_i(\hat{\lambda})} \right\} \sim \chi_{k-1-p}^2$$

if the simpler model is true. We may write  $W = 2 \sum O_i \log(O_i/E_i)$ , where  $O_i = y_i$  and  $E_i = n\pi_i(\hat{\lambda})$  are the  $i$ th observed and expected values under the fitted model;

We take  $0 \log 0 = \lim_{y \downarrow 0} y \log y = 0$ .

Karl Pearson (1857–1936) was a leader of the English biometrical school, which applied statistical ideas to heredity and evolution. His energy was astonishing: he practised law and wrote books on history and religion as well as the classic ‘The Grammar of Science’ and over 500 other publications. He coined the terms ‘standard deviation’, ‘histogram’ and ‘mode’. He invented the correlation coefficient and also the chi-square test. He feuded with Fisher, who pointed out that Pearson gave  $P$  too many degrees of freedom. The statistic  $P$  is sometimes denoted  $X^2$  or  $\chi^2$ .

as  $\sum \pi_i(\hat{\lambda}) = 1$ , it is true that  $\sum E_i = \sum O_i = n$ . Taylor series expansion shows that  $W \doteq \sum (O_i - E_i)^2 / E_i$  (Exercise 4.5.5), leading to *Pearson’s statistic*,

$$P = \sum_{i=1}^k \frac{\{y_i - n\pi_i(\hat{\lambda})\}^2}{n\pi_i(\hat{\lambda})};$$

this too has an approximate  $\chi_{k-1-p}^2$  distribution if the simpler model is true.

Both  $W$  and  $P$  provide checks on the adequacy of the restricted multinomial compared to the most general multinomial possible, which requires only that the probabilities sum to one. The approximate distributions of  $W$  and  $P$  apply when there are large counts, and experience suggests that the chi-squared approximations are more accurate if most of the fitted values exceed five. Though asymptotically equivalent to  $W$ ,  $P$  behaves better in small samples because it does not involve logarithms.

**Example 4.36 (Birth data)** Figure 2.2 shows the Poisson density with mean  $\hat{\theta} = 12.9$  fitted to the numbers of daily arrivals for the delivery suite data. How good is the fit? Here  $p = 1$  parameters are estimated under the Poisson model. With the  $n = 92$  daily counts split among the  $k = 13$  categories  $[0, 7.5)$ ,  $[7.5, 8.5)$ ,  $\dots$ ,  $[18.5, \infty)$ , the values for  $O$  and  $E$  are

|   |      |      |      |      |      |       |       |      |      |      |      |      |      |
|---|------|------|------|------|------|-------|-------|------|------|------|------|------|------|
| O | 6    | 3    | 3    | 8    | 13   | 10    | 11    | 11   | 8    | 6    | 4    | 4    | 5    |
| E | 5.23 | 4.37 | 6.26 | 8.08 | 9.48 | 10.19 | 10.11 | 9.32 | 8.01 | 6.46 | 4.91 | 3.52 | 6.07 |

and  $P$  takes value 4.39, to be treated as a  $\chi_{11}^2$  variable. As  $\Pr(\chi_{11}^2 \geq 4.39) \doteq 0.96$ , the Poisson model fits very well, perhaps surprisingly so.

A minor problem here is that  $\hat{\theta}$  is obtained from the original data rather than from the data grouped into the  $k$  categories. However the maximum likelihood estimate from the grouped data is 12.89, so the fit is hardly affected at all. Use of the parameter estimate from the ungrouped data increases the degrees of freedom for the test, because slightly fewer than  $p$  degrees of freedom must be subtracted from the  $k - 1$ . The estimates will usually be similar unless the grouping is very coarse. ■

**Example 4.37 (Two-way contingency table)** Suppose that each of  $n$  individuals chosen at random from a population is classified according to two sets of categories. The first corresponds to the  $r$  rows of the table, and the second to the  $c$  columns; there are  $k = rc$  cells indexed by  $(i, j)$ ,  $i = 1, \dots, r$ ,  $j = 1, \dots, c$ . Such a setup is known as an  $r \times c$  *contingency table* or *two-way contingency table*. The top part of Table 4.3 shows an example in which 422 people have been cross-classified according to presence or absence of the antigens ‘A’ and ‘B’ in their blood. There are 202 people without either antigen, 179 with antigen ‘A’ but not ‘B’, and so forth. This is the simplest cross-classification, a  $2 \times 2$  *table*.

Suppose that there are  $y_{ij}$  individuals in the  $(i, j)$  cell, so  $\sum_{i,j} y_{ij} = n$ . If the individuals are independently chosen at random from a population in which the proportion in cell  $(i, j)$  is  $\pi_{ij}$ , the joint density of the cell counts  $Y_{ij}$  is multinomial with

|             |         | Antigen 'B' |         | Total |
|-------------|---------|-------------|---------|-------|
|             |         | Absent      | Present |       |
| Antigen 'A' | Absent  | 'O': 202    | 'B': 35 | 237   |
|             | Present | 'A': 179    | 'AB': 6 | 185   |
| Total       |         | 381         | 41      | 422   |

**Table 4.3** Blood groups in England (Taylor and Prior, 1938). The upper part of the table shows a cross-classification of 422 persons by presence or absence of antigens 'A' and 'B', giving the groups 'A', 'B', 'AB', 'O' of the human blood group system. The lower part shows genotypes and corresponding probabilities under one- and two-locus models. See Example 4.38 for details.

| Group | Two-locus model                           |                           | One-locus model |                                     |
|-------|-------------------------------------------|---------------------------|-----------------|-------------------------------------|
|       | Genotype                                  | Probability               | Genotype        | Probability                         |
| 'A'   | (AA; bb), (Aa; bb)                        | $\alpha(1 - \beta)$       | (AA), (AO)      | $\lambda_A^2 + 2\lambda_A\lambda_O$ |
| 'B'   | (aa; BB), (aa; Bb)                        | $(1 - \alpha)\beta$       | (BB), (BO)      | $\lambda_B^2 + 2\lambda_B\lambda_O$ |
| 'AB'  | (AA; BB), (Aa; BB),<br>(AA; Bb), (Aa; Bb) | $\alpha\beta$             | (AB)            | $2\lambda_A\lambda_B$               |
| 'O'   | (aa; bb)                                  | $(1 - \alpha)(1 - \beta)$ | (OO)            | $\lambda_O^2$                       |

denominator  $n$  and probabilities  $\pi_{ij}$ , that is,

$$\frac{n!}{y_{11}!y_{12}!\cdots y_{rc}!} \pi_{11}^{y_{11}} \pi_{12}^{y_{12}} \cdots \pi_{rc}^{y_{rc}}, \quad y_{ij} = 0, \dots, n, \quad \sum_{i,j} y_{ij} = n,$$

where  $0 < \pi_{ij} < 1$  and  $\sum_{i,j} \pi_{ij} = 1$ . The log likelihood is

$$\ell(\pi) \equiv \sum_{i,j} y_{ij} \log \pi_{ij}, \quad 0 < \pi_{ij} < 1, \quad \sum_{i,j} \pi_{ij} = 1;$$

there are  $rc - 1$  parameters because of the constraint that the probabilities sum to one. The preceding general results imply that estimated proportion of the population in cell  $(i, j)$  is the sample proportion in that cell, that is,  $\hat{\pi}_{ij} = y_{ij}/n$ , so the maximized log likelihood is  $\sum_{i,j} y_{ij} \log(y_{ij}/n)$ .

Often the question arises whether the row and column classifications are independent. If so, and if the proportion of the population in row category  $i$  is  $\alpha_i$ , and that in column category  $j$  is  $\beta_j$ , then  $\pi_{ij} = \alpha_i\beta_j$ . As  $\sum_i \alpha_i = \sum_j \beta_j = 1$ , this model has  $p = (r - 1) + (c - 1)$  parameters. The log likelihood is  $\sum_{i,j} y_{ij} \log(\alpha_i\beta_j)$ , and to maximize it subject to the constraints on the  $\alpha_i$  and  $\beta_j$  we use Lagrange multipliers  $\zeta$  and  $\eta$  and seek extremal points of

$$\ell^*(\alpha, \beta, \zeta, \eta) = \sum_{i,j} y_{ij} \log(\alpha_i\beta_j) + \zeta \left( \sum_i \alpha_i - 1 \right) + \eta \left( \sum_j \beta_j - 1 \right).$$

We find that  $\hat{\alpha}_i = y_{i\cdot}/n$  and  $\hat{\beta}_j = y_{\cdot j}/n$ , where  $y_{i\cdot} = \sum_j y_{ij}$  and  $y_{\cdot j} = \sum_i y_{ij}$ ; these are respectively the observed proportions of observations in the  $i$ th row and  $j$ th column categories. The fitted value in cell  $(i, j)$  is  $n\hat{\alpha}_i\hat{\beta}_j = y_{i\cdot}y_{\cdot j}/n$ , and the maximized log likelihood is  $\sum_{i,j} y_{ij} \log(\hat{\alpha}_i\hat{\beta}_j)$ .

The likelihood ratio statistic for comparing the independence model with the more general model is

$$W = 2 \sum_{i,j} \left\{ y_{ij} \log \left( \frac{y_{ij}}{n} \right) - y_{ij} \log \left( \frac{y_{i \cdot} y_{\cdot j}}{n^2} \right) \right\} = 2 \sum_{i,j} y_{ij} \log \left( \frac{ny_{ij}}{y_{i \cdot} y_{\cdot j}} \right),$$

and when the independence model is true, the approximate distribution of  $W$  is  $\chi^2_{k-1-p}$ ; here  $k - 1 - p = rc - 1 - \{(r - 1) + (c - 1)\} = (r - 1)(c - 1)$ .

In this case Pearson's statistic may be expressed as

$$P = \sum_{i,j} \frac{(y_{ij} - y_{i \cdot} y_{\cdot j} / n)^2}{y_{i \cdot} y_{\cdot j} / n},$$

with an approximate  $\chi^2_{(r-1)(c-1)}$  distribution when the categorizations are independent. ■

**Example 4.38 (ABO blood group system)** The most important classification of human blood types is into the four groups 'A', 'B', 'AB', and 'O', corresponding to presence or absence of the antigens 'A' and 'B'; 'AB' refers to the presence of both and 'O' to their absence. In a set of data shown in Table 4.3, the frequencies of these groups were 179, 35, 6, and 202.

According to a model thought credible until the 1920s, the blood group of a person is controlled by two loci (1; 2) on a pair of chromosomes, one chromosome being inherited from each parent. At the loci they independently inherit alleles  $(x_1; y_1)$  from their mother and  $(x_2; y_2)$  from their father, where  $x_1$  and  $x_2$  are one of  $a$  or  $A$ , and  $y_1$  and  $y_2$  are one of  $b$  or  $B$ . Thus their genotype  $(x_1 x_2; y_1 y_2)$  is any one of  $(aa; bb), \dots, (AA, BB)$ , and they have the antigen 'A' only if allele  $A$  is present; similarly for antigen 'B'. In fact  $(Aa; Bb)$  is indistinguishable from  $(aA; bB)$  and so forth, so under this model there are nine genotypes shown in the second column of the lower part of Table 4.3. Since the loci are independent, the probabilities that a person randomly taken from the population will have blood groups 'A', 'B', 'AB' and 'O' may be written as  $\alpha(1 - \beta)$ ,  $(1 - \alpha)\beta$ ,  $\alpha\beta$ , and  $(1 - \alpha)(1 - \beta)$ , where  $\alpha$  and  $\beta$  are the probabilities that they have antigens 'A' and 'B'.

An alternative model posits a single locus at which three alleles,  $A$ ,  $B$ , and  $O$  may appear,  $A$  and  $B$  conferring the respective antigens, and  $O$  conferring nothing. If  $\lambda_A$ ,  $\lambda_B$  and  $\lambda_O$  denote the probabilities that a parent has the three alleles on one chromosome, and if the population is in equilibrium, then the probabilities that the child has blood types 'A', 'B', 'AB' and 'O' are

$$\pi_A = \lambda_A^2 + 2\lambda_A\lambda_O, \quad \pi_B = \lambda_B^2 + 2\lambda_B\lambda_O, \quad \pi_{AB} = 2\lambda_A\lambda_B, \quad \pi_O = \lambda_O^2.$$

where  $\lambda_O = 1 - \lambda_A - \lambda_B$ .

Under the two-locus model, Example 4.37 implies that the maximum likelihood estimates of  $\alpha$  and  $\beta$  are the corresponding sample proportions,  $\hat{\alpha} = 185/422 = 0.438$  and  $\hat{\beta} = 41/422 = 0.097$ . The fitted values, 213.97, 167.03, 23.02, 17.97, are rather far from 202, 179, 35, 6. The values for  $W$  and  $P$  are 17.66 and 15.73, to be treated as  $\chi^2_{k-1-p}$  if the two-locus model is adequate; here  $k - 1 - p = 4 - 1 - 2 = 1$ . As  $c_1(0.95) = 3.84$ , the fit is poor.

Under the single-locus model, the log likelihood is

$$179 \log(\lambda_A^2 + 2\lambda_A\lambda_O) + 35 \log(\lambda_B^2 + 2\lambda_B\lambda_O) + 6 \log(2\lambda_A\lambda_B) + 202 \log(\lambda_O^2),$$

where  $\lambda_O = 1 - \lambda_A - \lambda_B$ , and maximization in terms of  $(\log \lambda_A, \log \lambda_B)$  gives  $\hat{\lambda}_A = 0.252$ ,  $\hat{\lambda}_B = 0.050$ . The fitted values for the blood groups are 205.85, 174.99, 30.54, and 10.62, and the values of  $W$  and  $P$  are 3.17 and 2.82. The single-locus model is much better supported by the data. ■

#### Derivations of (4.39) and (4.43)

In the regular case when the model is correct and the true values of the  $p \times 1$  and  $q \times 1$  vectors  $\psi$  and  $\lambda$  are  $\psi^0$  and  $\lambda^0$ , we denote the score vector and observed and expected information matrices by

$$U(\psi^0, \lambda^0) = \begin{pmatrix} U_\psi \\ U_\lambda \end{pmatrix}, \quad J(\psi^0, \lambda^0) = \begin{pmatrix} J_{\psi\psi} & J_{\psi\lambda} \\ J_{\lambda\psi} & J_{\lambda\lambda} \end{pmatrix}, \quad I(\psi^0, \lambda^0) = \begin{pmatrix} I_{\psi\psi} & I_{\psi\lambda} \\ I_{\lambda\psi} & I_{\lambda\lambda} \end{pmatrix},$$

where, for example,  $U_\lambda$  is the  $q \times 1$  vector  $\partial \ell / \partial \lambda$ ,  $J_{\lambda\psi}$  is the  $q \times p$  matrix  $-\partial^2 \ell / \partial \lambda \partial \psi^T$ , and  $I_{\lambda\psi} = E(-\partial^2 \ell / \partial \lambda \partial \psi^T)$ , evaluated at  $(\psi^0, \lambda^0)$ . The components of  $U$  are  $O_p(n^{1/2})$ , those of  $J$  are  $O_p(n)$ , and those of  $I$  are  $O(n)$ .

To establish (4.43), we expand the likelihood equations  $U(\hat{\psi}, \hat{\lambda}) = 0$  and  $\partial \ell(\psi^0, \hat{\lambda}_{\psi^0}) / \partial \lambda = 0$  about  $(\psi^0, \lambda^0)$ , giving

$$\begin{aligned} \begin{pmatrix} U_\psi \\ U_\lambda \end{pmatrix} &= J(\psi^0, \lambda^0) \begin{pmatrix} \hat{\psi} - \psi^0 \\ \hat{\lambda} - \lambda^0 \end{pmatrix} + o_p(n^{1/2}) \\ &= I(\psi^0, \lambda^0) \begin{pmatrix} \hat{\psi} - \psi^0 \\ \hat{\lambda} - \lambda^0 \end{pmatrix} + o_p(n^{1/2}), \\ U_\lambda &= J_{\lambda\lambda}(\hat{\lambda}_{\psi^0} - \lambda^0) + o_p(n^{1/2}) = I_{\lambda\lambda}(\hat{\lambda}_{\psi^0} - \lambda^0) + o_p(n^{1/2}). \end{aligned}$$

Thus

$$\hat{\lambda}_{\psi^0} - \lambda^0 = I_{\lambda\lambda}^{-1} U_\lambda + o_p(n^{-1/2}) = \hat{\lambda} - \lambda^0 + I_{\lambda\lambda}^{-1} I_{\lambda\psi}(\hat{\psi} - \psi^0) + o_p(n^{-1/2}).$$

Taylor series expansion gives

$$\frac{\partial \ell(\psi^0, \hat{\lambda}_{\psi^0})}{\partial \psi} = U_\psi - I_{\psi\lambda}(\hat{\lambda}_{\psi^0} - \lambda^0) + o_p(n^{-1/2}) = U_\psi - I_{\psi\lambda} I_{\lambda\lambda}^{-1} U_\lambda + o_p(n^{-1/2}),$$

and the joint limiting normal distribution

$$\begin{pmatrix} U_\psi \\ U_\lambda \end{pmatrix} \sim N_{p+q}\{0, I(\psi^0, \lambda^0)\}$$

implies that

$$\frac{\partial \ell(\psi^0, \hat{\lambda}_{\psi^0})}{\partial \psi} \sim N_p(0, I_{\psi\psi} - I_{\psi\lambda} I_{\lambda\lambda}^{-1} I_{\lambda\psi}), \quad (4.48)$$

so

$$\frac{\partial \ell(\psi^0, \hat{\lambda}_{\psi^0})}{\partial \psi^T} (I_{\psi\psi} - I_{\psi\lambda} I_{\lambda\lambda}^{-1} I_{\lambda\psi})^{-1} \frac{\partial \ell(\psi^0, \hat{\lambda}_{\psi^0})}{\partial \psi} \sim \chi_p^2.$$

This may be skipped at a first reading.

To establish (4.39), we write the likelihood ratio statistic (4.38) as

$$W_p(\psi^0) = 2\{\ell(\hat{\psi}, \hat{\lambda}) - \ell(\psi^0, \lambda^0)\} - 2\{\ell(\psi^0, \hat{\lambda}_{\psi^0}) - \ell(\psi^0, \lambda^0)\},$$

and then replace  $\ell(\hat{\psi}, \hat{\lambda})$  and  $\ell(\psi^0, \hat{\lambda}_{\psi^0})$  with second-order Taylor series expansions about  $(\psi^0, \lambda^0)$ . The results above imply that  $W_p(\psi^0)$  is approximately

$$\begin{pmatrix} \hat{\psi} - \psi^0 \\ \hat{\lambda} - \lambda^0 \end{pmatrix}^T I(\psi^0, \lambda^0) \begin{pmatrix} \hat{\psi} - \psi^0 \\ \hat{\lambda} - \lambda^0 \end{pmatrix} - \begin{pmatrix} 0 \\ \hat{\lambda}_{\psi^0} - \lambda^0 \end{pmatrix}^T I(\psi^0, \lambda^0) \begin{pmatrix} 0 \\ \hat{\lambda}_{\psi^0} - \lambda^0 \end{pmatrix},$$

and replacement of  $\hat{\lambda}_{\psi^0}$  with its expression in terms of  $(\hat{\psi}, \hat{\lambda})$  gives

$$W_p(\psi^0) \doteq (\hat{\psi} - \psi^0)^T (I_{\psi\psi} - I_{\psi\lambda} I_{\lambda\lambda}^{-1} I_{\lambda\psi}) (\hat{\psi} - \psi^0) + o_p(1). \quad (4.49)$$

But as our previous asymptotics for the maximum likelihood estimators under the full model give

$$\begin{pmatrix} \hat{\psi} \\ \hat{\lambda} \end{pmatrix} \sim N_{p+q} \left\{ \begin{pmatrix} \psi^0 \\ \lambda^0 \end{pmatrix}, I(\psi^0, \lambda^0)^{-1} \right\}, \quad (4.50)$$

the asymptotic covariance matrix of  $\hat{\psi}$  is  $(I_{\psi\psi} - I_{\psi\lambda} I_{\lambda\lambda}^{-1} I_{\lambda\psi})^{-1}$ , and (4.49) and (3.23) give

$$W_p(\psi^0) = 2\{\ell(\hat{\psi}, \hat{\lambda}) - \ell(\psi^0, \hat{\lambda}_{\psi^0})\} \sim \chi_p^2:$$

the asymptotic distribution of the likelihood ratio statistic for comparison of two nested models is chi-squared with degrees of freedom equal to the number of parameters that are restricted by the less general model. This result applies only to nested models, and the expansions leading to it are valid only when  $(\hat{\lambda}, \hat{\psi})$  converges to  $(\psi^0, \lambda^0)$ . This may need checking in applications.

## Exercises 4.5

- 1 If  $Y_1, \dots, Y_n$  is a random sample from the  $N(\mu, \sigma^2)$  distribution with known  $\sigma^2$ , show that the likelihood ratio statistic for comparing  $\mu = \mu^0$  with general  $\mu$  is  $W(\mu^0) = n(\bar{Y} - \mu^0)^2/\sigma^2$ . Show that  $W(\mu^0)$  is a pivot, and give the likelihood ratio confidence region for  $\mu$ .
- 2 Independent values  $y_1, \dots, y_n$  arise from a distribution putting probabilities  $\frac{1}{4}(1 + 2\theta)$ ,  $\frac{1}{4}(1 - \theta)$ ,  $\frac{1}{4}(1 - \theta)$ ,  $\frac{1}{4}$  on the values 1, 2, 3, 4, where  $-\frac{1}{2} < \theta < 1$ . Show that the likelihood for  $\theta$  is proportional to  $(1 + 2\theta)^{m_1}(1 - \theta)^{m_2}$  and express  $m_1$  and  $m_2$  in terms of  $y_1, \dots, y_n$ . Find the maximum likelihood estimate of  $\theta$  in terms of  $m_1$  and  $m_2$ . Obtain the maximum likelihood estimate and the likelihood ratio statistic for  $\theta = 0$  based on data in which the frequencies of 1, 2, 3, 4 were 55, 11, 8, 26. Is it plausible that  $\theta = 0$ ?
- 3 Consider Examples 4.27 and 4.33. Show that the standard error for  $\eta = \beta_0 + 31\beta_1$  is  $(9.289 - 2 \times 31 \times 0.142 + 31^2 \times 0.00220)^{1/2}$ , and hence obtain a 95% confidence interval for  $\eta$ . Use this to construct an interval for  $\phi = e^\eta/(1 + e^\eta)$ , and compare it with the interval based on the profile log likelihood for  $\phi$ .
- 4 Use (4.46) to show that  $\hat{\pi}_j = y_j/n$ , and verify the contents of the corresponding observed, expected, and inverse expected information matrices.

- 5 Verify that the Taylor expansion  $O \log(O/E) \doteq O - E + \frac{1}{2}(O - E)^2/E + \dots$  is valid for small  $O - E$ , and hence check that provided  $O_i - E_i$  is small relative to  $E_i$ , Pearson's statistic  $P$  is close to the likelihood ratio statistic  $W$ .
- 6 Let  $Y_1, \dots, Y_n$  and  $Z_1, \dots, Z_m$  be two independent random samples from the  $N(\mu_1, \sigma_1^2)$  and  $N(\mu_2, \sigma_2^2)$  distributions respectively. Consider comparison of the model in which  $\sigma_1^2 = \sigma_2^2$  and the model in which no restriction is placed on the variances, with no restriction on the means in either case. Show that the likelihood ratio statistic  $W_p$  to compare these models is large when the ratio  $T = \sum(Y_j - \bar{Y})^2 / \sum(Z_j - \bar{Z})^2$  is large or small, and that  $T$  is proportional to a random variable with the  $F$  distribution.
- 7 In an experiment to assess the effectiveness of a treatment to reduce blood pressure in heart patients,  $n$  independent pairs of heart patients are matched according to their sex, weight, smoking history, initial blood pressure, and so forth. Then one of each pair is selected at random and given the treatment. After a set time the blood pressures are again recorded, and it is desired to assess whether the treatment had any effect. A simple model for this is that the  $j$ th pair of final measurements,  $(Y_{j1}, Y_{j2})$  is two independent normal variables with means  $\mu_j$  and  $\mu_j + \beta$ , and variances  $\sigma^2$ . It is desired to assess whether  $\beta = 0$  or not. One approach is a  $t$  confidence interval based on  $Z_j = Y_{j2} - Y_{j1}$ . Explain this, and give the degrees of freedom for the  $t$  statistic. Show that the likelihood ratio statistic for  $\beta = 0$  is equivalent to  $\bar{Z}^2 / \sum(Z_j - \bar{Z})^2$ .

## 4.6 Non-Regular Models

The large-sample normal and chi-squared approximations (4.26) and (4.39) apply to many important models. There are exceptions, however, due to failure of regularity conditions for the parameter space, the likelihood and its derivatives, and convergence of information quantities. A model can be non-regular in many ways, and rather than attempt a general discussion we give some examples intended to flag possible problems.

### *Parameter space*

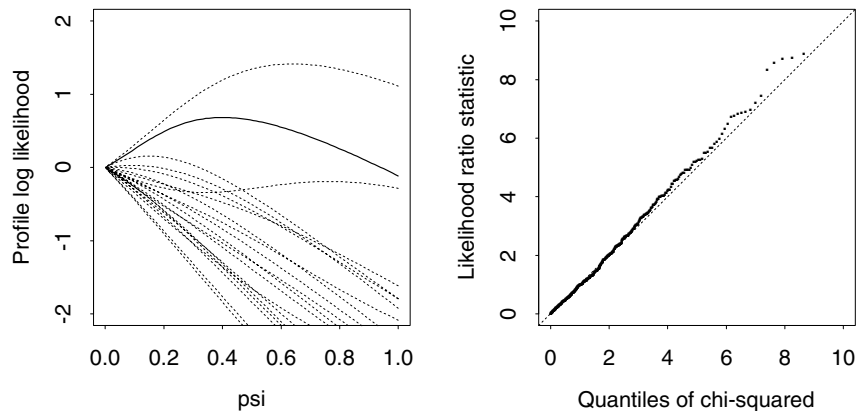
If standard asymptotics are to apply, the true parameter value must be interior to the parameter space  $\Theta$ . One way to ensure this is to insist that  $\Theta$  be an open subset of  $\mathbb{R}^p$  endowed with its usual topology. If not, and if the true  $\theta^0$  lies on the edge of the parameter space, then the maximum likelihood estimator cannot fall on 'both sides' of  $\theta^0$ , and therefore cannot have a limiting normal distribution with mean  $\theta^0$ . Alternatively, if one or more components of  $\theta$  are discrete, we cannot expect the maximum likelihood estimator to be approximately normal.

**Example 4.39 ( $t$  distribution)** One model for heavy-tailed data is

$$f(y; \mu, \sigma^2, \psi) = \frac{\Gamma\{(\psi^{-1} + 1)/2\} \psi^{1/2}}{(\sigma^2 \pi)^{1/2} \Gamma\{1/(2\psi)\}} \{1 + \psi(y - \mu)^2 / \sigma^2\}^{-(\psi^{-1} + 1)/2},$$

where  $\psi, \sigma > 0$  and  $-\infty < \mu, y < \infty$ . This generalizes the Student  $t$  density with  $\psi^{-1} = \nu$  degrees of freedom to continuous  $\psi$ . Its tails are heavier than those of the normal density, obtained when  $\psi \rightarrow 0$ ;  $f(y; \mu, \sigma^2, 1)$  is Cauchy. The left panel of Figure 4.8 shows the profile log likelihood for  $\psi$  based on the  $n = 15$  differences between heights of plants in the fourth column of Table 1.1;  $\psi = 0$  is of particular interest. The likelihood ratio statistic for comparing  $t$  and normal models

**Figure 4.8** Likelihood inference for  $t_\nu$  distribution. Left: profile log likelihoods for  $\psi = \nu^{-1}$  for maize data (solid), and for 19 simulated normal samples (dots);  $\psi = 0$  corresponds to the  $N(\mu, \sigma^2)$  density. Right:  $\chi_1^2$  probability plot for the 1237 positive values of the likelihood ratio statistic  $W_p(0)$  observed in 5000 simulated normal samples of size 15; the rest had  $W_p(0) = 0$ .



is  $W_p(0) = 2\{\ell(\hat{\mu}, \hat{\sigma}^2, \hat{\psi}) - \ell(\hat{\mu}_0, \hat{\sigma}_0^2, 0)\}$ , where  $\hat{\mu}_0$  and  $\hat{\sigma}_0^2$  are maximum likelihood estimates for the  $N(\mu, \sigma^2)$  density. Its observed value of 1.366 suggests that the  $t$  fit is only marginally better, but  $\psi = 0$  is on the boundary of the parameter space and standard asymptotics do not apply, as we see from profile log likelihoods for simulated normal samples of size 15. In many cases  $\hat{\psi} = 0$ , so  $W_p(0) = 0$ : its distribution cannot be  $\chi_1^2$ .

To understand this, we expand  $\log f(\mu, \sigma^2, \psi)$  about  $\psi = 0$ , giving

$$\begin{aligned} & -\frac{1}{2}\{z^2 + \log(2\pi\sigma^2)\} + \frac{\psi}{4}(z^4 - 2z^2 - 1) + \frac{\psi^2}{2}\left(\frac{1}{2}z^4 - \frac{1}{3}z^6\right) \\ & + \frac{\psi^3}{24}(3z^8 - 4z^6 - 1) + O(\psi^4), \end{aligned}$$

where  $z = (y - \mu)/\sigma$ . The first and second derivatives that involve  $\psi$  are  $\partial \log f / \partial \psi = (z^4 - 2z^2 - 1)/4$  and

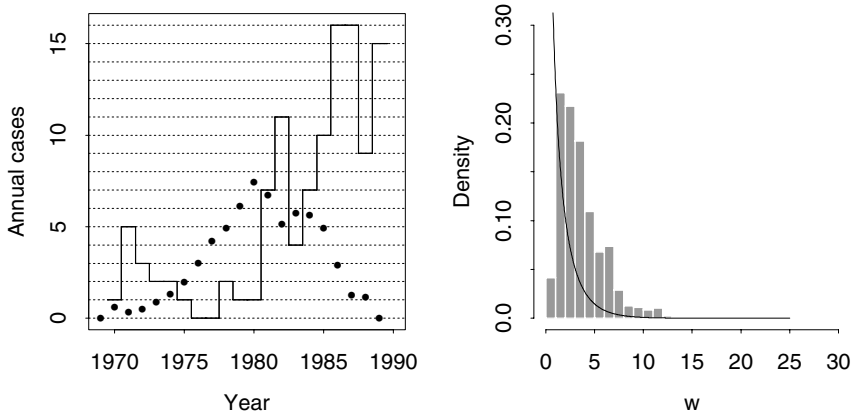
$$\frac{\partial^2 \log f}{\partial \psi^2} = \frac{1}{2}z^4 - \frac{1}{3}z^6, \quad \frac{\partial^2 \log f}{\partial \psi \partial \mu} = (z - z^3)/\sigma, \quad \frac{\partial^2 \log f}{\partial \psi \partial \sigma^2} = (z^2 - z^4)/(2\sigma^2)$$

evaluated at  $\psi = 0$ , while Example 4.18 gives the other derivatives needed. When  $\psi = 0$ ,  $Z = (Y - \mu)/\sigma \sim N(0, 1)$ , with odd moments zero and first three even moments 1, 3, and 15, so  $\text{cov}(Z^4, Z^4) = 96$ ,  $\text{cov}(Z^2, Z^4) = 12$ , and  $\text{var}(Z^2) = 2$ . The expected information matrix,

$$i(\mu, \sigma^2, 0) = \begin{pmatrix} \sigma^{-2} & 0 & 0 \\ 0 & \frac{1}{2}\sigma^{-4} & \sigma^{-2} \\ 0 & \sigma^{-2} & \frac{7}{2} \end{pmatrix},$$

equals the covariance matrix of the score statistic, and the third derivatives of  $\log f$  are well-behaved, so the large-sample distribution of the score vector when  $\psi = 0$  is normal with mean zero and covariance matrix  $ni(\mu, \sigma^2, 0)$ . On setting  $\lambda = (\mu, \sigma^2)$  and  $\psi = 0$ , (4.48) entails

$$\frac{\partial \ell(\hat{\mu}_0, \hat{\sigma}_0^2, 0)}{\partial \psi} \sim N(0, 3n/2).$$



**Figure 4.9** Change point analysis for data on diarrhoea-associated haemolytic uraemic syndrome (HUS) (Henderson and Matthews, 1993). Left: counts of cases of HUS treated in Birmingham, 1970–1989 (solid), and scaled likelihood ratio statistic  $W_p(\tau)/10$  (blobs). Right: density of  $W$ , estimated from 10,000 simulations, and  $\chi_1^2$  density (solid).

In large samples this derivative is negative with probability  $\frac{1}{2}$ , and then  $W_p(0) = 0$ ; while if it is positive the usual Taylor series expansion applies and  $W_p(0) \sim \chi_1^2$ . Thus the limiting distribution of  $W_p(0)$  is  $\frac{1}{2} + \frac{1}{2}\chi_1^2$ , giving

$$\Pr\{W_p(0) \leq 1.366\} = \frac{1}{2} + \frac{1}{2}\Pr(\chi_1^2 \leq 1.366) = 0.88.$$

The asymptotic distribution of  $\hat{\psi}$  puts mass  $\frac{1}{2}$  at  $\psi = 0$ , with the remaining probability spread as a normal density confined to the positive half-line.

To assess the quality of such approximations, 5000 normal samples of size  $n = 15$  were generated. Just 1237 of the  $W_p(0)$  were positive, but those that were had distribution close to  $\chi_1^2$ , as the right panel of Figure 4.8 shows. Hence

$$\Pr\{W_p(0) \leq 1.366\} \doteq (3763/5000) + (1237/5000)\Pr(\chi_1^2 \leq 1.366) = 0.94,$$

stronger though not decisive evidence for the  $t$  model. Large-sample results are unreliable even with  $n = 100$ , when  $\Pr\{W_p(0) = 0\} \doteq 0.37$ .

Such problems also arise if the favoured model is close to the boundary. For example, despite being normal in large samples, when  $n$  is small the distribution of  $\hat{\psi}$  would have a point mass at  $\psi = 0$ . If several parameters lie on their boundaries, then asymptotics become yet more cumbersome. Simulation seems preferable. ■

**Example 4.40 (HUS data)** The left panel of Figure 4.9 shows annual numbers of cases of ‘diarrhoea-associated haemolytic uraemic syndrome’ (HUS) treated at a clinic in Birmingham from 1970 to 1989. HUS is a disease that can threaten the lives of small children; physicians have speculated that it is linked to levels of *E. coli*. The data suggest a sharp rise in incidence at about 1980.

A simple model for this increase is that the annual counts  $y_1, \dots, y_n$  are realizations of independent Poisson variables  $Y_1, \dots, Y_n$  with positive means

$$E(Y_j) = \begin{cases} \lambda_1, & j = 0, \dots, \tau, \\ \lambda_2, & j = \tau + 1, \dots, n. \end{cases}$$

Here the change point  $\tau$  is a discrete parameter with possible values  $0, \dots, n$ . The simpler model of no change appears when  $\tau = 0$  or  $n$ , and then  $\lambda_1$  or  $\lambda_2$  vanishes

from the model. Obviously these two situations are indistinguishable. Moreover, there would be no changepoint to detect if  $\lambda_1 = \lambda_2$ .

In terms of  $s_i = y_1 + \cdots + y_i$  the log likelihood may be written

$$\ell(\tau, \lambda_1, \lambda_2) \equiv s_\tau \log \lambda_1 - \tau \lambda_1 + (s_n - s_\tau) \log \lambda_2 - (n - \tau) \lambda_2,$$

and given  $\tau$ , the maximum likelihood estimates are  $\hat{\lambda}_1(\tau) = s_\tau/\tau$  and  $\hat{\lambda}_2(\tau) = (s_n - s_\tau)/(n - \tau)$ . Hence the profile log likelihood for  $\tau$  is

$$\ell_p(\tau) = s_\tau \log(s_\tau/\tau) + (s_n - s_\tau) \log\{(s_n - s_\tau)/(n - \tau)\} - s_n, \quad \tau = 0, \dots, n,$$

and the likelihood ratio statistic for comparing the model of change at  $\tau$  with that of constant  $\lambda$  is

$$W_p(\tau) = 2 \left[ S_\tau \log \left( \frac{S_\tau/\tau}{S_n/n} \right) + (S_n - S_\tau) \log \left\{ \frac{(S_n - S_\tau)/(n - \tau)}{S_n/n} \right\} \right],$$

where  $S_i$  is the random variable corresponding to  $s_i$ . As  $S_i$  is a sum of independent Poisson variables, its distribution is Poisson. For completeness we set  $W_p(0) = W_p(n) = 0$ . The values of  $W_p(\tau)/10$  shown in the left panel of Figure 4.9 give strong evidence of change in the rate.

If we wish to test for change at a known value of  $\tau$ , the usual asymptotics will apply provided  $\lambda_1$  and  $\lambda_2$  can be estimated consistently from the independent Poisson variables  $S_\tau$  and  $S_n - S_\tau$ , and this will be so if their means  $\tau\lambda_1$  and  $(n - \tau)\lambda_2$  both tend to infinity. Two asymptotic frameworks for this are:

- $\lambda_1, \lambda_2 \rightarrow \infty$  with  $n$  and  $\tau$  fixed; and
- $n \rightarrow \infty$  and  $\tau/n \rightarrow a$ , with  $0 < a < 1$  and  $\lambda_1, \lambda_2$  positive and fixed.

The practical implication is that if  $\tau$  is so close to one of the endpoints that  $\tau\lambda_1$  or  $(n - \tau)\lambda_2$  is small, a  $\chi^2_1$  approximation for the null distribution of  $W_p(\tau)$  will be poor, and its quality should be checked; otherwise no new issues arise. They do, however, if  $\tau$  is unknown.

The likelihood ratio statistic for existence of a changepoint, regardless of its location, is

$$W = \max\{W_p(\tau) : \tau = 1, \dots, n - 1\}.$$

The values of  $W_p(\tau)$  in the left panel of Figure 4.9 show that  $\hat{\tau} = 11$ , corresponding to a change between 1980 and 1981; the observed value of  $W$  is  $w = 74.14$ . This seems to be the strong evidence for change that we would have anticipated from plotting the data, but can we be sure?

To find the distribution of  $W$  when  $\lambda_1 = \lambda_2 = \lambda$ , we first note that  $Y_1, \dots, Y_n$  are then a Poisson random sample with mean  $\lambda$ . For reasons given in Sections 5.2.3 it is appropriate to treat  $W$  conditional on  $S_n = m$ , and Example 2.36 implies that the joint distribution of  $Y_1, \dots, Y_n$  conditional on  $S_n = m$  is multinomial with denominator  $m$  and probability vector  $\pi = (n^{-1}, \dots, n^{-1})^T$ . We can simulate the exact distribution of  $W$  under this setup, because no parameters are involved. The right panel of Figure 4.9 shows a histogram of 10,000 simulated values of  $W$ . Clearly  $W$  is stochastically

larger than the  $\chi_1^2$  density, that is,  $\Pr(W > v) > \Pr(\chi_1^2 > v)$  for any  $v > 0$ . Even so,  $w = 74.14$  is much too large to have occurred by chance: there is overwhelming evidence for a change.

Here the maximum likelihood estimator  $\hat{\tau}$  has a discrete distribution on  $0, \dots, 20$  and normal approximation would be foolish. Other approaches have more appeal, and we revisit these data in Example 11.13. ■

### Parameter identifiability

There must be a 1–1 mapping between models and elements of the parameter space, otherwise there may be no unique value of  $\theta$  for  $\hat{\theta}$  to converge to. A model in which each  $\theta$  generates a different distribution is called *identifiable*. We saw a failure of this in Example 4.40, where setting  $\lambda_1 = \lambda_2$  gave the same model for any changepoint  $\tau$ . A rarer possibility is that a parameter cannot be estimated from a particular set of data. In the changepoint example, for instance, the profile likelihood for  $\tau$  is flat when  $y_1 = \dots = y_n$ . The probability of such an event vanishes asymptotically, but such likelihoods do occasionally occur in practice; they demand a simpler model, more data or external knowledge about parameter values.

Sometimes a model has been set up in such a way that its parameters are non-identifiable from any dataset. Suppose we have data  $y_1, \dots, y_n$  with corresponding parameters  $\eta_1, \dots, \eta_n$ , and that we may write both  $\eta_j = \eta_j(\theta)$  and  $\eta_j = \eta_j(\beta)$ , where  $\theta$  and  $\beta = \beta(\theta)$  are  $p \times 1$  and  $q \times 1$  vectors of parameters, with  $q < p$ . Then the model with  $\eta(\theta)$  is said to be *parameter redundant*. The chain rule gives

$$\frac{\partial \eta^T}{\partial \theta} = \frac{\partial \beta^T}{\partial \theta} \frac{\partial \eta^T}{\partial \beta},$$

where both matrices on the right have rank  $q$  or lower for any  $\theta$ . Hence the matrix on the left is symbolically rank-deficient: there is a  $1 \times p$  vector function  $\gamma(\theta)$ , non-zero for all  $\theta$ , such that  $\gamma(\theta) \partial \eta^T / \partial \theta = 0$  for all  $\theta$ . It is fairly straightforward to see that the converse is true, so the model is parameter redundant if and only if  $\partial \eta^T / \partial \theta$  is symbolically rank-deficient. Computer algebra can be used to check the symbolic rank of  $\partial \eta^T / \partial \theta$  for a complex model.

**Example 4.41 (Exponential density)** Let  $Y_1, \dots, Y_n$  be independent exponential variables with mean  $\eta$ , and set  $\eta = \theta_1 \theta_2$ , where  $\theta_1 = \beta$  and  $\theta_2 = \beta$ . Evidently  $\theta_1$  and  $\theta_2$  cannot be estimated separately, and this is reflected by the  $n \times 2$  matrix  $\partial \eta^T / \partial \theta$ , which consists of a row of  $\theta_2$ 's above a row of  $\theta_1$ 's. It has symbolic rank one, as is seen on premultiplying it by  $\gamma(\theta) = (\theta_1, -\theta_2)$ .

The likelihood  $L(\theta)$  is constant on the curves  $(\theta_1, \theta_2) = (\psi\beta, \beta^{-1})$  in  $\mathbb{R}_+^2$  and is maximized not at a single point but everywhere on the curve  $(\theta_1, \theta_2) = (\bar{y}t, t^{-1})$ ,  $t > 0$ . A ridge such as this is a feature of parameter-redundant likelihoods. ■

### Score and information

For regular inference the log likelihood and its derivatives must be well-behaved enough to allow Taylor series expansions and the neglect of their higher-order terms, and the score must have the asymptotic normal distribution at (4.34). For a random

sample,  $I(\theta^0) = ni(\theta^0)$ , and so the expected information increases without limit as  $n \rightarrow \infty$ ; in order to have a normal limit in more complicated situations we also need  $I(\theta^0) \rightarrow \infty$ . Furthermore the observed information must converge in probability as at (4.34).

**Example 4.42 (Normal mixture)** For an example of a non-smooth likelihood, let  $L(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \gamma)$  be the likelihood for a random sample  $y_1, \dots, y_n$  from the mixture of normal densities

$$\frac{\gamma}{(2\pi)^{1/2}\sigma_1} \exp\left\{-\frac{(y-\mu_1)^2}{2\sigma_1^2}\right\} + \frac{1-\gamma}{(2\pi)^{1/2}\sigma_2} \exp\left\{-\frac{(y-\mu_2)^2}{2\sigma_2^2}\right\}, \quad 0 \leq \gamma \leq 1,$$

with the means and variances in their usual ranges. This corresponds to taking observations in proportions  $\gamma, 1-\gamma$  from two normal populations, not knowing from which they come. If  $\gamma \neq 0, 1$ , then for each  $y_j$

$$\lim_{\sigma_1 \rightarrow 0} L(y_j, \mu_2, \sigma_1^2, \sigma_2^2, \gamma) = \lim_{\sigma_2 \rightarrow 0} L(\mu_1, y_j, \sigma_1^2, \sigma_2^2, \gamma) = +\infty,$$

so  $L$  is a smooth surface pocked with singularities, each of which corresponds to estimating the mean and variance of one of the populations from a single observation. For large  $n$  the strong consistency result guarantees the existence of a smooth local maximum of  $L$  near the true parameter values. When finding this numerically a careful choice of starting values can help one avoid ending up at a spike instead, but it is worth asking why they occur.

The issue is rounding. As we saw in Example 4.21, the fiction that data are continuous is usually harmless and convenient. Here it is not harmless, however, because it results in infinite likelihoods. The spikes can be removed by accounting for the rounding of the  $y_j$ . If they are rounded to multiples of  $\delta$ , then  $\Pr(Y = k\delta) = F(k\delta + \delta/2) - F(k\delta - \delta/2)$ , where

$$F(y) = \gamma \Phi\left(\frac{y-\mu_1}{\sigma_1}\right) + (1-\gamma) \Phi\left(\frac{y-\mu_2}{\sigma_2}\right).$$

As  $0 < F(y_j) < 1$ , the largest possible contribution to  $L$  is then finite. See Example 5.36 for further discussion. ■

**Example 4.43 (Shifted exponential density)** To see a failure of regularity conditions for the score statistic, let  $y_1, \dots, y_n$  be an exponential random sample with lower endpoint  $\phi$  and mean  $\theta + \phi$ , so

$$f(y; \phi, \theta) = \theta^{-1} \exp\{-(y-\phi)/\theta\}, \quad y > \phi, \theta > 0.$$

The corresponding random variables  $Y_1, \dots, Y_n$  have the same distribution as  $\phi + \theta E_1, \dots, \phi + \theta E_n$ , where  $E_1, \dots, E_n$  is a random sample from the standard exponential density. The log likelihood contribution from a single observation  $y > \phi$  is  $\ell(\phi, \theta) = -\log \theta - (y-\phi)/\theta$ , so

$$\frac{\partial \ell(\phi, \theta)}{\partial \phi} = \begin{cases} \theta^{-1}, & y > \phi, \\ 0, & \text{otherwise.} \end{cases}$$

For a regular model this would have mean zero, but here the interchange of differentiation and integration that yields (4.32) fails because the support of the density depends on  $\phi$ , and  $E(\partial \ell / \partial \phi) = \theta^{-1}$ .

The support of  $g(y)$  is the set  $\{y : g(y) > 0\}$ .

The likelihood is  $L(\phi, \theta) = \theta^{-n} \exp \{-n(\bar{y} - \phi)/\theta\}$  for  $y_1, \dots, y_n > \phi$  and  $\theta > 0$ , and for any  $\theta$  this increases as  $\phi \uparrow \min y_j$  and is zero thereafter. Thus  $\phi$  has maximum likelihood estimate  $\hat{\phi} = y_{(1)}$ , while  $\hat{\theta} = \bar{y} - \hat{\phi} = \bar{y} - y_{(1)}$ .

To find limiting distributions of  $\hat{\phi}$  and  $\hat{\theta}$ , recall from Example 2.28 that the  $r$ th order statistic  $E_{(r)}$  of a standard exponential random sample may be written  $\sum_{j=1}^r (n+1-j)^{-1} E_j$ , where  $E_1, \dots, E_n$  is an exponential random sample. As  $Y_{(r)} \stackrel{D}{=} \phi + \theta E_{(r)}$ , we see that  $Y_{(1)} \stackrel{D}{=} \phi + n^{-1} \theta E_1$ , implying that  $n\theta^{-1}(\hat{\phi} - \phi) \stackrel{D}{=} E_1$ : the rescaled endpoint estimate  $\hat{\phi}$  has a non-normal limit distribution. Moreover it converges faster than usual because  $\hat{\phi} - \phi$  must be multiplied by  $n$  rather than  $n^{1/2}$  in order to give a non-degenerate limit.

For the distribution of  $\hat{\theta}$ , note that as  $\bar{Y} - Y_{(1)} = n^{-1} \sum_{r=1}^n Y_{(r)} - Y_{(1)}$ ,

$$\hat{\theta} \stackrel{D}{=} n^{-1} \left\{ n\phi + \theta \sum_{r=1}^n \sum_{j=1}^r \frac{E_j}{n+1-j} - n\phi - \theta E_1 \right\} = n^{-1}(n-1)\theta \bar{E},$$

with  $\bar{E}$  the average of  $E_2, \dots, E_n$ . The central limit theorem implies that  $n^{1/2}(\hat{\theta} - \theta)/\theta \xrightarrow{D} N(0, 1)$ , so standard asymptotics apply to  $\hat{\theta}$  despite their failure for  $\hat{\phi}$ , which converges so fast that its randomness has no impact on the limiting distribution of  $\hat{\theta}$ .

In this problem exact inference is possible for any  $n$  (Exercise 4.6.4), but the general conclusion is that endpoints must be treated gingerly. ■

Though artificial, our next example illustrates how trouble in stochastic process problems can stem from the information quantities.

**Example 4.44 (Poisson birth process)** Consider a sequence  $Y_0, \dots, Y_n$  such that given the values of  $Y_0, \dots, Y_{j-1}$ , the variable  $Y_j$  has a Poisson density with mean  $\theta Y_{j-1}$ , and  $E(Y_0) = \theta$ . The likelihood for  $\theta$  based on such data was given in Example 4.6, and the log likelihood and observed information are

$$\ell(\theta) \equiv \sum_{j=0}^n Y_j \log \theta - \theta \left( 1 + \sum_{j=0}^{n-1} Y_j \right), \quad J(\theta) = \theta^{-2} \sum_{j=0}^n Y_j.$$

The expected value of  $Y_j$ , given  $Y_{j-1}$ , is  $\theta Y_{j-1}$ , so its unconditional expectation is  $\theta^{j+1}$ . Hence the expected information is  $I(\theta) = \theta^{-2}(\theta + \dots + \theta^{n+1})$ . If  $\theta \geq 1$ , then  $I(\theta) \rightarrow \infty$  as  $n \rightarrow \infty$ , but if not,  $I(\theta)$  is asymptotically bounded. In fact, as  $n \rightarrow \infty$ , the process is certain to become extinct — that is, there will be an  $n_0$  such that  $Y_{n_0} = Y_{n_0+1} = \dots = 0$  — unless  $\theta > 1$ , and even then there is a non-zero probability of extinction. Hence  $J(\theta)$  remains finite with probability one unless  $\theta > 1$ , and remains finite with non-zero probability for any  $\theta$ . Thus the maximum likelihood estimator  $\hat{\theta} = (Y_0 + \dots + Y_n)/(1 + Y_0 + \dots + Y_{n-1})$  is neither consistent nor asymptotically normal if  $\theta \leq 1$ .

From a practical viewpoint, this failure of standard asymptotics is less critical than it might appear. The limit (4.26) is used to obtain finite-sample approximations such as (4.27), but we can still use these if they can be justified by other means. Inference is not impossible, merely more difficult than with independent data. ■

### Wrong model

Up to now we have supposed that the model fitted to the data is correct, with only parameter values unknown. To explore some consequences of fitting the wrong model, suppose the true model is  $g(y)$ , but that ignorant of this we attempt to fit  $f(y; \theta)$  to a random sample  $y_1, \dots, y_n$ . Under mild conditions the log likelihood  $\ell(\theta) = \sum \log f(y_j; \theta)$  will be maximized at  $\hat{\theta}$ , say, and as  $n \rightarrow \infty$  the quantity  $\bar{\ell}(\hat{\theta}) = n^{-1} \ell(\hat{\theta})$  will tend to

$$\int \log f(y; \theta_g) g(y) dy,$$

where  $\theta_g$  is the value of  $\theta$  that minimizes the *Kullback–Leibler discrepancy*

$$D(f_\theta, g) = \int \log \left\{ \frac{g(y)}{f(y; \theta)} \right\} g(y) dy$$

with respect to  $\theta$ . Thus  $\theta_g$  is the ‘least bad’ value of  $\theta$  given our wrong model; of course  $\theta_g$  depends on  $g$ . Differentiation gives

$$0 = \int \frac{\partial \log f(y; \theta_g)}{\partial \theta} g(y) dy,$$

with  $\hat{\theta}$  determined by the finite-sample version of this,

$$0 = n^{-1} \sum_{j=1}^n \frac{\partial \log f(y_j; \hat{\theta})}{\partial \theta}. \quad (4.51)$$

Expansion of (4.51) about  $\theta_g$  yields

$$\hat{\theta} \doteq \theta_g + \left\{ -n^{-1} \sum_{j=1}^n \frac{\partial^2 \log f(y_j; \theta_g)}{\partial \theta \partial \theta^\tau} \right\}^{-1} \left\{ n^{-1} \sum_{j=1}^n \frac{\partial \log f(y_j; \theta_g)}{\partial \theta} \right\}$$

and a modification of the derivation that starts on page 124 gives

$$\hat{\theta} \sim N_p\{\theta_g, I(\theta_g)^{-1} K(\theta_g) I(\theta_g)^{-1}\}, \quad (4.52)$$

where the *information sandwich* variance matrix depends on

$$K(\theta_g) = n \int \frac{\partial \log f(y; \theta)}{\partial \theta} \frac{\partial \log f(y; \theta)}{\partial \theta^\tau} g(y) dy, \quad (4.53)$$

$$I_g(\theta_g) = -n \int \frac{\partial^2 \log f(y; \theta)}{\partial \theta \partial \theta^\tau} g(y) dy.$$

If  $g(y) = f(y; \theta)$ , so that the supposed density is correct, then  $\theta_g$  is the true  $\theta$ , the multivariate version of (4.33) gives  $K(\theta_g) = I_g(\theta_g) = I(\theta)$ , and (4.52) reduces to the usual approximation.

In practice  $g(y)$  is of course unknown, and then  $K(\theta_g)$  and  $I_g(\theta_g)$  may be estimated by

$$\widehat{K} = \sum_{j=1}^n \frac{\partial \log f(y_j; \widehat{\theta})}{\partial \theta} \frac{\partial \log f(y_j; \widehat{\theta})}{\partial \theta^T}, \quad \widehat{J} = - \sum_{j=1}^n \frac{\partial^2 \log f(y_j; \widehat{\theta})}{\partial \theta \partial \theta^T}; \quad (4.54)$$

the latter is just the observed information matrix. We may then construct confidence intervals for  $\theta_g$  using (4.52) with variance matrix  $\widehat{J}^{-1} \widehat{K} \widehat{J}^{-1}$ .

For future reference we give the approximate distribution of the likelihood ratio statistic. Taylor series approximation gives

$$\begin{aligned} 2\{\ell(\widehat{\theta}) - \ell(\theta_g)\} &\doteq (\widehat{\theta} - \theta_g)^T \left\{ - \sum_{j=1}^n \frac{\partial^2 \log f(y_j; \theta_g)}{\partial \theta \partial \theta^T} \right\} (\widehat{\theta} - \theta_g) \\ &\doteq n(\widehat{\theta} - \theta_g)^T I_g(\theta_g) (\widehat{\theta} - \theta_g) \end{aligned}$$

and the normal distribution (4.52) of  $\widehat{\theta}$  implies that the likelihood ratio statistic has a distribution proportional to  $\chi_p^2$ , but with mean  $\text{tr}\{I_g(\theta_g)^{-1} K(\theta_g)\}$ . If the model is correct,  $I_g(\theta_g) = K(\theta_g)$ , giving the previous mean,  $p$ .

**Example 4.45 (Exponential and log-normal models)** Let  $f(y; \theta)$  be the exponential density with mean  $\theta$ , while in fact  $Y = e^{\sigma Z}$ , where  $Z$  is standard normal. Then  $Y$  is log-normal, with mean  $e^{\sigma^2/2}$  and variance  $e^{\sigma^2}(e^{\sigma^2} - 1)$ .

The presumed log likelihood is  $-\log \theta - y/\theta$ , so that

$$\int \log f(y; \theta) g(y) dy = -\log \theta - \theta^{-1} \int y g(y) dy = -\log \theta - \theta^{-1} e^{\sigma^2/2},$$

and differentiation of this with respect to  $\theta$  gives  $\theta_g = e^{\sigma^2/2}$ . Here the ‘least bad’ exponential model has the same mean as the true log-normal distribution, which must always exceed one. Further calculation gives  $I(\theta_g) = \theta_g^{-2}$  and  $K(\theta_g) = 1 - \theta_g^{-2}$ ,

The maximum likelihood estimate of  $\theta$  is  $\widehat{\theta} = \bar{Y}$ , and either directly or using the information sandwich we see that  $\text{var}(\widehat{\theta}) = n^{-1} \theta_g^2 (\theta_g^2 - 1)$ . Note that replacement of  $\theta_g$  with its estimate  $\widehat{\theta}$  could result in a negative variance. This is not the case if we use the empirical variance — simple calculations give  $\widehat{J} = n/\bar{y}^2$  and  $\widehat{K} = \bar{y}^{-4} \sum (y_j - \bar{y})^2$ , so  $\widehat{J}^{-2} \widehat{K} = n^{-2} \sum (y_j - \bar{y})^2$ . Reassuringly, this is a consistent estimate of the variance of the average of a random sample from any distribution with finite variance (Example 2.20).

As  $I_g(\theta_g)^{-1} K(\theta_g) = e^{\sigma^2} - 1 = \theta_g^2 - 1$ , the likelihood ratio statistic may be over- or under-dispersed relative to the  $\chi_1^2$  distribution. ■

The discussion above is too crude to be the last word. In practice the model fitted will often be elaborate enough to be reasonably close to the data, in the sense that only glaring departures from the model are certain to be detected. Thus it would be better to examine the properties of  $\widehat{\theta}$  and related quantities when  $f(y; \theta)$  is near  $g(y)$  in a suitable sense.

## Exercises 4.6

- 1 Data arise from a mixture of two exponential populations, one with probability  $\pi$  and parameter  $\lambda_1$ , and the other with probability  $1 - \pi$  and parameter  $\lambda_2$ . The exponential parameters are both positive real numbers and  $\pi$  lies in the range  $[0, 1]$ , so  $\Theta = [0, 1] \times \mathbb{R}_+^2$  and

$$f(y; \pi, \lambda_1, \lambda_2) = \pi \lambda_1 e^{-\lambda_1 y} + (1 - \pi) \lambda_2 e^{-\lambda_2 y}, \quad y > 0, \quad 0 \leq \pi \leq 1, \quad \lambda_1, \lambda_2 > 0.$$

Are the parameters identifiable?

Does standard likelihood theory apply when (i) using a likelihood ratio statistic to test if  $\pi = 0$ ? (ii) estimating  $\pi$  when  $\lambda_1 = \lambda_2$ ?

- 2 One model for outliers in a normal sample is the mixture

$$f(y; \mu, \pi) = (1 - \pi) \phi(y - \mu) + \pi g(y - \mu), \quad 0 \leq \pi \leq 1, \quad -\infty < \mu < \infty,$$

where  $g(z)$  has heavier tails than the standard normal density  $\phi(z)$ ; take  $g(z) = \frac{1}{2} e^{-|z|}$ , for example. Typically  $\pi$  will be small or zero. Show that when  $\pi = 0$  the likelihood derivative for  $\pi$  has zero mean but infinite variance, and discuss the implications for the likelihood ratio statistic comparing normal and mixture models.

- 3 Show that the capture-recapture model in Example 4.13 is not parameter redundant, but that it is if different survival probabilities are allowed in each year. Why is this obvious?
- 4 In Example 4.43, use relations between the exponential, gamma, chi-squared and  $F$  distributions (Section 3.2.1) to show that

$$\frac{2n\hat{\theta}}{\theta} \sim \chi_{2(n-1)}^2, \quad \frac{n(\hat{\phi} - \phi)}{\hat{\theta}} \sim \frac{n}{n-1} F_{2, 2(n-1)};$$

hence give exact  $(1 - 2\alpha)$  confidence intervals for the parameters.

- 5 Show that the score statistic for a variable  $Y$  from the uniform density on  $(0, \theta)$  is  $U(\theta) = -\theta^{-1}$  in the range  $0 < Y < \theta$  and zero otherwise, and deduce that  $E\{U(\theta)\} = -1$  and  $i(\theta) = -\theta^{-1}$ . Why is this model non-regular? Sketch the likelihood based on a random sample  $Y_1, \dots, Y_n$ , and verify that  $\hat{\theta} = Y_{(n)}$ . To find its limiting distribution, note that

$$\Pr(\hat{\theta} \leq a) = \begin{cases} 0, & a < 0, \\ (a/\theta)^n, & 0 \leq a \leq \theta, \\ 1, & a > \theta. \end{cases}$$

Show that as  $n \rightarrow \infty$ ,  $Z_n = n(\theta - \hat{\theta})/\theta \xrightarrow{D} E$ , where  $E$  is exponential.

- 6 Suppose that  $\partial \eta^T / \partial \theta$  is symbolically rank-deficient, that is, there exist  $\gamma_r(\theta)$ , non-zero for all  $\theta$ , such that

$$\sum_{r=1}^p \gamma_r(\theta) \frac{\partial \eta_j}{\partial \theta_r} = 0, \quad j = 1, \dots, n.$$

Show that the auxiliary equations

$$\frac{d\theta_1}{\gamma_1(\theta)} = \dots = \frac{d\theta_p}{\gamma_p(\theta)}$$

have  $p - 1$  solutions given implicitly by  $\beta_i(\theta) = c_i$  for constants  $c_1, \dots, c_{p-1}$ . Deduce that the model is parameter redundant. (Catchpole and Morgan, 1997)

This requires basic knowledge of partial differential equations.

## 4.7 Model Selection

Model formulation involves judgement, experience, trial, and error. Evidently models should be consistent with knowledge of the system under study, extrapolate to related sets of data, and if possible have reasonable mathematical and statistical properties. Thus, for example, we prefer discrete distributions for discrete quantities and continuous for continuous, while if a probability  $\pi(x)$  depends on a quantity  $x$ , the relation  $\pi(x) = e^{\beta x} / (1 + e^{\beta x})$  is preferable to  $\pi(x) = \beta x$ , because the latter may lie outside the interval  $(0, 1)$ ; see Example 4.5. Often subject-matter considerations suggest a stochastic argument for a range of suitable models, which typically have primacy over purely *ad hoc* ones. Even after such principles have been applied, however, there are usually several competing models, and a basis is needed for comparing them.

A principle already used but as yet unstated is the *principle of parsimony* or Ockham's razor: 'it is vain to do with more what can be done with fewer'. According to this, given several explanations of the same phenomenon, we should prefer the simplest, or, in our terms, favour simple models over complex ones that fit our data about equally well. But what does this last phrase mean? If we have models with 1, 2, and 3 parameters and maximized log likelihoods of 0, 10, and 11, the second clearly improves on the first, but do the second and third fit 'about equally well'? For regular nested models, standard asymptotics could be applied, but more generally there are difficulties. First, model selection usually involves many fits to the same set of data, so our previous discussion focussing on comparing two prespecified models may be wildly inappropriate. Second, useful asymptotics may be unavailable, for example because the models to be compared are not nested. Third, we may wish to treat none of the models as the truth. An example is in prediction, where a fitted model is sometimes treated as a 'black box' whose contents have no intrinsic interest but are merely used to generate predictions; we should then adopt the agnostic position described at the end of Section 4.6. Here we outline how those ideas may be applied to model selection.

Suppose we have a random sample  $Y_1, \dots, Y_n$  from the unknown *true model*  $g(y)$ . We fit a *candidate model*  $f(y; \theta)$  by maximizing  $\ell(\theta) = \sum \log f(y_j; \theta)$ , giving  $p \times 1$  parameter estimate  $\hat{\theta}$ ; equivalently we could minimize  $-\ell(\theta)$ . The fact that the Kullback–Leibler discrepancy is positive,

$$D(f_\theta, g) = \int \log \left\{ \frac{g(y)}{f(y; \theta)} \right\} g(y) dy \geq 0,$$

with equality if and only if  $f(y; \theta) = g(y)$ , suggests that we aim to choose the candidate that minimizes  $D(f_\theta, g)$ . Let  $\theta_g$  denote the corresponding value of  $\theta$ . Unfortunately this approach to model selection is not sufficiently discriminating. The catch is that an infinity of candidate models have  $D(f_{\theta_g}, g) = 0$ . To see why, suppose that by a lucky chance the candidate model contains the true one. Then  $f(y; \theta_g) = g(y)$  and we call  $f_{\theta_g}$  *correct*. As  $g$  has fewer parameters we prefer it to  $f_\theta$ , but  $D(f_\theta, g) \geq 0$  with equality when  $\theta = \theta_g$ . Hence on this basis any correct model is indistinguishable from the true one. We want to pick out the simplest correct model, so we should favour models with few rather than many parameters, provided they fit about equally

William of Ockham or Occam (?1285–1347/1349) was an English Franciscan who studied at Oxford and Paris, was imprisoned by Pope John XXII for arguing that the Franciscan ideal of poverty was prefigured in the Gospels, and then escaped to Bavaria where he wrote in defense of Emperor Louis IV against papal claims; Eco (1984) gives some idea of these controversies. Regarded as the most important scholastic philosopher after Thomas Aquinas, his insistence that logic and human knowledge could be studied without reference to theology and metaphysics encouraged scientific research. He probably died in the Black Death of 1349.

well. For example, if  $g$  is the exponential density with unit mean,  $f_\theta$  might be the Weibull density with unknown shape and scale parameters. This is correct because it reduces to  $g$  when both its parameters take value one, but given the choice we would prefer  $g$ . A example of a wrong model is the log normal density, which does not become exponential for any values of its parameters.

The expected likelihood ratio statistic for comparing  $g$  with  $f_\theta$  at  $\theta = \hat{\theta}$  for another random sample  $Y_1^+, \dots, Y_n^+$  from  $g$ , independent of  $Y_1, \dots, Y_n$ , is

$$E_g^+ \left[ \sum_{j=1}^n \log \left\{ \frac{g(Y_j^+)}{f(Y_j^+; \hat{\theta})} \right\} \right] = nD(f_{\hat{\theta}}, g) \geq nD(f_{\theta_g}, g),$$

where  $E_g^+(\cdot)$  denotes expectation over the density  $g$  of  $Y^+$ . If  $f_\theta$  is close to  $g$ , then  $nD(f_{\theta_g}, g)$  will be close to  $nD(g, g)$ , and we may hope that  $nD(f_{\hat{\theta}}, g)$  is close to both. But if further parameters do not give a worthwhile reduction in  $D(f_{\theta_g}, \theta)$ , adding degrees of freedom gives  $\hat{\theta}$  more latitude to miss  $\theta_g$ , and the corresponding increase in  $D(f_{\hat{\theta}}, g)$  will tend to outweigh any decrease in  $D(f_{\theta_g}, g)$ . To remove dependence on  $\hat{\theta}$ , we average over its distribution, giving

$$E_g \left( E_g^+ \left[ \sum_{j=1}^n \log \left\{ \frac{g(Y_j^+)}{f(Y_j^+; \hat{\theta})} \right\} \right] \right) = nE_g \{ D(f_{\hat{\theta}}, g) \}; \quad (4.55)$$

the outer expectation is over the distribution of  $\hat{\theta}$ , independent of  $Y^+$ . Taylor series expansion shows that  $\log f(y; \hat{\theta})$  approximately equals

$$\log f(y; \theta_g) + (\hat{\theta} - \theta_g)^T \frac{\partial \log f(y; \theta_g)}{\partial \theta} + \frac{1}{2} (\hat{\theta} - \theta_g)^T \frac{\partial^2 \log f(y; \theta_g)}{\partial \theta \partial \theta^T} (\hat{\theta} - \theta_g),$$

and as  $\theta_g$  minimizes  $D(f_\theta, g)$ ,

$$\int \frac{\partial \log f(y; \theta_g)}{\partial \theta} g(y) dy = 0.$$

Hence

$$\begin{aligned} nD(f_{\hat{\theta}}, g) &= n \int \log \left\{ \frac{g(y)}{f(y; \hat{\theta})} \right\} g(y) dy \\ &\doteq nD(f_{\theta_g}, g) + \frac{1}{2} \text{tr} \{ (\hat{\theta} - \theta_g)(\hat{\theta} - \theta_g)^T I_g(\theta_g) \}, \end{aligned}$$

where  $I_g(\theta_g)$  is given at (4.53) and we have used the fact that the trace of a scalar is itself. At the end of Section 4.6 we discussed likelihood estimation under the wrong model, and saw that for regular models  $\hat{\theta}$  is asymptotically normal with mean  $\theta_g$  and variance matrix  $I_g(\theta_g)^{-1} K(\theta_g) I_g(\theta_g)^{-1}$ , where  $K(\theta_g)$  too is given at (4.53); both  $I_g(\theta_g)$  and  $K(\theta_g)$  are positive definite. Hence

$$nE_g \{ D(f_{\hat{\theta}}, g) \} \doteq nD(f_{\theta_g}, g) + \frac{1}{2} \text{tr} \{ I_g(\theta_g)^{-1} K(\theta_g) \}, \quad (4.56)$$

where the second term penalizes the dimension  $p$  of  $\theta$ . The first term here is  $O(n)$ , but as both  $I_g(\theta)$  and  $K(\theta)$  are  $O(n)$ , the second term is  $O(p)$ . When  $f_\theta$  is correct and regular,  $I_g(\theta_g) = K(\theta_g)$  so  $\text{tr} \{ I_g(\theta_g)^{-1} K(\theta_g) \} = p$ .

To build an estimator of (4.56), note first that the term  $\int \log g(y) g(y) dy$  is constant and can be ignored. Now  $\ell(\hat{\theta}) = \ell(\theta_g) + \{\ell(\hat{\theta}) - \ell(\theta_g)\}$ , so

$$\begin{aligned} E_g\{-\ell(\hat{\theta})\} &= -E_g\left\{\ell(\theta_g) + \frac{1}{2}W(\theta_g)\right\} \\ &\doteq nD(f_{\theta_g}, g) - \frac{1}{2}\text{tr}\{I(\theta_g)^{-1}K(\theta_g)\} - n \int \log g(y) g(y) dy, \end{aligned}$$

where we have used the fact that under the wrong model, the likelihood ratio statistic  $W(\theta_g)$  has mean approximately  $\text{tr}\{I(\theta_g)^{-1}K(\theta_g)\}$ . Hence  $-\ell(\hat{\theta})$  tends to underestimate  $nD(f_{\theta_g}, g) - n \int \log g(y) g(y) dy$ . On reflection this is obvious, because  $\ell(\hat{\theta}) \geq \ell(\theta_g)$  by definition of  $\hat{\theta}$ . As  $p$  increases, so will the extent of overestimation.

An estimator of (4.56) is  $-\ell(\hat{\theta}) + c$ , where  $c$  estimates  $\text{tr}\{I(\theta_g)^{-1}K(\theta_g)\}$ . Two possible choices of  $c$  are  $p$  and  $\text{tr}(\hat{J}^{-1}\hat{K})$ , where  $\hat{J}$  and  $\hat{K}$  are defined at (4.54), and these lead to

$$\text{AIC} = 2\{-\ell(\hat{\theta}) + p\}, \quad \text{NIC} = 2\{-\ell(\hat{\theta}) + \text{tr}(\hat{J}^{-1}\hat{K})\}; \quad (4.57)$$

another possibility derived in Section 11.3.1 is  $\text{BIC} = -2\ell(\hat{\theta}) + p \log n$ . The model is chosen to minimize AIC, say, with the factor 2 putting differences of AIC on the same scale as likelihood ratio statistics. In practice AIC, BIC, and NIC are used far beyond random samples.

For insight into properties of AIC, suppose that by rare good fortune we fit the true and a correct model, getting maximized log likelihoods  $\hat{\ell}_g$  and  $\ell(\hat{\theta})$  with  $q$  and  $p$  parameters respectively, and  $p > q$ . We prefer  $f_\theta$  to  $g$  if  $\ell(\hat{\theta}) - p > \hat{\ell}_g - q$ , but as  $g$  is nested within  $f_\theta$ , properties of the likelihood ratio statistic give

$$\Pr\{\ell(\hat{\theta}) - p > \hat{\ell}_g - q\} \doteq \Pr\{\chi_{p-q}^2 > 2(p - q)\}.$$

For every large  $n$ , and with  $p - q = 1, 2, 4$  and  $10$ ,  $g$  is selected with probability 0.84, 0.86, 0.91 and 0.97. Hence model selection using AIC is inconsistent:

$$\Pr(\text{true model selected}) \not\rightarrow 1 \quad \text{as } n \rightarrow \infty.$$

In applications many models would be fitted, and the probability of selecting the true one might be much lower than these calculations suggest.

Modification of this argument shows that NIC also gives an inconsistent procedure. For consistent model selection differences of the penalty must lie between  $O(1)$  and  $O(n)$  — for example,  $O(\log n)$  — but in practice the true model is rarely among those fitted and finite-sample properties are more important. BIC does give consistent model selection when  $f_\theta$  is correct, but in finite samples it typically leads to underfitting because it tends to suggest too parsimonious a model.

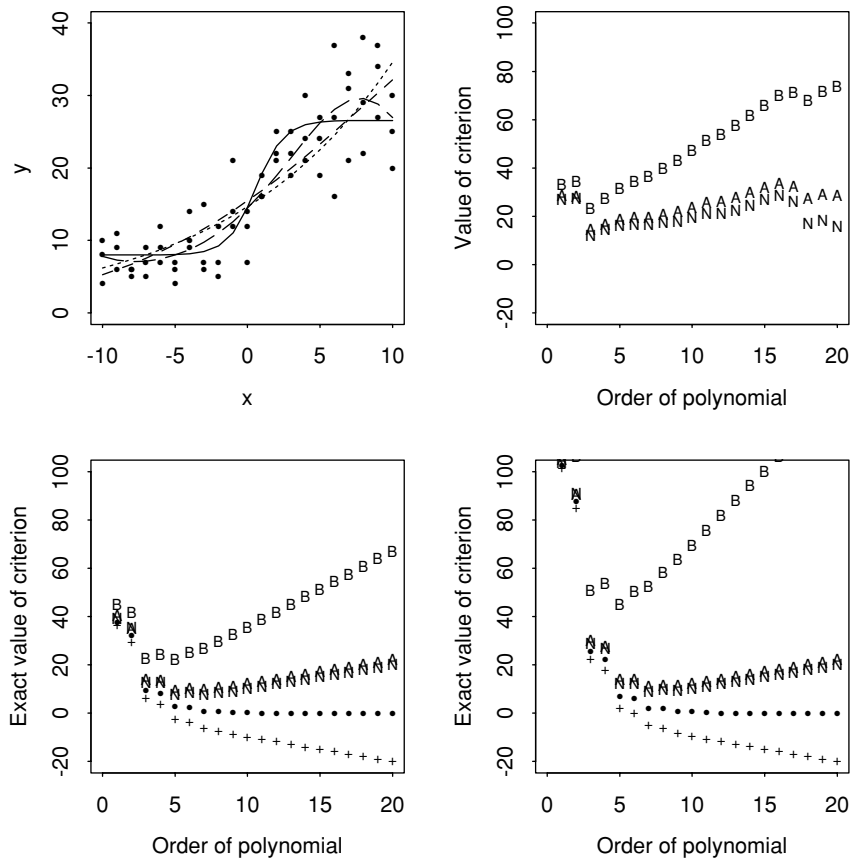
If the candidate model  $f_\theta$  is not correct, then

$$E_g\{\hat{\ell}_g - \ell(\hat{\theta})\} \doteq nD(f_{\theta_g}, g) > 0,$$

so the weak law of large numbers implies that  $\Pr\{\hat{\ell}_g - q > \ell(\hat{\theta}) - p\} \rightarrow 1$  as  $n \rightarrow \infty$  for fixed  $p$ . Hence with enough data we can always distinguish the true model from a fixed incorrect one.

AIC was introduced by Akaike (1973) and is known as Akaike's information criterion. Hirotugu Akaike (1927–) was educated in Tokyo and worked at the Institute of Statistical Mathematics. He has made important contributions to time series and model selection, and also to production engineering; see Findley and Parzen (1995). NIC and BIC are the *network information criterion*, and *Bayes' information criterion*. They may be modified to improve their behaviour for particular models.

**Figure 4.10** Model selection using likelihood criteria. Upper left:  $21n$  observations (blobs) with true mean (solid) and polynomial fits  $r = 1, 2, 3$  (dots, small dashes, large dashes);  $n = 3$ . Upper right: empirical versions of AIC, BIC and NIC for data on left. All are maximized with  $r = 3$ . Lower left: twice expected log likelihood  $2E_g\{\ell(\theta_g)\}$  (blobs) and theoretical versions of AIC, BIC and NIC for the panel above. The crosses show how  $2E_g\{\ell(\theta)\}$  increases with the dimension of the fitted model. Lower right: as lower left panel, but with  $n = 8$  observations at each value of  $x$ .



**Example 4.46 (Poisson model)** We illustrate this discussion with data whose mean  $\mu(x) = 8 \exp q(x)$  is shown in the upper left panel of Figure 4.10, together with observations generated by taking  $n = 3$  independent Poisson variables with means  $\mu(-10), \mu(-9), \dots, \mu(10)$ ;  $21n$  variables in all. This is the true model  $g$ .

We fit candidate models  $f_\theta$  with Poisson variables having means  $\lambda(x) = \exp(\theta_0 + \theta_1 x + \dots + \theta_r x^r)$ . The dimension is  $p = r + 1$ , and taking  $r = 1, \dots, 19$  gives increasingly complex incorrect models, because  $q(x) = 1.2e^x/(1 + e^x)$  is not polynomial. A polynomial with  $r = 20$  terms can mimic  $q(x)$  exactly at  $x = -10, -9, \dots, 10$ , however, so taking  $r = 20$  is correct but hardly parsimonious. The difference between the linear and the quadratic fits shown in the upper left panel of Figure 4.10 is small, but adding a cubic term seems to improve the fit.

The upper right panel shows AIC, NIC, and BIC for these data. All three suggest the choice of  $r = 3$ , but BIC penalizes complexity much more drastically than the others. In practice one should not only look at such a graph, but also examine any models for which the chosen criterion is close to the optimum.

To see the theoretical quantities estimated by AIC, BIC, and NIC, note that the data here comprise  $n$  variables  $Y_{1,x}, \dots, Y_{n,x}$  at each value of  $x$ . The log likelihood for an

incorrect model which takes  $Y_{j,x}$  to be Poisson with mean  $\lambda(x)$  is

$$\ell(\theta) \equiv \sum_{j=1}^n \sum_{x=-10}^{10} \{Y_{j,x} \log \lambda(x) - \lambda(x)\}.$$

Now  $E_g(Y_{j,x}) = \mu(x)$ , so  $E_g\{\ell(\theta_g)\} = n \sum_x \{\mu(x) \log \lambda(x) - \lambda(x)\}$ ; the values of  $\theta_0, \dots, \theta_r$  that maximize this give  $E_g\{\ell(\theta_g)\}$ . The blobs in the lower left panel of the figure show how  $-2E_g\{\ell(\theta_g)\}$  depends on  $r$ . Initially there are big decreases, but after  $r = 5$  adding further parameters is barely worthwhile. The crosses show how  $-2E_g\{\ell(\hat{\theta})\}$  depends on  $r$ : not penalizing the log likelihood would lead to choosing  $r = 20$ . The exact values of AIC, BIC, and NIC all indicate  $r = 5$ . However BIC indicates fits about equally good for  $r = 5$  and the simpler model  $r = 3$ , whereas for AIC and NIC the best fit is similar to that with the more complex model  $r = 7$ . The penalty applied by BIC is substantially larger than for the others, which are very similar. These functions are what is being estimated in the upper right panel.

To see the effect of increased sample size, the lower right panel of the figure shows exact values of  $-E_g\{\ell(\theta_g)\}$ , AIC, BIC and NIC when  $n = 8$ . The jumps in  $-E_g\{\ell(\theta_g)\}$  are larger than with  $n = 3$ , and with this larger sample  $r = 7$  seems appreciably better than  $r = 5$ : more data make it worthwhile to fit more complex models, because we can distinguish them more clearly. Enormous values of  $n$ , however, are required to separate  $r = 10$  and  $r = 20$  reliably:  $-E_g\{\ell(\theta_g)\} \doteq -0.08$  when  $n = 3$  and  $r = 10$ , so even a sample with  $n = 100$  might indicate that  $r = 10$ . With  $n = 8$ , BIC is much more peaked than when  $m = 3$ , so the value  $r = 5$  it indicates is better determined, even though the more complex choice  $r = 7$  seems sensible on the basis of  $-E_g\{\ell(\theta_g)\}$ . By contrast the penalties applied by AIC and NIC are unchanged. Both indicate  $r = 7$ , but evidently their empirical counterparts might have minima anywhere in the range  $r = 5, \dots, 20$ .

The closeness of NIC to AIC in this context leads us to ignore NIC below. ■

**Example 4.47 (Spring failure data)** To analyze the full set of spring failure data in Example 1.2, suppose that the data have Weibull densities whose parameters  $\alpha$  and  $\theta$  may depend on stress  $x$ , and consider the models:

- $M_1$ : unconnected values of  $\alpha$  and  $\theta$  at each stress, with  $p = 12$  parameters;
- $M_2$ : a common value of  $\alpha$  but unconnected  $\theta$  at each stress, with  $p = 7$ ;
- $M_3$ : a common value of  $\alpha$ , and  $\theta = (\beta x)^{-1}$ , with  $p = 2$ ; and
- $M_4$ : common values of  $\alpha$  and  $\theta$  at every stress, with  $p = 2$ .

The nesting structure of these models is  $M_4, M_3 \subseteq M_2 \subseteq M_1$ , where  $\subseteq$  means ‘is nested within’; neither  $M_3$  nor  $M_4$  is nested within the other. We anticipate from Figure 1.2 that  $M_4$  will fit the data very poorly.

To deal with the censoring at lower stresses, note that Example 4.20 implies that the likelihood for a censored Weibull random sample  $y_1, \dots, y_n$  is

$$\prod_u \frac{\alpha}{\theta} \left( \frac{y_j}{\theta} \right)^{\alpha-1} \exp \left\{ - \left( \frac{y_j}{\theta} \right)^\alpha \right\} \prod_c \exp \left\{ - \left( \frac{y_j}{\theta} \right)^\alpha \right\},$$

**Table 4.4** Model selection for spring failure data.

| Model | $p$ | Maximized log likelihood | AIC   | BIC   |
|-------|-----|--------------------------|-------|-------|
| $M_1$ | 12  | −360.40                  | 744.8 | 769.9 |
| $M_2$ | 7   | −378.90                  | 771.8 | 786.5 |
| $M_3$ | 2   | −411.50                  | 827.0 | 831.2 |
| $M_4$ | 2   | −460.56                  | 925.1 | 929.3 |

**Table 4.5** Parameter estimates and standard errors based on observed information for model  $M_1$  for the spring failure data, fitting separate parameters at each stress.

| Stress $x_s$        | 700          | 750         | 800         | 850         | 900         | 950         |
|---------------------|--------------|-------------|-------------|-------------|-------------|-------------|
| $\hat{\alpha}$ (SE) | 1.59 (0.82)  | 1.44 (0.39) | 1.69 (0.39) | 7.36 (1.85) | 5.37 (1.23) | 5.97 (2.13) |
| $\hat{\theta}$ (SE) | 18044 (7295) | 6609 (1566) | 907 (180)   | 372 (16.9)  | 232 (14.5)  | 181 (10.2)  |

where  $\prod_u$  and  $\prod_c$  denote products over uncensored and censored data. We regard all observations as independent, with parameters  $\alpha_s$  and  $\theta_s$  at stress  $x_s$ , and with indicator  $d_{sj}$  equalling one if the  $j$ th observation at stress  $x_s$ ,  $y_{sj}$ , is uncensored and equalling zero otherwise. The overall likelihood is then

$$\prod_{s=1}^6 \prod_{j=1}^{10} \left\{ \frac{\alpha_s}{\theta_s} \left( \frac{y_{sj}}{\theta_s} \right)^{\alpha_s - 1} \right\}^{d_{sj}} \exp \left\{ - \left( \frac{y_{sj}}{\theta_s} \right)^{\alpha_s} \right\}.$$

Table 4.4 shows that  $M_4$  fits much worse than any of the other models, and  $M_3$ , which has the same number of parameters, is more promising. Evidently  $M_1$  is best by a large margin.

Table 4.5 gives estimates for  $M_1$ , with standard errors based on observed information. The values of  $\hat{\alpha}$  depend strongly on the stress, and suggest one value of  $\alpha$  at the three lower stresses and another at the higher ones. The standard errors are useless at the lower stresses, with heavy censoring: with so little information any inference will be very uncertain.

The model with six separate values of  $\theta_s$  and two values of  $\alpha$ , one for the three upper and one for the three lower levels of  $x_s$ , has maximized log likelihood −360.92, AIC = 737.8, and BIC = 754.6, so it beats  $M_1$ . A plot of  $\log \hat{\theta}_s$  against log stress is close to a straight line, suggesting a three-parameter model with  $\theta = 1/(\beta x)$  and two different levels for  $\alpha$ , but smooth dependence of  $\alpha$  on  $x$  is both more plausible and more useful for prediction: what value of  $\alpha$  is suitable at stress 825 N/mm<sup>2</sup>? Absent more knowledge about the purpose of the experiment, we proceed no further. ■

Further discussion of model selection and the related topic of model uncertainty may be found in Sections 8.7.3 and 11.2.4.

## Exercises 4.7

- 1 Show that both sides of (4.56) are invariant to 1–1 reparametrizations  $\theta = \theta(\phi)$ . Why is this important?
- 2 Use AIC and BIC to compare the models fitted in Example 4.34.

- 3 Two densities for counts  $y = 0, 1, \dots$  are the Poisson  $\theta^y e^{-\theta} / y!$ ,  $\theta > 0$  and the geometric  $\pi(1 - \pi)^y$ ,  $0 < \pi < 1$ ; their means are  $\theta$  and  $\pi^{-1} - 1$ . Show that if the true model is one but the other is fitted, the 'least bad' parameter value matches the means. How easy is it to tell them apart when the data are Poisson with  $\theta = 1, 5, 10$ , and when the data are geometric?
- 4 Consider a regular penalized log likelihood  $\ell(\hat{\theta}) - c_n$ , where  $c_n \xrightarrow{P} c$  as  $n \rightarrow \infty$ ,  $\ell(\theta)$  is based on a correct model,  $\theta$  has dimension  $p$ , and  $\ell_g$  is the log likelihood for the true model. Show that  $2\{\ell(\hat{\theta}) - c_n - \ell_g\} \xrightarrow{D} \chi_p^2 - 2c$ , and deduce that the probability of selecting the true model is  $\Pr(\chi_p^2 \leq 2c)$ . Hence show that while model selection based on BIC is consistent, that based on AIC is not.

## 4.8 Bibliographic Notes

The ideas of likelihood, information, sufficiency and efficient estimation were developed in a remarkable series of papers by R. A. Fisher in the 1920s and 1930s. Most introductions to mathematical statistics contain this core material. A recent excellent account is Knight (2000). The approach here is influenced by Silvey (1970), Edwards (1972), Cox and Hinkley (1974) and Kalbfleisch (1985). See also Barndorff-Nielsen and Cox (1994) and Pace and Salvan (1997).

The literature on non-regular models is diffuse. See Self and Liang (1987), Smith (1985, 1989b, 1994) and Cheng and Traylor (1995), or Davison (2001) for a partial review. Parameter redundancy is discussed by Catchpole and Morgan (1997), with applications to capture-recapture models.

Model selection and uncertainty are topics of current research interest, with much heat generated by Chatfield (1995) and discussants. For a longer discussion, see Burnham and Anderson (2002).

## 4.9 Problems

- 1 The *logistic density* with location and scale parameters  $\mu$  and  $\sigma$  is

$$f(y; \mu, \sigma) = \frac{\exp\{(y - \mu)/\sigma\}}{\sigma[1 + \exp\{(y - \mu)/\sigma\}]^2}, \quad -\infty < y < \infty, \quad -\infty < \mu < \infty, \sigma > 0.$$

- (a) If  $Y$  has density  $f(y; \mu, 1)$ , show that the expected information for  $\mu$  is  $1/3$ .
- (b) Instead of observing  $Y$ , we observe the indicator  $Z$  of whether or not  $Y$  is positive. When  $\sigma = 1$ , show that the expected information for  $\mu$  based on  $Z$  is  $e^\mu / (1 + e^\mu)^2$ , and deduce that the maximum efficiency of sampling based on  $Z$  rather than  $Y$  is  $3/4$ . Why is this greatest at  $\mu = 0$ ?
- (c) Find the expected information  $I(\mu, \sigma)$  based on  $Y$  when  $\sigma$  is unknown. Without doing any calculations, explain why both parameters cannot be estimated based only on  $Z$ .
- 2 Let  $\psi(\theta)$  be a 1–1 transformation of  $\theta$ , and consider a model with log likelihoods  $\ell(\theta)$  and  $\ell^*(\psi)$  in the two parametrizations respectively;  $\ell$  has a unique maximum at which the likelihood equation is satisfied. Show that

$$\frac{\partial \ell^*(\psi)}{\partial \psi_r} = \frac{\partial \theta^T}{\partial \psi_r} \frac{\partial \ell(\theta)}{\partial \theta}, \quad \frac{\partial^2 \ell^*(\psi)}{\partial \psi_r \partial \psi_s} = \frac{\partial \theta^T}{\partial \psi_r} \frac{\partial^2 \ell(\theta)}{\partial \theta \partial \theta^T} \frac{\partial \theta}{\partial \psi_s} + \frac{\partial^2 \theta^T}{\partial \psi_r \partial \psi_s} \frac{\partial \ell(\theta)}{\partial \theta}$$

and deduce that

$$I^*(\psi) = \frac{\partial \theta^T}{\partial \psi} I(\theta) \frac{\partial \theta}{\partial \psi^T},$$

but that a similar equation holds for observed information only when  $\theta = \hat{\theta}$ .

- 3 A location-scale model with parameters  $\mu$  and  $\sigma$  has density

$$f(y; \mu, \sigma) = \frac{1}{\sigma} g\left(\frac{y - \mu}{\sigma}\right), \quad -\infty < y < \infty, \quad -\infty < \mu < \infty, \sigma > 0.$$

- (a) Show that the information in a single observation has form

$$i(\mu, \sigma) = \sigma^{-2} \begin{pmatrix} a & b \\ b & c \end{pmatrix},$$

and express  $a$ ,  $b$ , and  $c$  in terms of  $h(\cdot) = \log g(\cdot)$ . Show that  $b = 0$  if  $g$  is symmetric about zero, and discuss the implications for the joint distribution of the maximum likelihood estimators  $\hat{\mu}$  and  $\hat{\sigma}$  when  $g$  is regular.

(b) Find  $a$ ,  $b$ , and  $c$  for the normal density  $(2\pi)^{-1/2} e^{-u^2/2}$  and the log-gamma density  $\exp(\kappa u - e^u)/\Gamma(\kappa)$ , where  $\kappa > 0$  is known.

- 4 Let  $y_1, \dots, y_n$  be a random sample from  $f(y; \mu, \sigma) = (2\sigma)^{-1} \exp(-|y - \mu|/\sigma)$ ,  $-\infty < y, \mu < \infty, \sigma > 0$ ; this is the Laplace density.

- (a) Write down the log likelihood for  $\mu$  and  $\sigma$  and by showing that

$$\frac{d}{d\mu} \sum |y_j - \mu| = \#\{y_j < \mu\} - \#\{y_j > \mu\} = n - 2R,$$

where  $R = \#\{y_j > \mu\}$ , show that for any fixed  $\sigma > 0$  the maximum likelihood estimate of  $\mu$  is  $\hat{\mu} = \text{median}\{y_j\}$ , and deduce that the maximum likelihood estimate of  $\sigma$  is the mean absolute deviation  $\hat{\sigma} = n^{-1} \sum |y_j - \hat{\mu}|$ .

(b) Use the results of Section 2.3 to show that in large samples  $\hat{\mu} \sim N(\mu, \sigma^2/n)$  and  $\hat{\sigma} \xrightarrow{p} \sigma$ . Hence give an approximate confidence interval for the difference of means based on the data in Table 1.1.

(c) Is this a regular model for maximum likelihood estimation?

- 5 Show that the expected information for a random sample of size  $n$  from the Weibull density in Example 4.4 is

$$I(\theta, \alpha) = n \begin{pmatrix} \alpha^2/\theta^2 & -\psi(2)/\theta \\ -\psi(2)/\theta & \{1 + \psi'(2) + \psi(2)^2/\alpha^2\} \end{pmatrix},$$

where  $\psi(z) = d \log \Gamma(z)/dz$ .

Given that  $\psi(2) = 0.42278$  and  $\psi'(2) = 0.64493$ , show that

$$I^{-1}(\theta, \alpha) = n^{-1} \begin{pmatrix} 1.108\theta^2/\alpha^2 & 0.257\theta \\ 0.257\theta & 0.608\alpha^2 \end{pmatrix}.$$

Hence find standard errors based on expected information for the estimates in the last column of Table 4.5. What problem arises in a similar calculation for the column with stress  $x = 700$ ?

- 6 Persons who catch an infectious disease either die almost at once during its initial phase, or live an exponential time; denote the survival time  $Y$  and declare that  $Y = 0$  if death occurs in the initial phase. Explain why the likelihood can be written as a product of terms of form

$$(1 - p)^{1-I} \times \{p\theta^{-1} \exp(-Y/\theta)\}^I, \quad 0 < p < 1, \theta > 0,$$

where  $I$  is an indicator of survival beyond the initial phase. Give interpretations of  $p$  and  $\theta$ .

# means 'the number of times'.

|                                               |                                               |                                               |                                               |
|-----------------------------------------------|-----------------------------------------------|-----------------------------------------------|-----------------------------------------------|
| MMM<br>953<br>(1 + $\theta$ ) <sup>2</sup> /8 | MMF<br>914<br>(1 - $\theta$ ) <sup>2</sup> /8 | MFM<br>846<br>(1 - $\theta$ ) <sup>2</sup> /8 | MFF<br>845<br>(1 - $\theta$ ) <sup>2</sup> /8 |
| FMM<br>825<br>(1 - $\theta$ ) <sup>2</sup> /8 | FMF<br>748<br>(1 - $\theta$ ) <sup>2</sup> /8 | FFM<br>852<br>(1 - $\theta$ ) <sup>2</sup> /8 | FFF<br>923<br>(1 + $\theta$ ) <sup>2</sup> /8 |

**Table 4.6** Frequencies of eight possible sequences, with their probabilities based on a model in which the probability of a male at first birth is  $\frac{1}{2}$  but the probability that the next child has the same sex is  $(1 + \theta)/2$ , for 6906 three-child families.

Given data  $(i_1, y_1), \dots, (i_n, y_n)$  on the survival of  $n$  persons, show that the log likelihood has form

$$\ell(p, \theta) = r \log p + (n - r) \log(1 - p) - r \log \theta - \theta^{-1} \sum_{j=1}^n i_j y_j,$$

where  $r = \sum i_j$ , and hence find the maximum likelihood estimators of  $p$  and  $\theta$ , together with the observed and expected information matrices.

Comment on the form of the information matrices and give approximate 95% confidence intervals for the parameters.

- 7 The administrator of a private hospital system is comparing legal claims for damages against two of the hospitals in his system. In the last five years at hospital A the following 19 claims (\$, inflation-adjusted) have been paid:

|     |       |       |      |       |      |      |      |      |      |
|-----|-------|-------|------|-------|------|------|------|------|------|
| 59  | 172   | 4762  | 1000 | 2885  | 1905 | 7094 | 6259 | 1950 | 1208 |
| 882 | 22793 | 30002 | 55   | 32591 | 853  | 2153 | 738  | 311  |      |

At hospital B, in the same period, there were 16 claims settled out of court for \$800 or less, and 16 claims settled in court for

|       |       |      |      |      |      |      |       |
|-------|-------|------|------|------|------|------|-------|
| 36539 | 3556  | 1194 | 1010 | 5000 | 1370 | 1494 | 55945 |
| 19772 | 31992 | 1640 | 1985 | 2977 | 1304 | 1176 | 1385  |

The proposed model is that claims within a hospital follow an exponential distribution. How would you check this for hospital A?

Assuming that the exponential model is valid, set up the equations for calculating maximum likelihood estimates of the means for hospitals A and B. Indicate how you would solve the equation for hospital B.

The maximum likelihood estimate for hospital B is 5455.7. If a common mean is fitted for both hospitals, the maximum likelihood estimate is 5730.6. Use these results to calculate the likelihood ratio statistic for comparing the mean claims of the two hospitals, and interpret the answer.

- 8 Are the sexes of successive children within a family dependent? Table 4.6 gives for 6906 three-child families the frequencies of the eight possible sequences, with their probabilities based on a model in which the probability of a male at first birth is  $\frac{1}{2}$  but the probability that the next child has the same sex is  $(1 + \theta)/2$ ; here  $-1 < \theta < 1$ . What is special about the model in which  $\theta = 0$ ?
- (a) If  $y_{\text{MMM}}$ ,  $y_{\text{MMF}}$  and so forth denote the numbers of families with orders MMM, MMF, in a sample of  $m$  families, write down the likelihood for  $\theta$  and show that the numbers of consecutive pairs MM and FF is a sufficient statistic.
- (b) Obtain the score statistic and observed information, and verify that for the data above the maximum likelihood estimate is  $\hat{\theta} \doteq 0.04$  with standard error 0.0085. Give a 95% confidence interval for  $\theta$ . Discuss.
- (c) Is it true that the probability that the first child is male is  $\frac{1}{2}$ ? Suggest how you might generalize the model to allow for (i) this probability being unequal to  $\frac{1}{2}$ , and (ii) the probability that a female follows a female being unequal to the probability that a male follows a male. Write down the probabilities for Table 4.6. If you are feeling energetic, conduct a full likelihood analysis of the data.

- 9 Let  $Y_{ij}, j = 1, \dots, n_i, i = 1, \dots, k$ , be independent normal random variables with means  $\mu_i$  and variances  $\sigma_i^2$ , and  $n_i \geq 2$ ; set  $\bar{Y}_i = n_i^{-1} \sum_j Y_{ij}$ .  
 (a) Show that the likelihood ratio statistic for  $\sigma_1^2 = \dots = \sigma_k^2 = \sigma^2$ , with no restrictions on the  $\mu_i$ , is given by

$$W = \sum_{i=1}^k n_i \log \left( \frac{\bar{\sigma}^2}{\hat{\sigma}_i^2} \right), \quad \bar{\sigma}^2 = \sum_{i=1}^k n_i \hat{\sigma}_i^2 / \sum_{i=1}^k n_i, \quad \hat{\sigma}_i^2 = n_i^{-1} \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_i)^2, \quad (4.58)$$

and give its approximate distribution for large  $n_i$ .

- (b) A modification to  $W$  to improve its behaviour in small samples replaces the  $n_i$  in (4.58) with  $v_i = n_i - 1$ . Use the modified statistic to check the homogeneity of the variances for the data in Table 1.2 at the three highest stresses, and comment.  
 (c) If  $k = 2$  show that a test of  $\sigma_1^2 = \sigma_2^2$  may be based on  $\hat{\sigma}_1^2 / \hat{\sigma}_2^2$ , and give its exact distribution.  
 (d) If  $n_1 = \dots = n_k = 3$ , show that  $\hat{\sigma}_i^2$  may be written as  $2\sigma_i^2 E_i / 3$ , where the  $E_i$  are independent exponential random variables with unit means. Explain how a plot of the ordered  $\hat{\sigma}_i^2$  against exponential plotting positions can be used to check variance homogeneity and to assess the adequacy of the assumption of normality. What could be done if  $n_1 = \dots = n_k = 2$ ?
- 10 In a normal linear model through the origin, independent observations  $Y_1, \dots, Y_n$  are such that  $Y_j \sim N(\beta x_j, \sigma^2)$ . Show that the log likelihood for a sample  $y_1, \dots, y_n$  is

$$\ell(\beta, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{j=1}^n (y_j - \beta x_j)^2.$$

Deduce that the likelihood equations are equivalent to  $\sum x_j(y_j - \hat{\beta}x_j) = 0$  and  $\hat{\sigma}^2 = n^{-1} \sum (y_j - \hat{\beta}x_j)^2$ , and hence find the maximum likelihood estimates  $\hat{\beta}$  and  $\hat{\sigma}^2$  for data with  $x = (1, 2, 3, 4, 5)$  and  $y = (2.81, 5.48, 7.11, 8.69, 11.28)$ .

Show that the observed information matrix evaluated at the maximum likelihood estimates is diagonal and use it to obtain approximate 95% confidence intervals for the parameters. Plot the data and your fitted line  $y = \hat{\beta}x$ . Say whether you think the model is correct, with reasons. Discuss the adequacy of the normal approximations in this example.

- 11 In some measurements of  $\mu$ -meson decay by L. Janossy and D. Kiss the following observations were recorded from a four channel discriminator: in 844 cases the decay time was less than 1 second; in 467 cases the decay time was between 1 and 2 seconds; in 374 cases the decay time was between 2 and 3 seconds; and in 564 cases the decay time was greater than 3 seconds.

Assuming that decay time has density  $\lambda e^{-\lambda t}, t > 0, \lambda > 0$ , find the likelihood for  $\lambda$ . Find the maximum likelihood estimate,  $\hat{\lambda}$ , find its standard error, and give a 95% confidence interval for  $\lambda$ .

Check whether the data are consistent with an exponential distribution by comparing the observed and fitted frequencies.

- 12 A family has two children  $A$  and  $B$ . Child  $A$  catches an infectious disease  $\mathcal{D}$  which is so rare that the probability that  $B$  catches it other than from  $A$  can be ignored. Child  $A$  is infectious for a time  $U$  having probability density function  $\alpha e^{-\alpha u}, u \geq 0$ , and in any small interval of time  $[t, t + \delta t]$  in  $[0, U)$ ,  $B$  will catch  $\mathcal{D}$  from  $A$  with probability  $\beta \delta t + o(\delta t)$ , where  $\alpha, \beta > 0$ . Calculate the probability  $\rho$  that  $B$  does catch  $\mathcal{D}$ . Show that, in a family where  $B$  is actually infected, the density function of the time to infection is  $\gamma e^{-\gamma t}, t \geq 0$ , where  $\gamma = \alpha + \beta$ .

An epidemiologist observes  $n$  independent similar families, in  $r$  of which the second child catches  $\mathcal{D}$  from the first, at times  $t_1, \dots, t_r$ . Write down the likelihood of the data as the product of the probability of observing  $r$  and the likelihood of the fixed sample  $t_1, \dots, t_r$ . Find the maximum likelihood estimators  $\hat{\rho}$  and  $\hat{\gamma}$  of  $\rho$  and  $\gamma$ , and the asymptotic variance of  $\hat{\gamma}$ .

|        | Round      | Wrinkled   |
|--------|------------|------------|
| Yellow | 315 (9/16) | 101 (3/16) |
| Green  | 108 (3/16) | 32 (1/16)  |

**Table 4.7** Mendel's data on four kinds of pea seeds (theoretical probability) (Kendall and Stuart, 1973, p. 439).

- 13 Counts  $y_1, y_2, y_3$  are observed from a multinomial density

$$\Pr(Y_1 = y_1, Y_2 = y_2, Y_3 = y_3) = \frac{m!}{y_1!y_2!y_3!} \pi_1^{y_1} \pi_2^{y_2} \pi_3^{y_3}, \quad y_r = 0, \dots, m, \quad \sum y_r = m,$$

where  $0 < \pi_1, \pi_2, \pi_3 < 1$  and  $\pi_1 + \pi_2 + \pi_3 = 1$ . Show that the maximum likelihood estimate of  $\pi_r$  is  $y_r/m$ .

It is suspected that in fact  $\pi_1 = \pi_2 = \pi$ , say, where  $0 < \pi < 1$ . Show that the maximum likelihood estimate of  $\pi$  is then  $\frac{1}{2}(y_1 + y_2)/m$ .

Give the likelihood ratio statistic for comparing the models, and state its asymptotic distribution.

- 14 In experiments on cross-breeding peas, Mendel noted frequencies of seeds of different kinds when crossing plants with round yellow seeds and plants with wrinkled green seeds. His data and the theoretical probabilities according to his theory of inheritance are in Table 4.7.

Calculate the expected values under the model, and check the adequacy of the theory using the likelihood ratio and Pearson statistics  $W$  and  $P$ . How would the degrees of freedom change if the table was treated as a two-way contingency table with unknown probabilities?

- 15 The negative binomial density may be written

$$f(y; \mu, \psi) = \frac{\Gamma(y + \psi^{-1})}{\Gamma(\psi^{-1})y!} \frac{(\psi\mu)^y}{(1 + \psi\mu)^{y+1/\psi}}, \quad y = 0, 1, \dots, \quad \mu, \psi > 0;$$

its limit as  $\psi \rightarrow 0$  is the Poisson density. Taylor series expansion about  $\psi = 0$  shows that  $\log f(y; \mu, \psi)$  is

$$y \log \mu - \mu - \log y! + \frac{\psi}{2} \{(y - \mu)^2 - y\} + \frac{\psi^2}{12} \{6\mu^2 y - 4\mu^3 - y(1 - 3y + 2y^2)\} \\ + \frac{\psi^3}{12} \{3\mu^4 - 4\mu^3 y + y^2(y - 1)^2\} + O(\psi^4).$$

Find the expected information  $I(\mu, \psi)$  when  $\psi = 0$ , and show that the asymptotic distribution of the score  $\partial \ell(\hat{\mu}_\psi, \psi)/\partial \psi$  based on a sample of size  $n$  is then  $N(0, n\mu^2/2)$ . Discuss properties of the likelihood ratio statistic for comparison of Poisson and negative binomial models.

- 16 A possible model for the data in Table 11.7 is that pumps are independent, and that the failures for the  $j$ th pump have the Poisson distribution with mean  $\lambda x_j$ , where  $x_j$  is the operating hours (1000s). Find the maximum likelihood estimate of  $\lambda$  under this model and give its standard error. Construct the likelihood ratio statistic to compare this with the model in which all the pumps have different rates. Justifying your reasoning, say whether you expect this statistic to have an approximate  $\chi^2$  distribution.
- 17 If  $y_1, \dots, y_n$  is a random sample with density  $\sigma^{-1} f\{(y - \mu)/\sigma; \lambda\}$ , where  $f$  is the skew-normal density function (Problem 3.6), write down the log likelihood for  $\mu, \sigma$ , and  $\lambda$ , and investigate likelihood inference for this model.

Gregor Mendel (1823–1884) was the second child of farmers in Brunn, Moravia. He showed early promise but his family's poverty meant that he could continue his education only as an Augustinian monk. His work on pea plants was begun out of curiosity; it took seven years to amass enough data to formulate his theory of genetic inheritance based on discrete inheritable characteristics, which we know as genes.